Assignment 1

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03/10/2025

Setup

```r
library(tidyverse)
```

Q1. Loading the data

We must load the dataset as a tibble, and output all the rows to first understand the different column names and the dimensions of the dataset is 18 rows and 11 columns.

```r
#Question1
(1876802+2)%%3 #1
```

```
## [1] 1
```

```r
question1 <- read_csv("./datasets/peil_1.csv")
question1
```

```
## # A tibble: 18 x 11
##    Year  Team      PROVINCE inj   `Round 1`  `Rnd 2` `Round 3` PQF   QF    SF
##    <chr> <chr>     <chr>    <chr> <chr>      <chr>   <chr>     <chr> <chr> <chr>
##  1 1998  Meath     Leinster 2     away, pla~ home, ~ away, pl~ <NA>  <NA>  <NA>
##  2 1998  Monaghan  Ulster   4     away, pla~ home, ~ home, pl~ away~ <NA>  <NA>
##  3 1998  Louth     Leinster 7     home, pla~ away, ~ away, pl~ home~ away~ <NA>
##  4 *     *         *        *     sonrai ta~ *       *         *     *     *
##  5 1998  Cork      Munster  3     away, pla~ home, ~ home, pl~ away~ <NA>  <NA>
##  6 1998  Galway    Connacht 8     home, pla~ away, ~ away, pl~ home~ away~ away~
##  7 1998  Dublin    Leinster 7     home, pla~ away, ~ home, pl~ <NA>  home~ <NA>
##  8 1998  Westmeath Leinster 1     away, pla~ home, ~ away, pl~ <NA>  <NA>  <NA>
##  9 1998  Kerry     Munster  7     homehomeh~ away, ~ home, pl~ <NA>  home~ away~
## 10 1998  Mayo      CONNCHT  5     home, pla~ away, ~ away, pl~ home~ <NA>  <NA>
## 11 1998  Cavan     Ulster   -4    away, pla~ home, ~ home, pl~ <NA>  <NA>  <NA>
## 12 1998  Armagh    Ulster   6     home, pla~ away, ~ home, pl~ <NA>  home~ home~
## 13 1998  Clare     Munster  3     home, pla~ away, ~ home, pl~ <NA>  <NA>  <NA>
## 14 1998  Donegal   Ulster   5     home, pla~ away, ~ away, pl~ <NA>  home~ home~
## 15 1998  Donegal   Ulster   5     home, pla~ away, ~ away, pl~ <NA>  home~ home~
## 16 1998  Tyrone    Ulster   5     away, pla~ home, ~ away, pl~ home~ <NA>  <NA>
## 17 1998  Derry     Ulster   7     away, pla~ home, ~ home, pl~ away~ away~ <NA>
## 18 1998  Roscommon Connacht 6     away, pla~ home, ~ away, pl~ away~ away~ <NA>
## # i 1 more variable: Final <chr>
```

```r
dim(question1)
```

```
## [1] 18 11
```

Q2. Random permutation of the rows

Setting the correct seed is done to randomly sample all of the rows within the dataset that we assigned to the variable "question1".

```r
#Question2
set.seed(1876802)
question2 <- sample_n(question1, nrow(question1), replace = FALSE)
question2
```

```
## # A tibble: 18 x 11
##    Year  Team      PROVINCE inj   `Round 1`  `Rnd 2` `Round 3` PQF   QF    SF
##    <chr> <chr>     <chr>    <chr> <chr>      <chr>   <chr>     <chr> <chr> <chr>
##  1 1998  Tyrone    Ulster   5     away, pla~ home, ~ away, pl~ home~ <NA>  <NA>
##  2 1998  Kerry     Munster  7     homehomeh~ away, ~ home, pl~ <NA>  home~ away~
##  3 1998  Louth     Leinster 7     home, pla~ away, ~ away, pl~ home~ away~ <NA>
##  4 1998  Armagh    Ulster   6     home, pla~ away, ~ home, pl~ <NA>  home~ home~
##  5 1998  Donegal   Ulster   5     home, pla~ away, ~ away, pl~ <NA>  home~ home~
##  6 1998  Westmeath Leinster 1     away, pla~ home, ~ away, pl~ <NA>  <NA>  <NA>
##  7 1998  Galway    Connacht 8     home, pla~ away, ~ away, pl~ home~ away~ away~
##  8 1998  Clare     Munster  3     home, pla~ away, ~ home, pl~ <NA>  <NA>  <NA>
##  9 1998  Donegal   Ulster   5     home, pla~ away, ~ away, pl~ <NA>  home~ home~
## 10 1998  Mayo      CONNCHT  5     home, pla~ away, ~ away, pl~ home~ <NA>  <NA>
## 11 1998  Derry     Ulster   7     away, pla~ home, ~ home, pl~ away~ away~ <NA>
## 12 1998  Meath     Leinster 2     away, pla~ home, ~ away, pl~ <NA>  <NA>  <NA>
## 13 1998  Cork      Munster  3     away, pla~ home, ~ home, pl~ away~ <NA>  <NA>
## 14 *     *         *        *     sonrai ta~ *       *         *     *     *
## 15 1998  Monaghan  Ulster   4     away, pla~ home, ~ home, pl~ away~ <NA>  <NA>
## 16 1998  Cavan     Ulster   -4    away, pla~ home, ~ home, pl~ <NA>  <NA>  <NA>
## 17 1998  Dublin    Leinster 7     home, pla~ away, ~ home, pl~ <NA>  home~ <NA>
## 18 1998  Roscommon Connacht 6     away, pla~ home, ~ away, pl~ away~ away~ <NA>
## # i 1 more variable: Final <chr>
```

Q3. Adding a column of row numbers

Adding `S ID` column next to the team name column in the dataset.

```r
#Question3
question3 <- question2
question3 <- mutate(question3, `S ID` = 1:nrow(question3))
question3 <- relocate(question3, `S ID`,.before=PROVINCE)
question3
```

```
## # A tibble: 18 x 12
##    Year  Team      `S ID` PROVINCE inj   `Round 1` `Rnd 2` `Round 3` PQF   QF
##    <chr> <chr>      <int> <chr>    <chr> <chr>     <chr>   <chr>     <chr> <chr>
##  1 1998  Tyrone         1 Ulster   5     away, pl~ home, ~ away, pl~ home~ <NA>
##  2 1998  Kerry          2 Munster  7     homehome~ away, ~ home, pl~ <NA>  home~
##  3 1998  Louth          3 Leinster 7     home, pl~ away, ~ away, pl~ home~ away~
##  4 1998  Armagh         4 Ulster   6     home, pl~ away, ~ home, pl~ <NA>  home~
##  5 1998  Donegal        5 Ulster   5     home, pl~ away, ~ away, pl~ <NA>  home~
##  6 1998  Westmeath      6 Leinster 1     away, pl~ home, ~ away, pl~ <NA>  <NA>
##  7 1998  Galway         7 Connacht 8     home, pl~ away, ~ away, pl~ home~ away~
##  8 1998  Clare          8 Munster  3     home, pl~ away, ~ home, pl~ <NA>  <NA>
##  9 1998  Donegal        9 Ulster   5     home, pl~ away, ~ away, pl~ <NA>  home~
```

```
## 10 1998  Mayo          10 CONNCHT  5     home, pl~ away, ~ away, pl~ home~ <NA>
## 11 1998  Derry         11 Ulster   7     away, pl~ home, ~ home, pl~ away~ away~
## 12 1998  Meath         12 Leinster 2     away, pl~ home, ~ away, pl~ <NA>  <NA>
## 13 1998  Cork          13 Munster  3     away, pl~ home, ~ home, pl~ away~ <NA>
## 14 *     *             14 *        *     sonrai t~ *       *         *     *
## 15 1998  Monaghan      15 Ulster   4     away, pl~ home, ~ home, pl~ away~ <NA>
## 16 1998  Cavan         16 Ulster   -4    away, pl~ home, ~ home, pl~ <NA>  <NA>
## 17 1998  Dublin        17 Leinster 7     home, pl~ away, ~ home, pl~ <NA>  home~
## 18 1998  Roscommon     18 Connacht 6     away, pl~ home, ~ away, pl~ away~ away~
## # i 2 more variables: SF <chr>, Final <chr>
```

Row numbers are added to the dataset to be able to track the necessary changes. This is done by adding a column to the right of the team name called `S ID`. All the rows of the dataset have been displayed to see these changes.

Q4. Data cleaning

The process of cleaning the data is important for further analysis.

First we need to rename the columns to the correct column names as specified in the documentation.

```r
question4 <- question3
question4 <- rename(question4, "Inj" = "inj")
question4 <- rename(question4, "Round 2" = "Rnd 2")
```

We can see that renaming these column names creates more continuity within the dataset in correlation with the documentation.

Secondly, we removed all rows that need removing.

```r
filter(question4, `S ID`==14)
```

```
## # A tibble: 1 x 12
##   Year  Team  `S ID` PROVINCE Inj   `Round 1`    `Round 2` `Round 3` PQF   QF
##   <chr> <chr>  <int> <chr>    <chr> <chr>        <chr>     <chr>     <chr> <chr>
## 1 *     *         14 *        *     sonrai tast~ *         *         *     *
## # i 2 more variables: SF <chr>, Final <chr>
```

```r
question4 <- filter(question4, `S ID`!=14)
question4
```

```
## # A tibble: 17 x 12
##    Year  Team    `S ID` PROVINCE Inj   `Round 1` `Round 2` `Round 3` PQF   QF
##    <chr> <chr>    <int> <chr>    <chr> <chr>     <chr>     <chr>     <chr> <chr>
##  1 1998  Tyrone       1 Ulster   5     away, pl~ home, pl~ away, pl~ home~ <NA>
##  2 1998  Kerry        2 Munster  7     homehome~ away, pl~ home, pl~ <NA>  home~
##  3 1998  Louth        3 Leinster 7     home, pl~ away, pl~ away, pl~ home~ away~
##  4 1998  Armagh       4 Ulster   6     home, pl~ away, pl~ home, pl~ <NA>  home~
##  5 1998  Donegal      5 Ulster   5     home, pl~ away, pl~ away, pl~ <NA>  home~
##  6 1998  Westme~      6 Leinster 1     away, pl~ home, pl~ away, pl~ <NA>  <NA>
##  7 1998  Galway       7 Connacht 8     home, pl~ away, pl~ away, pl~ home~ away~
##  8 1998  Clare        8 Munster  3     home, pl~ away, pl~ home, pl~ <NA>  <NA>
##  9 1998  Donegal      9 Ulster   5     home, pl~ away, pl~ away, pl~ <NA>  home~
## 10 1998  Mayo        10 CONNCHT  5     home, pl~ away, pl~ away, pl~ home~ <NA>
## 11 1998  Derry       11 Ulster   7     away, pl~ home, pl~ home, pl~ away~ away~
## 12 1998  Meath       12 Leinster 2     away, pl~ home, pl~ away, pl~ <NA>  <NA>
## 13 1998  Cork        13 Munster  3     away, pl~ home, pl~ home, pl~ away~ <NA>
```

```
## 14 1998  Monagh~     15 Ulster   4     away, pl~ home, pl~ home, pl~ away~ <NA>
## 15 1998  Cavan       16 Ulster   -4    away, pl~ home, pl~ home, pl~ <NA>  <NA>
## 16 1998  Dublin      17 Leinster 7     home, pl~ away, pl~ home, pl~ <NA>  home~
## 17 1998  Roscom~     18 Connacht 6     away, pl~ home, pl~ away, pl~ away~ away~
## # i 2 more variables: SF <chr>, Final <chr>
```

```r
count(question4,Team)
```

```
## # A tibble: 16 x 2
##    Team          n
##    <chr>     <int>
##  1 Armagh        1
##  2 Cavan         1
##  3 Clare         1
##  4 Cork          1
##  5 Derry         1
##  6 Donegal       2
##  7 Dublin        1
##  8 Galway        1
##  9 Kerry         1
## 10 Louth         1
## 11 Mayo          1
## 12 Meath         1
## 13 Monaghan      1
## 14 Roscommon     1
## 15 Tyrone        1
## 16 Westmeath     1
```

```r
question4 <- filter(question4,`S ID`!=5)
question4
```

```
## # A tibble: 16 x 12
##    Year  Team    `S ID` PROVINCE Inj   `Round 1` `Round 2` `Round 3` PQF   QF
##    <chr> <chr>    <int> <chr>    <chr> <chr>     <chr>     <chr>     <chr> <chr>
##  1 1998  Tyrone       1 Ulster   5     away, pl~ home, pl~ away, pl~ home~ <NA>
##  2 1998  Kerry        2 Munster  7     homehome~ away, pl~ home, pl~ <NA>  home~
##  3 1998  Louth        3 Leinster 7     home, pl~ away, pl~ away, pl~ home~ away~
##  4 1998  Armagh       4 Ulster   6     home, pl~ away, pl~ home, pl~ <NA>  home~
##  5 1998  Westme~      6 Leinster 1     away, pl~ home, pl~ away, pl~ <NA>  <NA>
##  6 1998  Galway       7 Connacht 8     home, pl~ away, pl~ away, pl~ home~ away~
##  7 1998  Clare        8 Munster  3     home, pl~ away, pl~ home, pl~ <NA>  <NA>
##  8 1998  Donegal      9 Ulster   5     home, pl~ away, pl~ away, pl~ <NA>  home~
##  9 1998  Mayo        10 CONNCHT  5     home, pl~ away, pl~ away, pl~ home~ <NA>
## 10 1998  Derry       11 Ulster   7     away, pl~ home, pl~ home, pl~ away~ away~
## 11 1998  Meath       12 Leinster 2     away, pl~ home, pl~ away, pl~ <NA>  <NA>
## 12 1998  Cork        13 Munster  3     away, pl~ home, pl~ home, pl~ away~ <NA>
## 13 1998  Monagh~     15 Ulster   4     away, pl~ home, pl~ home, pl~ away~ <NA>
## 14 1998  Cavan       16 Ulster   -4    away, pl~ home, pl~ home, pl~ <NA>  <NA>
## 15 1998  Dublin      17 Leinster 7     home, pl~ away, pl~ home, pl~ <NA>  home~
## 16 1998  Roscom~     18 Connacht 6     away, pl~ home, pl~ away, pl~ away~ away~
## # i 2 more variables: SF <chr>, Final <chr>
```

We identified the test values, where in `S ID` 14, because test data isn't needed for further analysis therefore this row was removed. It is also necessary to cross reference the number of teams evident in the dataset. It was found that the "Donegal" was duplicated within the dataset and therefore we removed the first instance of this team.

Then we removed all of the negative values.

```
question4 <- mutate(question4, Inj=str_replace_all(Inj,"-",""))
```

It is necessary to remove the negative values because it doesn't make sense for a person to have a negative injury. Therefore we converted the negative values into positive.

Next we corrected the provinces in the dataset to make them consistent with the documentation.

```
unique(question4$PROVINCE)
```

```
## [1] "Ulster"   "Munster"  "Leinster" "Connacht" "CONNCHT"
```

```
question4 <- mutate(question4, PROVINCE=str_replace_all(PROVINCE,"CONNCHT",
                                                        "Connacht"))
question4
```

```
## # A tibble: 16 x 12
##    Year  Team    `S ID` PROVINCE Inj   `Round 1` `Round 2` `Round 3` PQF   QF
##    <chr> <chr>    <int> <chr>    <chr> <chr>     <chr>     <chr>     <chr> <chr>
##  1 1998  Tyrone       1 Ulster   5     away, pl~ home, pl~ away, pl~ home~ <NA>
##  2 1998  Kerry        2 Munster  7     homehome~ away, pl~ home, pl~ <NA>  home~
##  3 1998  Louth        3 Leinster 7     home, pl~ away, pl~ away, pl~ home~ away~
##  4 1998  Armagh       4 Ulster   6     home, pl~ away, pl~ home, pl~ <NA>  home~
##  5 1998  Westme~      6 Leinster 1     away, pl~ home, pl~ away, pl~ <NA>  <NA>
##  6 1998  Galway       7 Connacht 8     home, pl~ away, pl~ away, pl~ home~ away~
##  7 1998  Clare        8 Munster  3     home, pl~ away, pl~ home, pl~ <NA>  <NA>
##  8 1998  Donegal      9 Ulster   5     home, pl~ away, pl~ away, pl~ <NA>  home~
##  9 1998  Mayo        10 Connacht 5     home, pl~ away, pl~ away, pl~ home~ <NA>
## 10 1998  Derry       11 Ulster   7     away, pl~ home, pl~ home, pl~ away~ away~
## 11 1998  Meath       12 Leinster 2     away, pl~ home, pl~ away, pl~ <NA>  <NA>
## 12 1998  Cork        13 Munster  3     away, pl~ home, pl~ home, pl~ away~ <NA>
## 13 1998  Monagh~     15 Ulster   4     away, pl~ home, pl~ home, pl~ away~ <NA>
## 14 1998  Cavan       16 Ulster   4     away, pl~ home, pl~ home, pl~ <NA>  <NA>
## 15 1998  Dublin      17 Leinster 7     home, pl~ away, pl~ home, pl~ <NA>  home~
## 16 1998  Roscom~     18 Connacht 6     away, pl~ home, pl~ away, pl~ away~ away~
## # i 2 more variables: SF <chr>, Final <chr>
```

It was important to cross reference the list of Provinces in the dataset and we renamed the "CONNCHT" to "Connacht" because there was a typo.

We need to check that there are no inconsistencies in the data in "Round 1".

```
select(question4, Team,"Round 1")
```

```
## # A tibble: 16 x 2
##    Team      `Round 1`
##    <chr>     <chr>
##  1 Tyrone    away, played Donegal, scored 14 against 21, attendance 16607
##  2 Kerry     homehomehome, played Monaghan, scored 24 against 14, attendance 80~
##  3 Louth     home, played Meath, scored 19 against 9, attendance 14399
##  4 Armagh    home, played Westmeath, scored 16 against 11, attendance 5989
##  5 Westmeath away, played Armagh, scored 11 against 16, attendance 5989
##  6 Galway    home, played Derry, scored 20 against 15, attendance 7602
##  7 Clare     home, played Cork, scored 14 against 16, attendance 3262
##  8 Donegal   home, played Tyrone, scored 21 against 14, attendance 16607
##  9 Mayo      home, played Cavan, scored 20 against 13, attendance 9197
## 10 Derry     away, played Galway, scored 15 against 20, attendance 7602
```

```
## 11 Meath     away, played Louth, scored 9 against 19, attendance 14399
## 12 Cork      away, played Clare, scored 16 against 14, attendance 3262
## 13 Monaghan  away, played Kerry, scored 14 against 24, attendance 8092
## 14 Cavan     away, played Mayo, scored 13 against 20, attendance 9197
## 15 Dublin    home, played Roscommon, scored 25 against 13, attendance 11176
## 16 Roscommon away, played Dublin, scored 13 against 25, attendance 11176
```

```
filter(question4, Team =="Kerry")%>% select("Round 1")
```

```
## # A tibble: 1 x 1
##   `Round 1`
##   <chr>
## 1 homehomehome, played Monaghan, scored 24 against 14, attendance 8092
```

```
question4[2,6]
```

```
## # A tibble: 1 x 1
##   `Round 1`
##   <chr>
## 1 homehomehome, played Monaghan, scored 24 against 14, attendance 8092
```

```
question4[2,6]<- "home, played Monaghan, scored 24 against 14, attendance 8092"
```

We checked for Round 1 and found out where row with team "Kerry" had "homehomehome", which is seen as a typo. Therefore we replaced it with "home".

Next we checked Round 2 to make sure that there were no errors evident in the dataset.

```
select(question4, Team, "Round 2")
```

```
## # A tibble: 16 x 2
##    Team      `Round 2`
##    <chr>     <chr>
##  1 Tyrone    home, played Clare, scored XX against 10, attendance 3000
##  2 Kerry     away, played Meath, scored 24 against 9, attendance 8224
##  3 Louth     away, played Monaghan, scored 16 against 16, attendance 11329
##  4 Armagh    away, played Derry, scored 26 against 15, attendance 10000
##  5 Westmeath home, played Galway, scored 11 against 15, attendance 4315
##  6 Galway    away, played Westmeath, scored 15 against 11, attendance 4315
##  7 Clare     away, played Tyrone, scored 10 against 24, attendance 3000
##  8 Donegal   away, played Cork, scored 16 against 15, attendance 7251
##  9 Mayo      away, played Roscommon, scored 20 against 18, attendance 8597
## 10 Derry     home, played Armagh, scored 15 against 26, attendance 10000
## 11 Meath     home, played Kerry, scored 9 against 24, attendance 8224
## 12 Cork      home, played Donegal, scored 15 against 16, attendance 7251
## 13 Monaghan  home, played Louth, scored 16 against 16, attendance 11329
## 14 Cavan     home, played Dublin, scored 13 against 32, attendance 9028
## 15 Dublin    away, played Cavan, scored 32 against 13, attendance 9028
## 16 Roscommon home, played Mayo, scored 18 against 20, attendance 8597
```

```
filter(question4, Team == "Tyrone" | Team == "Clare")%>% select("Round 2")
```

```
## # A tibble: 2 x 1
##   `Round 2`
##   <chr>
## 1 home, played Clare, scored XX against 10, attendance 3000
## 2 away, played Tyrone, scored 10 against 24, attendance 3000
```

```r
question4[1,7]
```

```
## # A tibble: 1 x 1
##   `Round 2`
##   <chr>
## 1 home, played Clare, scored XX against 10, attendance 3000
```

```r
question4[1,7]<-"home, played Clare, scored 24 against 10, attendance 3000"
select(question4, Team, "Round 2")
```

```
## # A tibble: 16 x 2
##    Team      `Round 2`
##    <chr>     <chr>
##  1 Tyrone    home, played Clare, scored 24 against 10, attendance 3000
##  2 Kerry     away, played Meath, scored 24 against 9, attendance 8224
##  3 Louth     away, played Monaghan, scored 16 against 16, attendance 11329
##  4 Armagh    away, played Derry, scored 26 against 15, attendance 10000
##  5 Westmeath home, played Galway, scored 11 against 15, attendance 4315
##  6 Galway    away, played Westmeath, scored 15 against 11, attendance 4315
##  7 Clare     away, played Tyrone, scored 10 against 24, attendance 3000
##  8 Donegal   away, played Cork, scored 16 against 15, attendance 7251
##  9 Mayo      away, played Roscommon, scored 20 against 18, attendance 8597
## 10 Derry     home, played Armagh, scored 15 against 26, attendance 10000
## 11 Meath     home, played Kerry, scored 9 against 24, attendance 8224
## 12 Cork      home, played Donegal, scored 15 against 16, attendance 7251
## 13 Monaghan  home, played Louth, scored 16 against 16, attendance 11329
## 14 Cavan     home, played Dublin, scored 13 against 32, attendance 9028
## 15 Dublin    away, played Cavan, scored 32 against 13, attendance 9028
## 16 Roscommon home, played Mayo, scored 18 against 20, attendance 8597
```

It can be seen that in Team "Tyrone" points are marked "XX" in the game that has an attendance of 3000 and "Clare" team scored 10 points. From the corresponding game in the "Clare" row we can see that the missing value is 24 when "Tyrone" played at home. This has been corrected by replacing the "XX" value with 24.

Next we further cross referenced PQF and QF to make sure that there were no errors in the dataset. We have also corrected any errors evident in the dataset.

```r
select(question4, Team, PQF)
```

```
## # A tibble: 16 x 2
##    Team      PQF
##    <chr>     <chr>
##  1 Tyrone    home, played Roscommon, scored 12 against 14, attendance 6000
##  2 Kerry     <NA>
##  3 Louth     home, played Cork, scored 12 against 11, attendance 7292
##  4 Armagh    <NA>
##  5 Westmeath <NA>
##  6 Galway    home, played Monaghan, scored 14 against 11, attendance 6768
##  7 Clare     <NA>
##  8 Donegal   <NA>
##  9 Mayo      home, played Derry, scored 15 against 15, attendance 13955
## 10 Derry     away, played Mayo, scored 15 against 15, attendance 13955
## 11 Meath     <NA>
## 12 Cork      away, played Louth, scored 11 against 12, attendance 7292
## 13 Monaghan  away, played Galway, scored 11 against 14, attendance 6768
```

```
## 14 Cavan     <NA>
## 15 Dublin    <NA>
## 16 Roscommon away, played Tyrone, scored 14 against 12, attendance 6000
```

```
select(question4, Team, QF)
```

```
## # A tibble: 16 x 2
##    Team      QF
##    <chr>     <chr>
##  1 Tyrone    <NA>
##  2 Kerry     home, played Derry, scored 15 against 10, attendance 47406
##  3 Louth     away, played Donegal, scored 18 against 26, attendance 4444444444
##  4 Armagh    home, played Roscommon, scored 18 against 12, attendance 49896
##  5 Westmeath <NA>
##  6 Galway    away, played Dublin, scored 17 against 16, attendance 49896
##  7 Clare     <NA>
##  8 Donegal   home, played Louth, scored 26 against 18, attendance 47406
##  9 Mayo      <NA>
## 10 Derry     away, played Kerry, scored 10 against 15, attendance 47406
## 11 Meath     <NA>
## 12 Cork      <NA>
## 13 Monaghan  <NA>
## 14 Cavan     <NA>
## 15 Dublin    home, played Galway, scored 16 against 17, attendance 49896
## 16 Roscommon away, played Armagh, scored 12 against 18, attendance 49896
```

```
filter(question4, Team=="Louth" | Team =="Donegal")%>% select(QF)
```

```
## # A tibble: 2 x 1
##   QF
##   <chr>
## 1 away, played Donegal, scored 18 against 26, attendance 4444444444
## 2 home, played Louth, scored 26 against 18, attendance 47406
```

```
question4[3,10]
```

```
## # A tibble: 1 x 1
##   QF
##   <chr>
## 1 away, played Donegal, scored 18 against 26, attendance 4444444444
```

```
question4[3,10]<- "away, played Donegal, scored 18 against 26, attendance 47406"
question4
```

```
## # A tibble: 16 x 12
##    Year  Team    `S ID` PROVINCE Inj   `Round 1` `Round 2` `Round 3` PQF   QF
##    <chr> <chr>    <int> <chr>    <chr> <chr>     <chr>     <chr>     <chr> <chr>
##  1 1998  Tyrone       1 Ulster   5     away, pl~ home, pl~ away, pl~ home~ <NA>
##  2 1998  Kerry        2 Munster  7     home, pl~ away, pl~ home, pl~ <NA>  home~
##  3 1998  Louth        3 Leinster 7     home, pl~ away, pl~ away, pl~ home~ away~
##  4 1998  Armagh       4 Ulster   6     home, pl~ away, pl~ home, pl~ <NA>  home~
##  5 1998  Westme~      6 Leinster 1     away, pl~ home, pl~ away, pl~ <NA>  <NA>
##  6 1998  Galway       7 Connacht 8     home, pl~ away, pl~ away, pl~ home~ away~
##  7 1998  Clare        8 Munster  3     home, pl~ away, pl~ home, pl~ <NA>  <NA>
##  8 1998  Donegal      9 Ulster   5     home, pl~ away, pl~ away, pl~ <NA>  home~
##  9 1998  Mayo        10 Connacht 5     home, pl~ away, pl~ away, pl~ home~ <NA>
## 10 1998  Derry       11 Ulster   7     away, pl~ home, pl~ home, pl~ away~ away~
```

```
## 11 1998  Meath       12 Leinster 2     away, pl~ home, pl~ away, pl~ <NA>  <NA>
## 12 1998  Cork        13 Munster  3     away, pl~ home, pl~ home, pl~ away~ <NA>
## 13 1998  Monagh~     15 Ulster   4     away, pl~ home, pl~ home, pl~ away~ <NA>
## 14 1998  Cavan       16 Ulster   4     away, pl~ home, pl~ home, pl~ <NA>  <NA>
## 15 1998  Dublin      17 Leinster 7     home, pl~ away, pl~ home, pl~ <NA>  home~
## 16 1998  Roscom~     18 Connacht 6     away, pl~ home, pl~ away, pl~ away~ away~
## # i 2 more variables: SF <chr>, Final <chr>
```

It can be seen that there are no errors in PQF. There is an error in QF in row where team name is "Louth" and the attendance is "4444444444". This has been resolved by finding the corresponding game "Donegal" to match the attendance results and Louth's attendance has been replaced with the value "47406".

Finally, we selected the Season Final (SF), and the Final to check that there were no errors present.

```r
select(question4, Team, SF)
```

```
## # A tibble: 16 x 2
##    Team      SF
##    <chr>     <chr>
##  1 Tyrone    <NA>
##  2 Kerry     away, played Armagh, scored 19 against 21, attendance 55548
##  3 Louth     <NA>
##  4 Armagh    home, played Kerry, scored 21 against 19, attendance 55548
##  5 Westmeath <NA>
##  6 Galway    away, played Donegal, scored 17 against 15, attendance 67002
##  7 Clare     <NA>
##  8 Donegal   home, played Galway, scored 15 against 17, attendance 67002
##  9 Mayo      <NA>
## 10 Derry     <NA>
## 11 Meath     <NA>
## 12 Cork      <NA>
## 13 Monaghan  <NA>
## 14 Cavan     <NA>
## 15 Dublin    <NA>
## 16 Roscommon <NA>
```

```r
select(question4, Team, Final)
```

```
## # A tibble: 16 x 2
##    Team      Final
##    <chr>     <chr>
##  1 Tyrone    <NA>
##  2 Kerry     <NA>
##  3 Louth     <NA>
##  4 Armagh    home, played Galway, scored 14 against 13, attendance 82300
##  5 Westmeath <NA>
##  6 Galway    away, played Armagh, scored 13 against 14, attendance 82300
##  7 Clare     <NA>
##  8 Donegal   <NA>
##  9 Mayo      <NA>
## 10 Derry     <NA>
## 11 Meath     <NA>
## 12 Cork      <NA>
## 13 Monaghan  <NA>
## 14 Cavan     <NA>
## 15 Dublin    <NA>
```

```
## 16 Roscommon <NA>
```

It is also important to note that the NA values are necessary to be included in this dataset as they represent the teams who didn't make it to the SF or Final games when considering the attendance rates and in correlation with the best goal scorers.

Q5. Data tidying

Q5(a). Converting to long form

To begin the tidying process we must first melt the data, this is where we convert wide form to long form and turn columns into rows.

```
#Question5
#5a
question5 <- gather(question4, key = "RND", value = "Details", `Round 1`:
                      `Final`)
head(question5, 10)
```

```
## # A tibble: 10 x 7
##    Year  Team      `S ID` PROVINCE Inj   RND     Details
##    <chr> <chr>      <int> <chr>    <chr> <chr>   <chr>
##  1 1998  Tyrone         1 Ulster   5     Round 1 away, played Donegal, scored 1~
##  2 1998  Kerry          2 Munster  7     Round 1 home, played Monaghan, scored ~
##  3 1998  Louth          3 Leinster 7     Round 1 home, played Meath, scored 19 ~
##  4 1998  Armagh         4 Ulster   6     Round 1 home, played Westmeath, scored~
##  5 1998  Westmeath      6 Leinster 1     Round 1 away, played Armagh, scored 11~
##  6 1998  Galway         7 Connacht 8     Round 1 home, played Derry, scored 20 ~
##  7 1998  Clare          8 Munster  3     Round 1 home, played Cork, scored 14 a~
##  8 1998  Donegal        9 Ulster   5     Round 1 home, played Tyrone, scored 21~
##  9 1998  Mayo          10 Connacht 5     Round 1 home, played Cavan, scored 20 ~
## 10 1998  Derry         11 Ulster   7     Round 1 away, played Galway, scored 15~
```

We now know that each row in the long form is an individual measurement: * The year of the given season (Col 1) * The team name (Col 2) * The row numbers (Col 3) * The province in which the country team must play (Col 4) * The number of injuries sustained by the team during the given game (Col 5) * The Round number (Col 6) * The details of each round (Col 7) The long form is useful for further data analysis.

Q5(b). Adding new columns.

It is important to add new columns to better reflect their specific roles within the dataset, where each column represents a variable and each row represents an team (observation).

```
#5b
question5 <- mutate(question5,
                    PFor = str_match(question5$Details, "scored (\\d+)")[,2],
                    POpp = str_match(question5$Details, "against (\\d+)")[,2],
                    `Home Game` = str_match(question5$Details, "([:alpha:]+)")[,2],
                    Spec = str_match(question5$Details, "attendance (\\d+)")[,2])
question5
```

```
## # A tibble: 112 x 11
##    Year  Team  `S ID` PROVINCE Inj   RND   Details PFor  POpp  `Home Game` Spec
##    <chr> <chr>  <int> <chr>    <chr> <chr> <chr>   <chr> <chr> <chr>       <chr>
##  1 1998  Tyro~      1 Ulster   5     Roun~ away, ~ 14    21    away        16607
##  2 1998  Kerry      2 Munster  7     Roun~ home, ~ 24    14    home        8092
```

```
##   3 1998  Louth      3 Leinster 7     Roun~ home, ~ 19    9     home         14399
##   4 1998  Arma~      4 Ulster   6     Roun~ home, ~ 16    11    home         5989
##   5 1998  West~      6 Leinster 1     Roun~ away, ~ 11    16    away         5989
##   6 1998  Galw~      7 Connacht 8     Roun~ home, ~ 20    15    home         7602
##   7 1998  Clare      8 Munster  3     Roun~ home, ~ 14    16    home         3262
##   8 1998  Done~      9 Ulster   5     Roun~ home, ~ 21    14    home         16607
##   9 1998  Mayo      10 Connacht 5     Roun~ home, ~ 20    13    home         9197
##  10 1998  Derry     11 Ulster   7     Roun~ away, ~ 15    20    away         7602
## # i 102 more rows
```

To add new columns we used the mutate function in conjunction with the str_match function to extract the specific data from the Details column. After creating these columns the information in the table is exactly the same as seen in previous questions, it is just stored in a tidier way.

Q5(b). Deleting the Details column.

Now that all the important information from the details column have been extracted and made into different rows we have no use for this column any longer, hence it is deleted.

```r
#5c
question5 <- mutate(question5, Details = NULL )

head(question5, 10)
```

```
## # A tibble: 10 x 10
##    Year  Team      `S ID` PROVINCE Inj   RND     PFor  POpp  `Home Game` Spec
##    <chr> <chr>      <int> <chr>    <chr> <chr>   <chr> <chr> <chr>       <chr>
##  1 1998  Tyrone         1 Ulster   5     Round 1 14    21    away        16607
##  2 1998  Kerry          2 Munster  7     Round 1 24    14    home        8092
##  3 1998  Louth          3 Leinster 7     Round 1 19    9     home        14399
##  4 1998  Armagh         4 Ulster   6     Round 1 16    11    home        5989
##  5 1998  Westmeath      6 Leinster 1     Round 1 11    16    away        5989
##  6 1998  Galway         7 Connacht 8     Round 1 20    15    home        7602
##  7 1998  Clare          8 Munster  3     Round 1 14    16    home        3262
##  8 1998  Donegal        9 Ulster   5     Round 1 21    14    home        16607
##  9 1998  Mayo          10 Connacht 5     Round 1 20    13    home        9197
## 10 1998  Derry         11 Ulster   7     Round 1 15    20    away        7602
```

```r
dim(question5)
```

```
## [1] 112  10
```

As we assign Details to NULL, this is the process of removing the column to make the log form data more concise.

Q6. Desciption of the variables.

- **Year**: This variable is categorical ordinal, as it is presented as a label and has an order. This is seen where the year 1998 is a point of time between 12:00pm, 1 Jan 1998 - 11:59pm, 31 Dec 1998. Although the dataset does not include other years as a comparison, it is still seen as a categorical ordinal variable. This variable could be removed later if needed because it is unnecessary to keep this value as the season is only in the year 1998.
- **Team**: This is a categorical nominal as it is seen as a label that doesn't have a specific order in this dataset.

- `S ID`: This is a categorical ordinal variable as the row numbers are seen as a label that has a sequential order(1,2,3. . .).
- **PROVINCE**: This is a categorical nominal variable as a label with no order. This column only represents the location in which the team played the game, either in Ulster, Munster, Linster, or Connacht.
- **Inj**: This is a quantitative discrete variable as there can't be infinitely divided number of critical injuries per team member There can only exist one injury per team member.
- **RND**: This is a categorical ordinal variable as it is seen as a label that has a distinct order within the season, where Round 3 cannot come before Round 1.
- **Pfor**: This is a quantitative discrete variable as these are the points are scored by the team this is represented to have a smallest unit. In Gaelic Football this is measured as either (3 points) or (1 point). They are added together as whole numbers and can't be infinitely divided.
- **POpp**: This is a quantitative discrete variable and is represented as the same format as a the Pfor variable where it portrays the opposing teams scores and has a smallest unit that can only be measured as a whole number. It is also to note that a quantitative discrete variable doesn't necessary have to be a whole number, in this instance it is.
- **Home Game**: This is represented as a categorical nominal variable that does not have an order, either portrayed as "home" or "away" from the given Team in that row. It can also be seen as a logical variable "yes/no" or "TRUE/FALSE" in correlation to if the given teams status of the game.
- **Spec**: This is seen as a quantitative discrete value, where you can't have infinitely divided spectators at a game. For example, half a spectator cannot attend the game.

Q7. Taming the data.

It is important to tame the data to increase efficiency and effectiveness when understanding the likeliness of the AIFDC players will be injured.

We must adhere to naming conventions for the given variable names.

```
#Question7
question7 <- question5
question7 <- rename(question7, year= Year,
                  team = Team,
                  s_id = `S ID`,
                  province = PROVINCE,
                  inj = Inj,
                  rnd = RND,
                  pfor = PFor,
                  popp = POpp,
                  home_game = `Home Game`,
                  spec = Spec)
question7
```

```
## # A tibble: 112 x 10
##    year  team       s_id province inj   rnd     pfor  popp  home_game spec
##    <chr> <chr>     <int> <chr>    <chr> <chr>   <chr> <chr> <chr>     <chr>
##  1 1998  Tyrone        1 Ulster   5     Round 1 14    21    away      16607
##  2 1998  Kerry         2 Munster  7     Round 1 24    14    home      8092
##  3 1998  Louth         3 Leinster 7     Round 1 19    9     home      14399
##  4 1998  Armagh        4 Ulster   6     Round 1 16    11    home      5989
##  5 1998  Westmeath     6 Leinster 1     Round 1 11    16    away      5989
```

```
## 6 1998 Galway        7 Connacht 8     Round 1 20    15    home      7602
## 7 1998 Clare         8 Munster  3     Round 1 14    16    home      3262
## 8 1998 Donegal       9 Ulster   5     Round 1 21    14    home      16607
## 9 1998 Mayo         10 Connacht 5     Round 1 20    13    home      9197
## 10 1998 Derry       11 Ulster   7     Round 1 15    20    away      7602
## # i 102 more rows
```

Using the rename function we have renamed the variables to be fewer than 20 characters long, using lower case and underscores between words to avoid spaces. This is important to adhere to the style guides used in R, and to enhance the overall readability.

Next it is important to accurately represent the given variables as ordered factors, factors, logicals and integers.

```r
question7$year<-as.integer(question7$year)
question7$team<-as.factor(question7$team)
question7$s_id<-as.ordered(question7$s_id)
question7$province<- as.factor(question7$province)
question7$inj<-as.integer(question7$inj)
question7$rnd<-as.ordered(question7$rnd)
question7$pfor<-as.integer(question7$pfor)
question7$popp<-as.integer(question7$popp)
question7$home_game <- ifelse(question7$home_game == "home",TRUE, FALSE)
question7$home_game <- as.logical(question7$home_game)
question7$spec <- as.integer(question7$spec)
question7
```

```
## # A tibble: 112 x 10
##     year team      s_id  province   inj rnd      pfor  popp home_game  spec
##    <int> <fct>     <ord> <fct>    <int> <ord>   <int> <int> <lgl>     <int>
##  1  1998 Tyrone    1     Ulster       5 Round 1    14    21 FALSE     16607
##  2  1998 Kerry     2     Munster      7 Round 1    24    14 TRUE       8092
##  3  1998 Louth     3     Leinster     7 Round 1    19     9 TRUE      14399
##  4  1998 Armagh    4     Ulster       6 Round 1    16    11 TRUE       5989
##  5  1998 Westmeath 6     Leinster     1 Round 1    11    16 FALSE      5989
##  6  1998 Galway    7     Connacht     8 Round 1    20    15 TRUE       7602
##  7  1998 Clare     8     Munster      3 Round 1    14    16 TRUE       3262
##  8  1998 Donegal   9     Ulster       5 Round 1    21    14 TRUE      16607
##  9  1998 Mayo      10    Connacht     5 Round 1    20    13 TRUE       9197
## 10  1998 Derry     11    Ulster       7 Round 1    15    20 FALSE      7602
## # i 102 more rows
```

```r
head(question7, 10)
```

```
## # A tibble: 10 x 10
##     year team      s_id  province   inj rnd      pfor  popp home_game  spec
##    <int> <fct>     <ord> <fct>    <int> <ord>   <int> <int> <lgl>     <int>
##  1  1998 Tyrone    1     Ulster       5 Round 1    14    21 FALSE     16607
##  2  1998 Kerry     2     Munster      7 Round 1    24    14 TRUE       8092
##  3  1998 Louth     3     Leinster     7 Round 1    19     9 TRUE      14399
##  4  1998 Armagh    4     Ulster       6 Round 1    16    11 TRUE       5989
##  5  1998 Westmeath 6     Leinster     1 Round 1    11    16 FALSE      5989
##  6  1998 Galway    7     Connacht     8 Round 1    20    15 TRUE       7602
##  7  1998 Clare     8     Munster      3 Round 1    14    16 TRUE       3262
##  8  1998 Donegal   9     Ulster       5 Round 1    21    14 TRUE      16607
##  9  1998 Mayo      10    Connacht     5 Round 1    20    13 TRUE       9197
```

```
## 10  1998 Derry      11     Ulster        7 Round 1     15     20 FALSE      7602
```

```r
dim(question7)
```

```
## [1] 112  10
```

In correlation with Question 6 that describes the type of variables, they have been changed to be represented correctly as portrayed in the tame data conventions. It is important to note that the home_game column that represents the location of the given team, is seen as a "yes/no" character encoding of binary values and is converted to "TRUE/FALSE", where the home game is default to "TRUE".

Q8. Adding new columns.

We are adding new columns that represent the difference between the cores of the team and their opponent (pdiff), and the difference in score per spectator (pdiffspec).

```r
#Question8
#8a
question8 <- question7
question8 <- mutate(question8, pdiff= pfor-popp)
question8 <- mutate(question8, pdiffspec = pdiff/spec)
```

Using the mutate function, these columns are relevant to quantify how likely players are to get injured in top scoring teams. The difference in score per spectator is important to consider if the influence of a larger crowd has an affect on the teams score.

We are now relocating these columns.

```r
#8b
question8 <- relocate(question8, "pdiff", .after = popp)
question8 <- relocate(question8, "pdiffspec", .after = pdiff)
question8
```

```
## # A tibble: 112 x 12
##     year team   s_id  province   inj rnd    pfor  popp pdiff pdiffspec home_game
##    <int> <fct>  <ord> <fct>    <int> <ord> <int> <int> <int>     <dbl> <lgl>
##  1  1998 Tyrone 1     Ulster       5 Roun~    14    21    -7 -0.000422 FALSE
##  2  1998 Kerry  2     Munster      7 Roun~    24    14    10  0.00124  TRUE
##  3  1998 Louth  3     Leinster     7 Roun~    19     9    10  0.000694 TRUE
##  4  1998 Armagh 4     Ulster       6 Roun~    16    11     5  0.000835 TRUE
##  5  1998 Westm~ 6     Leinster     1 Roun~    11    16    -5 -0.000835 FALSE
##  6  1998 Galway 7     Connacht     8 Roun~    20    15     5  0.000658 TRUE
##  7  1998 Clare  8     Munster      3 Roun~    14    16    -2 -0.000613 TRUE
##  8  1998 Doneg~ 9     Ulster       5 Roun~    21    14     7  0.000422 TRUE
##  9  1998 Mayo   10    Connacht     5 Roun~    20    13     7  0.000761 TRUE
## 10  1998 Derry  11    Ulster       7 Roun~    15    20    -5 -0.000658 FALSE
## # i 102 more rows
## # i 1 more variable: spec <int>
```

Re-positioning these columns to the right of the column that record the opponent's score better depicts the influence of these values on the teams performance.

The first 10 rows and the dimensions of the dataset have been output to depict the changes made.

```r
#8c
head(question8, 10)
```

```
## # A tibble: 10 x 12
##     year team   s_id  province   inj rnd    pfor  popp pdiff pdiffspec home_game
```

```
##    <int> <fct>  <ord> <fct>    <int> <ord> <int> <int> <int>     <dbl> <lgl>
##  1  1998 Tyrone 1     Ulster       5 Roun~    14    21    -7 -0.000422 FALSE
##  2  1998 Kerry  2     Munster      7 Roun~    24    14    10  0.00124  TRUE
##  3  1998 Louth  3     Leinster     7 Roun~    19     9    10  0.000694 TRUE
##  4  1998 Armagh 4     Ulster       6 Roun~    16    11     5  0.000835 TRUE
##  5  1998 Westm~ 6     Leinster     1 Roun~    11    16    -5 -0.000835 FALSE
##  6  1998 Galway 7     Connacht     8 Roun~    20    15     5  0.000658 TRUE
##  7  1998 Clare  8     Munster      3 Roun~    14    16    -2 -0.000613 TRUE
##  8  1998 Doneg~ 9     Ulster       5 Roun~    21    14     7  0.000422 TRUE
##  9  1998 Mayo   10    Connacht     5 Roun~    20    13     7  0.000761 TRUE
## 10  1998 Derry  11    Ulster       7 Roun~    15    20    -5 -0.000658 FALSE
## # i 1 more variable: spec <int>
```

```r
dim(question8)
```

```
## [1] 112  12
```

We must consider these new variables and what they represent

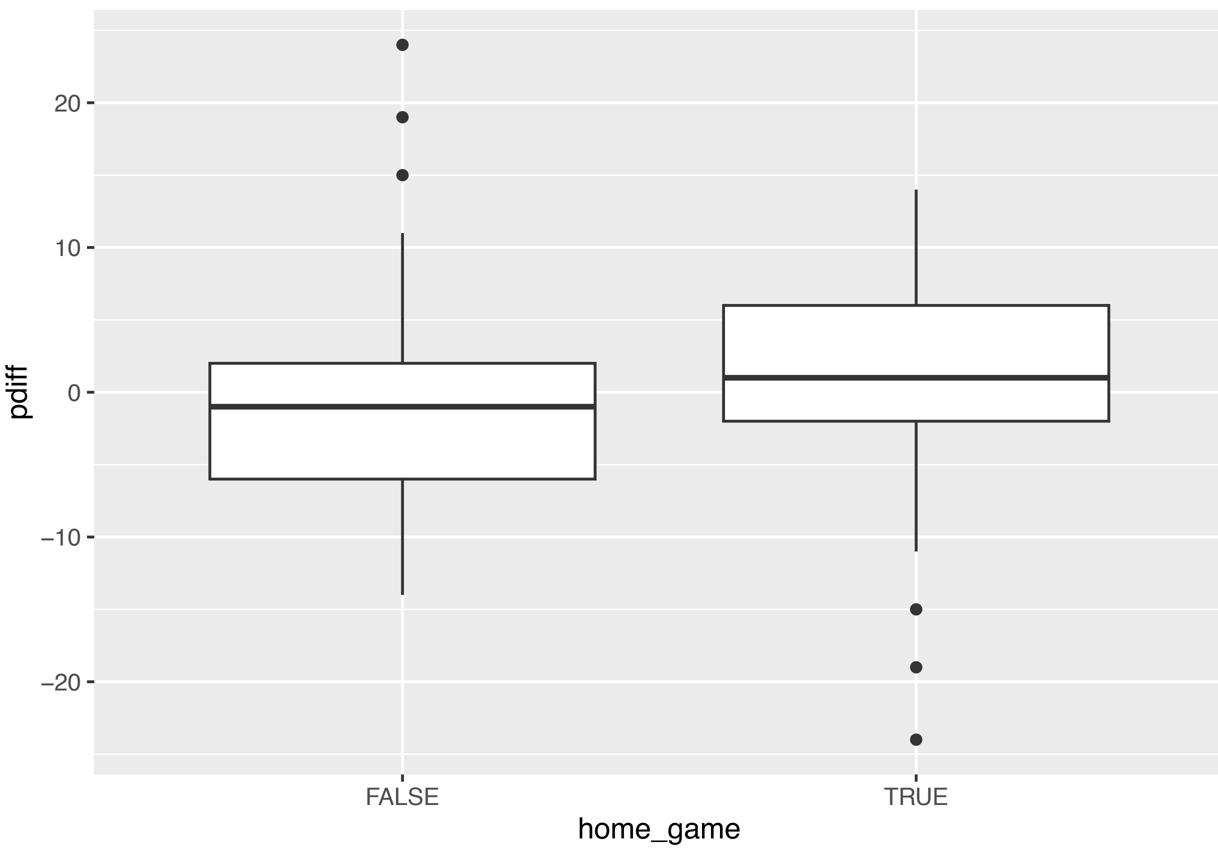
8d The new types of variables are represented below: * **pdiff**: This is a quantitative discrete variable that depicts the difference between scores of the team and the opponent. * **pdiffspec**: This depicts the difference in score per spectator and is important to consider the influence of spectators on the teams performance.

Q9.Boxplots depicting win/loss margin.

To further visualise the win/loss margin **pdiff** for home and away games, a boxplot has been generated.

```r
#Question9
question9 <- question8

ggplot(filter(question9, !is.na(home_game)), aes(home_game,pdiff))+
  geom_boxplot()
```



The median value in the away games (the FALSE boxplot) is lower than the median value in the home games (the TRUE boxplot), this means that the difference in scores increased when playing a home game. This reinforces that playing games at home may ensure a better performance in correlation with an away game. It is also important to recognise the outliers present in the dataset in the false boxplot, where some games have resulted in a win in an away game and vice versa in the true boxplot.

Q10.Investigate how the number of injuries is affected by the number of

points scored in the season.

We are now going to look into the number of injuries in correspondence with the number of points scored in the season.

```
#Question10
#10a
question10 <- question8
injury_record <- summarise(group_by(question10, team),
          total_inj = sum(inj, na.rm=TRUE),
          pfor_total = sum(pfor,na.rm=TRUE))
```

We are using the summarise() command to build a new tibble called injury_record that contains the team name, total number of injuries over the season, and the total number of points scored over the season. It is important to note that we are using question8 data instead of question9. This is because the NA values are important when evaluating the season and the teams that didn't make the Preliminary Quarter Final, Quarter Final, Quarter Final, Semi Final and Final.

Since we are trying to investigate how likely it is that the AIFDC players will be injured, creating another column called injury_record will help to quantify this.

```
#10b
injury_record <- mutate(injury_record, inj_per_pt = total_inj/pfor_total)
```

It is then important to rearrange this in correlation with the total number of points scored in the season.

```
#10c
injury_record <- arrange(injury_record, pfor_total)
```

The below displays the injury_record table.

```
#10d
injury_record
```

```
## # A tibble: 16 x 4
##    team      total_inj pfor_total inj_per_pt
##    <fct>         <int>      <int>      <dbl>
##  1 Clare            21         29      0.724
##  2 Westmeath         7         31      0.226
##  3 Meath            14         35      0.4
##  4 Cavan            28         49      0.571
##  5 Cork             21         59      0.356
##  6 Monaghan         28         64      0.438
##  7 Derry            49         68      0.721
##  8 Tyrone           35         71      0.493
##  9 Mayo             35         72      0.486
## 10 Louth            49         78      0.628
## 11 Roscommon        42         86      0.488
## 12 Dublin           49         90      0.544
## 13 Donegal          35        107      0.327
## 14 Kerry            49        109      0.450
## 15 Armagh           42        110      0.382
## 16 Galway           56        111      0.505
```

Q11 New Types of Variables .

The new columns in Q10 are depicted as the following variables:

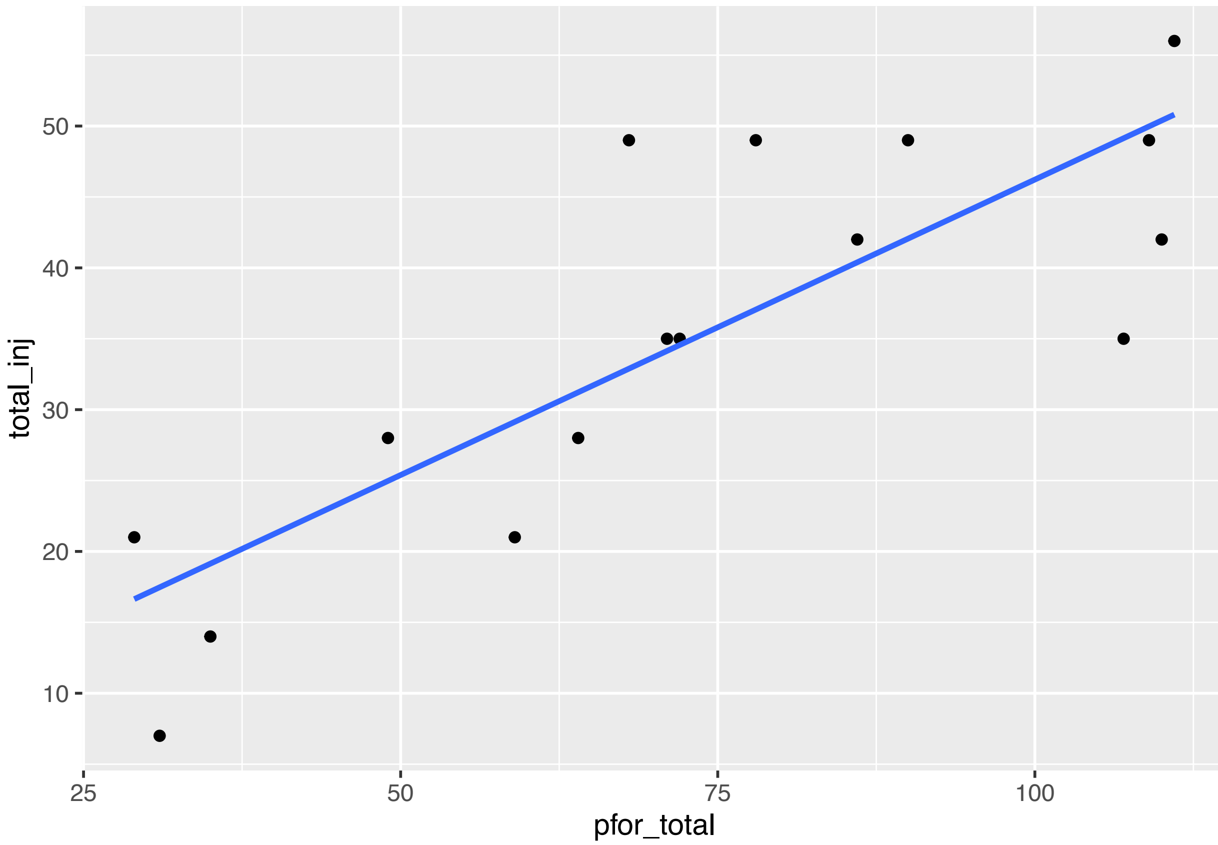
- **total_inj**: This is quantitative discrete, representing the total number of injuries per game. It is discrete because a singular injury cannot be infinity divided per team player.
- **pfor_total**:This is a quantitative discrete variable and represents the number of points the team scored in the season.
- **inj_per_pt**: This is a quantitative continuous because it represents the average number of injuries per point.

Q12 Generating scatterplots.

It is important to visualise this data and the total number of points scored over the season against the total number of injuries.

```
#Question12
#12a
question12 <- injury_record
```

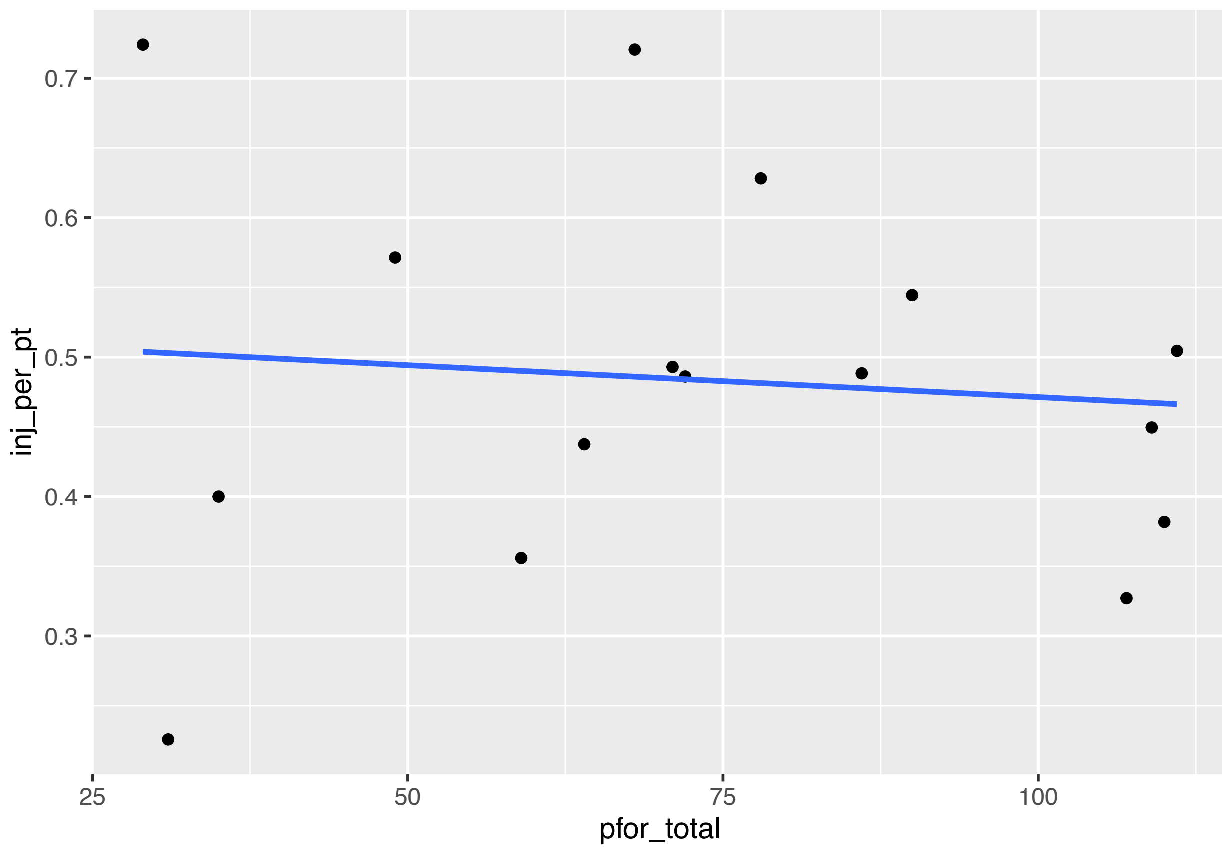
```r
ggplot(question12, aes(x=pfor_total ,y=total_inj))+
  geom_point()+
  geom_smooth(method="lm",se=FALSE)
```



The explanatory variable is the pfor_total variable because as the number of points increase (number of games played), the number of injuries also increase as the players are more susceptible to injury.

It is also important to evaluate the number of points scored over the season against the average number of injuries per point

```r
#12b
ggplot(question12, aes(x=pfor_total ,y=inj_per_pt))+
  geom_point()+
  geom_smooth(method="lm",se=FALSE)
```



The explanatory variable is the pfor_total because as the average number of injuries per point decreases the points scored over the season decreases. This is because the better players are less likely to be injured as there are more highly qualified games.

Q13 Final analysis.

The plots in Question 12 can tell us that the first plot show us the total number of points scored over the season (x axis) against the total number of injuries (y axis). This plot portrays that as the number of points scored over the season increase, the number of injuries also increase as a strong positive is depicted where the points generally increase. The AIFDC can expect if they enter a team in the All-Ireland Football Champion they might be likely to expect that as the season continues on from Round 1, Round 2… Final. The players who compete in these later rounds are more likely to have an injury. It is advised that the AIFDC focus on the on field injury management of the players, and create more places on the team to allow for substitute players to reduce exertion and stress resulting in serious injury.

The second plot in Question 12 show us that the total number of point scored over the season (x axis) against the average number of injuries per point (y axis). This plot depicts a weak negative relationship and suggests that the better players are less likely to be injured during the later rounds in the seasons. AIFDC can expect that these games may result in a better performance as there may be less serious injuries that occur.

Assignment 3

Jessica Evans

28/11/2025

Setup

```
library(tidyverse)
library(tidymodels)
library(inspectdf)
library(moments)
library(caret)
library(car)
```

Q1. Loading the data

In line with the deliverable specifications, I chose the correct dataset using my student number + 1 modulo 3, resulting in dataset 0, as seen below.

```
(1876802 + 1)%%3
```

```
## [1] 0
```

```
question1 <- read_csv("./a3_data/gortaithe_0.csv")
```

Secondly, the dataset is loaded as a tibble to access the data in the first 10 rows, this is done by using the tidyverse package (this is a form of metapackage), it contains a number of different libraries. The dimensions of the tibble are 20250 rows by 7 columns.

```
head(question1, 10)
```

```
## # A tibble: 10 x 7
##      INJ   AGE HEIGHT WEIGHT POS      REGION HOURS
##    <dbl> <dbl>  <dbl>  <dbl> <chr>    <chr>  <dbl>
##  1     1    21   191.   81.1 back     C      10.2
##  2     0    29   177.   85   forward  A      11.3
##  3     1    24   177.   83.6 goalie   D       6.86
##  4     0    27   183.   90.6 back     B       5.84
##  5     0    27   190.   91   midfield A       8.94
##  6     1    29   176.   89.8 back     C       7.32
##  7     1    27   180.   80.3 back     A      10.8
##  8     0    26   179.   91.1 back     D      13.9
##  9     0    32   178    72.5 midfield B       8.16
## 10     0    26   178.   84.8 forward  C      12.3
```

```
dim(question1)
```

```
## [1] 20250     7
```

Q2. Randomly Permute all Rows in the Dataset

Random permutation is essential to eliminate poor generalisation and bias.

```
question2 <- question1
set.seed(1876802)
question2 <- sample_n(question2, nrow(question2), replace = FALSE)
question2
```

```
## # A tibble: 20,250 x 7
##      INJ   AGE HEIGHT WEIGHT POS      REGION HOURS
##    <dbl> <dbl>  <dbl>  <dbl> <chr>    <chr>  <dbl>
##  1     0    26   184.   93.1 forward  C       6.81
##  2     0    27   189.   89.3 goalie   D      11.6
##  3     0    22   190.   92.2 forward  C       8.88
##  4     0    25   178.   82.5 back     D       5.28
##  5     1    29   182.   87.9 forward  A       6.02
##  6     1    32   181.   76.8 midfield A       8.18
##  7     1    34   184.   88.9 goalie   C      11.7
##  8     1    34   177.   70.3 forward  D       9.76
##  9     1    30   181.   81.5 back     C      10.1
## 10     1    24   184.   90   forward  C       6.16
## # i 20,240 more rows
```

Q3. Variable Types Present in the Dataset

It is important to identify the types of variables present for later interpretation.

- *INJ* - This is ***categorical nominal***, where 1 represents if the player has a serious injury, and 0 means that they do not. It is a label with no distinct order, where 1 does not represent a higher number of injuries within itself.
- *AGE* - This is a ***quantitative discrete*** variable, it represents the age of each player at the beginning of the current season. Determined by the number of birthdays the player has had up until the current season start date *01/01/2024*.
- *HEIGHT* - This is a ***quantitative discrete*** variable, as it represents the height of a player, where it is not continuous because it is rounded off to the nearest millimeter.
- *WEIGHT* - This is also a ***quantitative discrete*** variable, as it represents the weight of a player, where it is not continuous because it is rounded off to the nearest tenth of a kilogram.
- *POS* - This is a ***categorical nominal*** variable, as it represents a labeled field position that a player has, there are only four positions including the following;
  - back.
  - forward.
  - goalie.
  - midfield.

Within these positions, there is no distinct order, therefore it cannot be ordered based on ranking.

- *REGION* - This is classified as ***categorical ordinal*** variable, a label with a specific order. The names of the regions were removed and categorised with letters “A”, “B”, and “C” into regional groups as seen below:
  - ***region “A”***: This recorded the least player injuries.
  - ***region “B”***: This recorded the second fewest injuries.
  - ***region “C”***: This records the most injuries.

- *HOURS* - This is ***quantitative continuous*** representing the total number of hours the players spent on the field during the season. It is not specified how this is measured, therefore we can classify it as continuous.

Q4. Cleaning, Tidying, and Taming the Data

The cleaning that needs to be done is seen below:

It can be seen that there are no NA values present in the dataset.

```
question4 <- question2
check_na <- inspect_na(question4)
check_na
```

```
## # A tibble: 7 x 3
##   col_name   cnt  pcnt
##   <chr>    <int> <dbl>
## 1 INJ          0     0
## 2 AGE          0     0
## 3 HEIGHT       0     0
## 4 WEIGHT       0     0
## 5 POS          0     0
## 6 REGION       0     0
## 7 HOURS        0     0
```

The INJ column has some values that are not 0 or 1. The computer cannot tell the difference between 0 and any number smaller than $\approx 1 \times 10^{-15}$.

```
check_inj <- unique(question4$INJ)
check_inj
```

```
##  [1] 0.000000e+00 1.000000e+00 2.495478e-17 8.251577e-17 5.711577e-17
##  [6] 5.023714e-17 3.744046e-17 8.099098e-17 3.934469e-17 6.675037e-17
## [11] 8.747426e-17 5.287773e-17 2.037378e-17 2.072887e-17 3.590886e-17
## [16] 4.053093e-17 9.199241e-17 9.549577e-17 3.056932e-17 9.323723e-17
## [21] 6.194121e-17 8.285383e-17 6.000340e-17 4.586855e-17 1.580671e-17
## [26] 5.486727e-17 2.492803e-17 8.250391e-17 9.819727e-18 8.410564e-17
## [31] 4.651021e-17 3.418745e-17
```

For reference, I checked the unique function on all the other columns, they did not have any additional values.

Changing the variable name INJ:

The variable ***INJ*** is currently labeled as 1, meaning injured and 0 meaning not injured.

To make this consistent for the AIFDC to read involves converting these variables to the following:

- "ta" if the player is injured, in Irish language meaning "yes".
- "nil" otherwise meaning "is not".

```
question4$INJ <- ifelse(question4$INJ==1, "ta","nil")
```

It is important to tame all variable names, to reduce error and further follow the tidyverse conventions to maintain continuity.

```
question4 <- rename(question4, inj = INJ)
question4 <- rename(question4, age = AGE)
question4 <- rename(question4, height = HEIGHT)
question4 <- rename(question4, weight = WEIGHT)
question4 <- rename(question4, pos = POS)
```

```
question4 <- rename(question4, region = REGION)
question4 <- rename(question4, hours = HOURS)
```

It is important to treat any quantitative discrete variables as *quantitative continuous*, this is because we will be fitting geometric objects later, this assumes that all the predictors are continuous.

This applies to the following variables:

- age
- height
- weight

It is important to cross reference that the variable types are in the correct format. The variables *age*, *height*, *weight*, and *hours* are already in the correct type as numeric .

The variables *inj*, *pos*, and *region* must be converted to the factor types as they are categorical nominal.

```
question4 <- mutate(question4,
                    inj = as.factor(inj),
                    pos = as.factor(pos),
                    region = as.factor(region),
)
```

Finally the first 10 rows of the output of the cleaned, tamed, and tidied dataset is seen below.

```
head(question4, 10)
```

```
## # A tibble: 10 x 7
##    inj     age height weight pos      region hours
##    <fct> <dbl>  <dbl>  <dbl> <fct>    <fct>  <dbl>
##  1 nil      26   184.   93.1 forward  C       6.81
##  2 nil      27   189.   89.3 goalie   D      11.6
##  3 nil      22   190.   92.2 forward  C       8.88
##  4 nil      25   178.   82.5 back     D       5.28
##  5 ta       29   182.   87.9 forward  A       6.02
##  6 ta       32   181.   76.8 midfield A       8.18
##  7 ta       34   184.   88.9 goalie   C      11.7
##  8 ta       34   177.   70.3 forward  D       9.76
##  9 ta       30   181.   81.5 back     C      10.1
## 10 ta       24   184.   90   forward  C       6.16
```

```
dim(question4)
```

```
## [1] 20250     7
```

Q5. Removing Categorical Predictors

For further analysis, we want to determine the likelihood if Charming will be injured or not. This involves fitting a logistic regression model, which can only contain quantitative variables.

We must keep the response variable *inj*, and all 4 quantitative variables including the *age*, *height*, *weight*, and *hours*.

This dataset will be used for the rest of the assignment.

```
question5 <- question4
gortu_q <- bind_cols(
  question5["inj"],
  question5["age"],
```

```
  question5["height"],
  question5["weight"],
  question5["hours"],
)
gortu_q
```

```
## # A tibble: 20,250 x 5
##    inj     age height weight hours
##    <fct> <dbl>  <dbl>  <dbl> <dbl>
##  1 nil      26   184.   93.1  6.81
##  2 nil      27   189.   89.3 11.6
##  3 nil      22   190.   92.2  8.88
##  4 nil      25   178.   82.5  5.28
##  5 ta       29   182.   87.9  6.02
##  6 ta       32   181.   76.8  8.18
##  7 ta       34   184.   88.9 11.7
##  8 ta       34   177.   70.3  9.76
##  9 ta       30   181.   81.5 10.1
## 10 ta       24   184.   90    6.16
## # i 20,240 more rows
```

Q6. Splitting the Data into a Training and Testing Set

It is important to split the data into a training with 16,000 rows and testing set with the remaining rows. This is because it helps us create an unbiased estimate of the model for seasons in the future.

Setting the seed to *1876802* is important to do so that the data is able to be run the same way.

```
set.seed(1876802)
question6_split <- initial_split(gortu_q, prop = 16000/20250)

question6_train <- training(question6_split)
question6_test <- testing(question6_split)
```

This is then output into the first 10 rows to check with the dimensions of the new training and testing datasets.

```
head(question6_train, 10)
```

```
## # A tibble: 10 x 5
##    inj     age height weight hours
##    <fct> <dbl>  <dbl>  <dbl> <dbl>
##  1 nil      29   183.   84.7  7.32
##  2 nil      28   182.   93.1  7.94
##  3 nil      29   192    81.7  9.16
##  4 ta       28   182.   79.7  9.10
##  5 ta       28   180.   68.7 13.6
##  6 ta       24   183.   73.5  7.63
##  7 ta       26   185.   79.4  6.02
##  8 nil      26   176.   74   11.0
##  9 nil      32   192.   74.4  7.85
## 10 ta       34   192.   89.6 11.6
```

```
dim(question6_train)
```

```
## [1] 16000     5
```

```r
head(question6_test, 10)
```

```
## # A tibble: 10 x 5
##    inj     age height weight hours
##    <fct> <dbl>  <dbl>  <dbl> <dbl>
##  1 nil      27   189.   89.3 11.6
##  2 ta       29   182.   87.9  6.02
##  3 ta       34   177.   70.3  9.76
##  4 ta       30   181.   81.5 10.1
##  5 nil      29   175.   88.6  9.52
##  6 ta       28   176.   77.5  9.15
##  7 nil      30   175.   79.7  9.31
##  8 ta       25   181    82.4 12.1
##  9 nil      28   189.   85    5.97
## 10 nil      34   170.   85.7  6.64
```

```r
dim(question6_test)
```

```
## [1] 4250    5
```

Q7. Fitting a Logistic Regression Model

The logistic regression model is fitted using the training dataset *question6_train*. The response variable is *inj*, as we want to predict if a player will be injured. We are using a logistic regression model because it is used to predict a binary outcomes (if the player is injured or not).

It helps to determine how the independent variables being *age*, *height*, *weight*, and *hours*.

```r
question7_lr <- logistic_reg()%>%
  set_engine("glm")%>%
  fit(inj ~ ., data = question6_train)
```

The summary command output in the estimate column can tell us more about our predictions.

```r
summary(question7_lr$fit)
```

```
##
## Call:
## stats::glm(formula = inj ~ ., family = stats::binomial, data = data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.723412   0.590488  -2.919  0.00352 **
## age          0.133548   0.004948  26.990  < 2e-16 ***
## height       0.015458   0.002947   5.245 1.56e-07 ***
## weight      -0.074204   0.002337 -31.756  < 2e-16 ***
## hours        0.109212   0.006452  16.926  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 22027  on 15999  degrees of freedom
## Residual deviance: 19910  on 15995  degrees of freedom
## AIC: 19920
##
```

```
## Number of Fisher Scoring iterations: 4
```

- age: this is positive, this means that as a players age increases, it leaves them more vulnerable to injury over time.
- height: this is also positive, indicating that as a players height increases they are also less likely to recieve injuries.
- weight: this estimate is negative, this indicates that as a players weight increases, they are less likely to recieve injury in comparison to other players.
- hours: this is positive, as an increased amount of hours on the field is likely to lead to injury.

Q8. Geometric Object

Geometric objects are determined by the number of quantitative predictors in our models. There are 4 quantitative predictors including *age*, *height*, *weight*, and *hours*.

This means that we have a 4 dimensional surface in a 5 dimensional space. The extra dimension represents the response variable, for every predictor this increases the dimension of space.

Q9. Analysis of Variance

To determine the significance of any interaction term to predict the likelihood of injury we fitted a logistic regression model which included all 4 quantitative predictors and the interaction terms. The Analysis of Variance (ANOVA) is used to find the p values for each of the variables.

```
question9_lr <- logistic_reg() %>%
  set_engine("glm")%>%
  fit(inj~.^2, data=question6_train)
summary(question9_lr$fit)
```

```
##
## Call:
## stats::glm(formula = inj ~ .^2, family = stats::binomial, data = data)
##
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)    31.1099473  7.9603234   3.908 9.30e-05 ***
## age            -0.8728252  0.1693535  -5.154 2.55e-07 ***
## height          0.0043024  0.0419401   0.103   0.9183
## weight         -0.4764612  0.0802724  -5.936 2.93e-09 ***
## hours           0.0553067  0.2228071   0.248   0.8040
## age:height     -0.0001597  0.0008672  -0.184   0.8539
## age:weight      0.0117932  0.0007103  16.603  < 2e-16 ***
## age:hours       0.0048136  0.0019187   2.509   0.0121 *
## height:weight   0.0002769  0.0004156   0.666   0.5052
## height:hours   -0.0007721  0.0011278  -0.685   0.4936
## weight:hours    0.0006742  0.0009024   0.747   0.4550
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 22027  on 15999  degrees of freedom
## Residual deviance: 19605  on 15989  degrees of freedom
```

```
## AIC: 19627
##
## Number of Fisher Scoring iterations: 4
```

```
Anova(question9_lr$fit)
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: inj
##                LR Chisq Df Pr(>Chisq)
## age              777.61  1  < 2.2e-16 ***
## height            26.17  1  3.132e-07 ***
## weight          1114.88  1  < 2.2e-16 ***
## hours            293.61  1  < 2.2e-16 ***
## age:height         0.03  1    0.85390
## age:weight       291.14  1  < 2.2e-16 ***
## age:hours          6.30  1    0.01211 *
## height:weight      0.44  1    0.50514
## height:hours       0.47  1    0.49353
## weight:hours       0.56  1    0.45498
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Q10. Backwards Stepwise Regression

All of the interactions term were identified at a 96% significance level. The only interaction terms that were significant included age:weight, and age:hours.

The remainder of the terms had p values larger than 0.04, were removed sequentially and the model was refitted after each time as seen below.

```
question10_glm <- logistic_reg()%>%
  set_engine("glm")%>%
  fit(inj ~ age + height + weight + hours + age:height + age:weight
      + age:hours + height:weight + height:hours + weight:hours,
      data = question6_train)

Anova(question10_glm$fit)
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: inj
##                LR Chisq Df Pr(>Chisq)
## age              777.61  1  < 2.2e-16 ***
## height            26.17  1  3.132e-07 ***
## weight          1114.88  1  < 2.2e-16 ***
## hours            293.61  1  < 2.2e-16 ***
## age:height         0.03  1    0.85390
## age:weight       291.14  1  < 2.2e-16 ***
## age:hours          6.30  1    0.01211 *
## height:weight      0.44  1    0.50514
## height:hours       0.47  1    0.49353
## weight:hours       0.56  1    0.45498
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The interaction term age:height was first removed as it was not significant, as it had the largest p value.

```
question10_glm <- logistic_reg()%>%
  set_engine("glm")%>%
  fit(inj ~ age + height + weight + hours + age:weight
      + age:hours + height:weight + height:hours + weight:hours,
      data = question6_train)

Anova(question10_glm$fit)
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: inj
##                LR Chisq Df Pr(>Chisq)
## age              777.61  1  < 2.2e-16 ***
## height            26.17  1  3.132e-07 ***
## weight          1114.73  1  < 2.2e-16 ***
## hours            293.63  1  < 2.2e-16 ***
## age:weight       291.20  1  < 2.2e-16 ***
## age:hours          6.30  1    0.01205 *
## height:weight      0.42  1    0.51704
## height:hours       0.46  1    0.49995
## weight:hours       0.56  1    0.45438
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The interaction term height:weight was next removed as there was no evidence of significance.

```
question10_glm <- logistic_reg()%>%
  set_engine("glm")%>%
  fit(inj ~ age + height + weight + hours + age:weight
      + age:hours + height:hours + weight:hours,
      data = question6_train)

Anova(question10_glm$fit)
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: inj
##               LR Chisq Df Pr(>Chisq)
## age             778.04  1  < 2.2e-16 ***
## height           26.17  1  3.132e-07 ***
## weight         1114.73  1  < 2.2e-16 ***
## hours           293.60  1  < 2.2e-16 ***
## age:weight      291.06  1  < 2.2e-16 ***
## age:hours         6.27  1    0.01227 *
## height:hours      0.42  1    0.51871
## weight:hours      0.54  1    0.46262
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Next, the interaction term height:hours was removed as it was not significant.

```
question10_glm <- logistic_reg()%>%
  set_engine("glm")%>%
  fit(inj ~ age + height + weight + hours + age:weight
      + age:hours +  weight:hours,
```

```r
      data = question6_train)

Anova(question10_glm$fit)
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: inj
##                LR Chisq Df Pr(>Chisq)
## age              778.00  1  < 2.2e-16 ***
## height            26.17  1  3.132e-07 ***
## weight          1114.78  1  < 2.2e-16 ***
## hours            293.60  1  < 2.2e-16 ***
## age:weight       291.35  1  < 2.2e-16 ***
## age:hours          6.31  1    0.01202 *
## weight:hours       0.50  1    0.47931
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

After refitting the model it revealed that weight:hours was also insignificant.

```r
question10_glm <- logistic_reg()%>%
  set_engine("glm")%>%
  fit(inj ~ age + height + weight + hours + age:weight
      + age:hours ,
      data = question6_train)

Anova(question10_glm$fit)
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: inj
##              LR Chisq Df Pr(>Chisq)
## age            778.15  1  < 2.2e-16 ***
## height          26.16  1  3.137e-07 ***
## weight        1114.78  1  < 2.2e-16 ***
## hours          293.60  1  < 2.2e-16 ***
## age:weight     290.87  1  < 2.2e-16 ***
## age:hours        6.93  1   0.008472 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

During this process the principal of marginality was applied here, where all the higher order interactions remaining, also include the corresponding lower order terms.

After removing all of the interaction terms that weren't significant, our model only has two interaction terms that passed the 96% significance test on a consistent basis including age:weight and age:hours.

Q11. Interaction Terms Significance

11a) The Significant terms of interaction

In the last model, the interaction terms that are significant at 96% and the p value is less than 0.04 includes the following:

- age:weight - the p value is p< 2.2 x 10-16
- age: hours - the p value is 0.0085

11b) Why might these interaction terms might represent real effects

- age:weight: This interaction term may represent that older players that have an larger body weight are more likely to result in injury. This is because their weight on their joints and increase in age may result in a higher chance that they may be more susceptible to injury.
- age:hours: It is evident that more time on the field could lead to injury. This interaction term might indicate that older players who have a high amount of hours playing on the field may lack the endurance or stamina to keep up with the other players, and therefore are more likely to get injured.

Q12. Logistic Regression Model

$$\log\left(\frac{\hat{p}_i}{1-\hat{p}_i}\right) = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{\beta}_3 x_{3i} + \hat{\beta}_4 x_{4i} + \hat{\beta}_5 x_{5i} + \hat{\beta}_6 (x_{1i} \times x_{3i}) + \hat{\beta}_7 (x_{1i} \times x_{4i})$$

$\hat{p}_i$: The estimated chance that player is injured.

x_{1i}: The age of the player.

x_{2i}: The height of the player.

x_{3i}: The weight of the player.

x_{4i}: The total hours a player has played on the field during the season.

$x_{1i} \times x_{3i}$: The combined effect of the players age and weight.

$x_{2i} \times x_{4i}$: The combined effect of age and the number of hours a player spent on the field during that season.

This is important because it allows for correct interpenetration and further allows for analysis of the geometric situation in question 13 below.

Q13. Geometric Object Final Model

The geometric model that the final model describes include 4 quantitative predictors including age, height, weight, and hours. This means that within these four variables, there is a 4 dimensions identified with the players age, height, weight and hours played.

The logistic regression model involves writing the equation in the form of a log-odds scale that includes an extra dimension that is the predicted log-odds of the likelihood that a player will receive an injury. This means that the model will be in a 5 dimensional space including the four predictors and the fact that we are making a prediction of something.

It is evident that there are also two interaction terms including age:weight, and age:hours. These interaction terms do not add up to the dimensions as it is not a new independent axis, it is a collection of the current existing axes. They only changes the shape of the hyperplane, and does not create a new independent direction.

Q14. Summary of the Model

```r
question14 <- summary(question10_glm$fit)
question14
```

```
##
## Call:
## stats::glm(formula = inj ~ age + height + weight + hours + age:weight +
##     age:hours, family = stats::binomial, data = data)
```

```
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept) 28.5324495  1.8939157  15.065  < 2e-16 ***
## age         -0.9017017  0.0613648 -14.694  < 2e-16 ***
## height       0.0151683  0.0029695   5.108 3.26e-07 ***
## weight      -0.4179724  0.0210381 -19.867  < 2e-16 ***
## hours       -0.0354709  0.0557447  -0.636  0.52457
## age:weight   0.0117647  0.0007088  16.599  < 2e-16 ***
## age:hours    0.0050034  0.0019007   2.632  0.00848 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 22027  on 15999  degrees of freedom
## Residual deviance: 19606  on 15993  degrees of freedom
## AIC: 19620
##
## Number of Fisher Scoring iterations: 4
```

From the summary output we can determine the logistic regression equation for the log odds equation as seen below:

$$\log\left(\frac{\hat{p}_i}{1-\hat{p}_i}\right) = 28.53 - 0.9017x_{1i} + 0.0152x_{2i} - 0.418x_{3i} - 0.0355x_{4i} + 0.0118(x_{1i} \times x_{3i}) + 0.0050034(x_{1i} \times x_{4i})$$

This is where:

$$\begin{aligned} x_{1i} &= age \\ x_{2i} &= height \\ x_{3i} &= weight \\ x_{4i} &= hours \end{aligned}$$

hours is included because of the principal of marginality.

Q15. Estimate for the Injury Log-Odds

#a) A heavy player who weighs 100kgs, and spends 10 hours on the field over the season.

$$\begin{aligned} x_{3i} &= 100(weight) \\ x_{4i} &= 10(hours) \end{aligned}$$

$$
\begin{aligned}
\log\left(\frac{\hat{p}_i}{1-\hat{p}_i}\right) &= 28.53 - 0.901x_{1i} + 0.015x_{2i} - 0.418(100) - 0.0355(10) + 0.0118(x_{1i} \times 100) + 0.00500(x_{1i} \times 10) \\
&= 28.53 - 0.901x_{1i} + 0.015x_{2i} - 41.8 - 0.355 + 0.011(x_{1i} \times 100) + 0.00500(x_{1i} \times 10) \\
&= -13.625 - 0.901x_{1i} + 0.015x_{2i} + 1.1(x_{1i}) + 0.05(x_{1i}) \\
&= -13.625 + 0.249x_{1i} + 0.015x_{2i}
\end{aligned}
$$

It is evident through the positive constant 28.53, this means that the player that is 100kgs and plays 10 hours on the field has a higher baseline injury. This means that they have higher liklihood of being injured as each hour passes.

#b) A small player who is only straight out of high school and 18 years old, weighing 67kgs, and is 170cm tall.

$$
\begin{aligned}
x_{1i} &= 18(age) \\
x_{2i} &= 170(height) \\
x_{3i} &= 67(weight)
\end{aligned}
$$

$$
\begin{aligned}
\log\left(\frac{\hat{p}_i}{1-\hat{p}_i}\right) &= 28.53 - 0.901(18) + 0.0152(170) - 0.418(67) - 0.0355x_{4i} + 0.0118(18 \times 67) + 0.00500(18 \times x_{4i}) \\
&= -13.1 - 0.0355x_{4i} + 14.2 + 0.00500(18 \times x_{4i}) \\
&= 1.1 - 0.035x_{4i} + 0.0900x_{4i} \\
&= 1.1 + 0.126x_{4i}
\end{aligned}
$$

A player who is only 18 years old, weighing 67kgs, and is 170cm tall, as a positive baseline injury log odds of -13.1 this means that as each hour passes when the player is on the field there is an less chance of that player getting injured.

Q16. Applying the Final Model to the Testing Data

The testing data was applied to the final model and the results were used to estimate the likelihood of a player to be injured.

```
question16_preds <- predict(question10_glm, new_data = question6_test,
                            type = 'class')
question16_preds
```

```
## # A tibble: 4,250 x 1
##    .pred_class
##    <fct>
##  1 nil
##  2 nil
##  3 ta
##  4 ta
##  5 nil
##  6 ta
##  7 ta
```

```
##   8 nil
##   9 nil
## 10 nil
## # i 4,240 more rows
```

```r
question16_probs <- predict(question10_glm, new_data = question6_test,
                            type = 'prob')
question16_preds
```

```
## # A tibble: 4,250 x 1
##    .pred_class
##    <fct>
##   1 nil
##   2 nil
##   3 ta
##   4 ta
##   5 nil
##   6 ta
##   7 ta
##   8 nil
##   9 nil
## 10 nil
## # i 4,240 more rows
```

```r
question16_preds <- bind_cols(question16_preds, question16_probs)
question16_preds <- mutate(question16_preds, inj = question6_test$inj)

question16_preds <- relocate(question16_preds, inj, .before = .pred_nil)
```

The first 10 rows indicate the prediction of the class to the one predicted. It also portrays the likelihood of the model for each prediction. The entire table has the dimensions of 4,250 rows and 4 columns, this match those of the test dataset.

```r
head(question16_preds,10)
```

```
## # A tibble: 10 x 4
##    .pred_class inj   .pred_nil .pred_ta
##    <fct>       <fct>     <dbl>    <dbl>
##   1 nil         nil       0.681    0.319
##   2 nil         ta        0.719    0.281
##   3 ta          ta        0.353    0.647
##   4 ta          ta        0.479    0.521
##   5 nil         nil       0.673    0.327
##   6 ta          ta        0.489    0.511
##   7 ta          nil       0.494    0.506
##   8 nil         ta        0.596    0.404
##   9 nil         nil       0.678    0.322
## 10 nil         nil       0.550    0.450
```

```r
dim(question16_preds)
```

```
## [1] 4250    4
```

Q17. Evaluate the Model

17a) The confusion matrix and accuracy of the model.

The confusion portrays the performance of the model for predicting if a player recieves an injury or no injury. There are 1179 true positives (TP), 733 false negatives(FN), 727 false positives (FP), and 1571 true negatives(TN). The accuracy of this is 64.7, this means that it is only accurate 64.7% of the time.

```
question17 <- question16_preds %>% conf_mat(.pred_class, truth = inj)
question17
```

```
##           Truth
## Prediction  nil   ta
##        nil 1571  773
##        ta   727 1179
```

```
TP <- 1179
FN <- 773
FP <- 727
TN <- 1571

accuracy <- (TP + TN)/(TP+TN+FN+FP)
accuracy
```

```
## [1] 0.6470588
```

17b) Finding the Sensitivity and Specificity of the Model if "being injured"

is classified as a success.

The sensitivity is 0.6039959 equating to 60.4% of players actually sustained an injury. This means that there are some flaws in our model, where it is only able to identify injuries more that 50% of the time.

```
sensitivity <- TP / (TP+FN)
sensitivity
```

```
## [1] 0.6039959
```

The specificity is 0.6836379, this means that the model predicts 68.4% of players who weren't injured. This shows that the specificity is more accurate in comparison to the sensitivity, where it is more accurate in predicting if players are not injured.

```
specificity <- TN / (TN+FP)
specificity
```

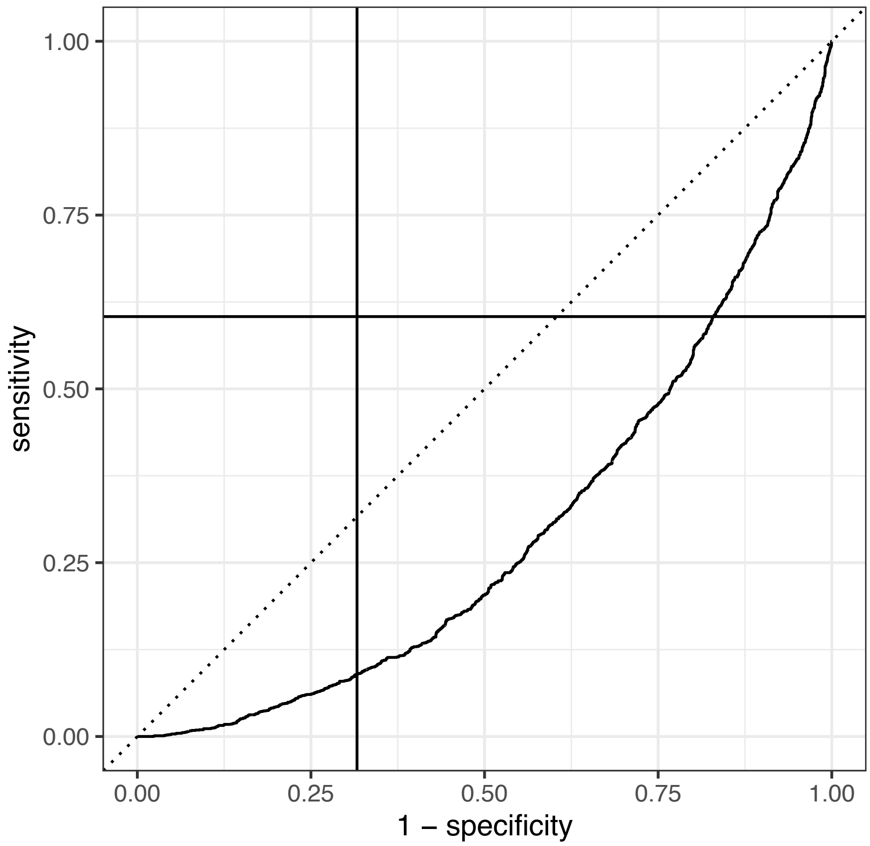
```
## [1] 0.6836379
```

17c) Plotting all Possible ROC curves.

The four possible ROC curves are seen below with alternating event_levels and .pred_nill and .pred_ta.

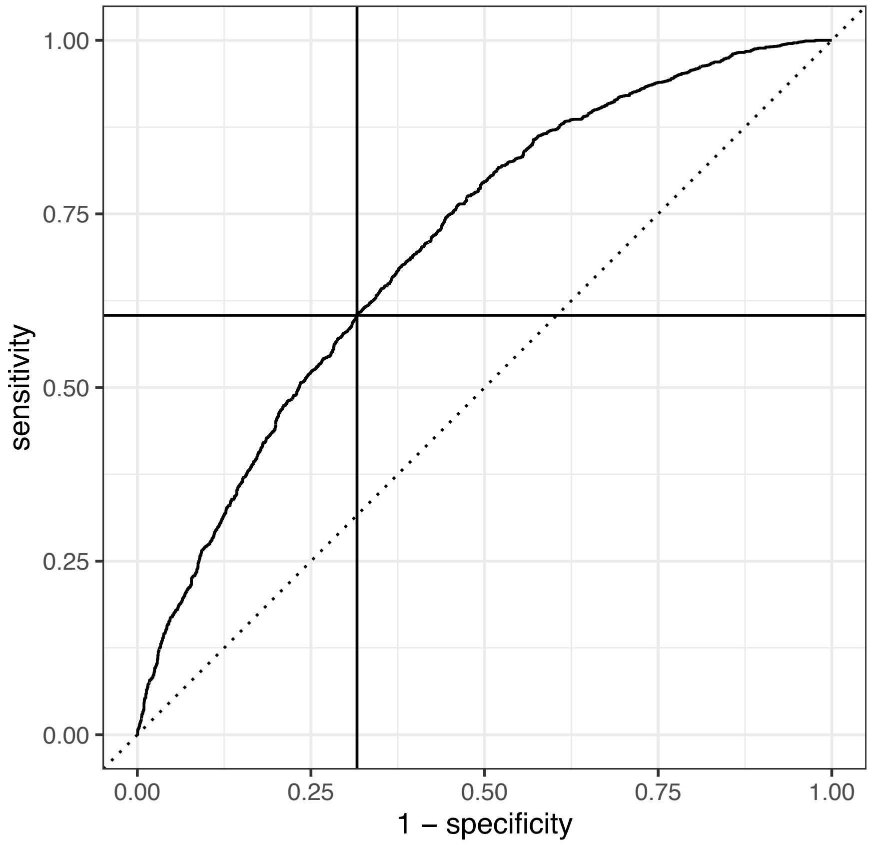
```
question16_preds %>%
  roc_curve(
    .pred_nil,
    truth = inj,
    event_level = "second"
) %>%
autoplot ()+
```

```r
  geom_vline(xintercept = 1-specificity)+
  geom_hline(yintercept = sensitivity)
```

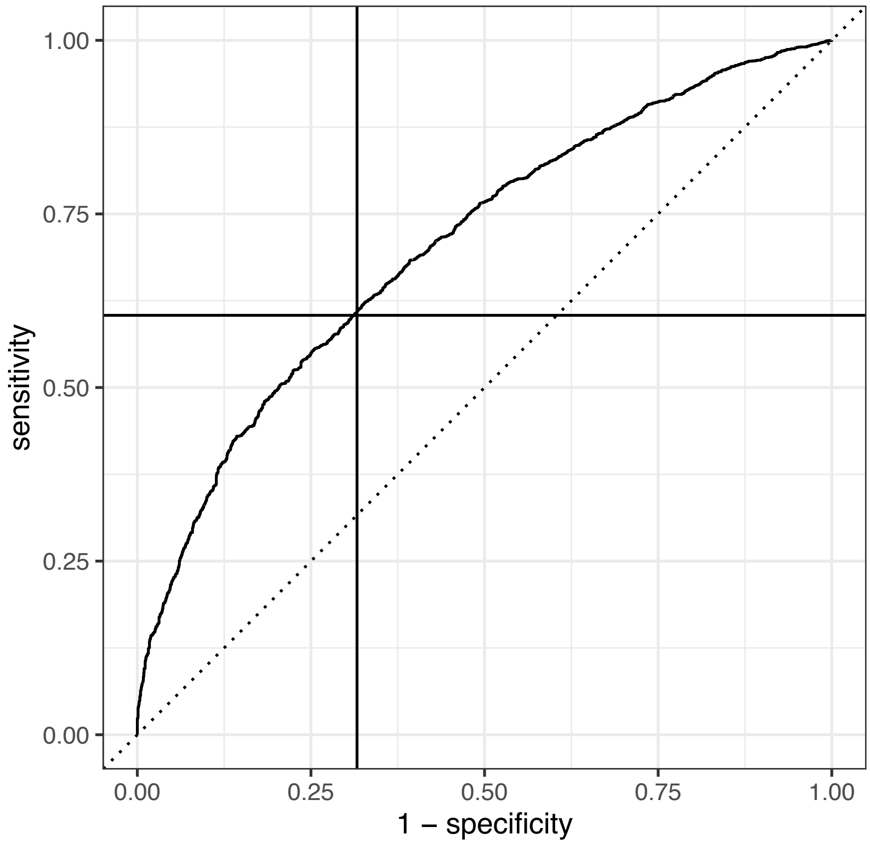


This is the correct ROC Curve is using .pred_ta, and event level second, this is where the horizontal line meets the curve at the model's sensitivity. The vertical line meets the curve at 1- specificity. Where these two lines meet depicts how the model is most optimal to perform at a 0.5 probability threshold.

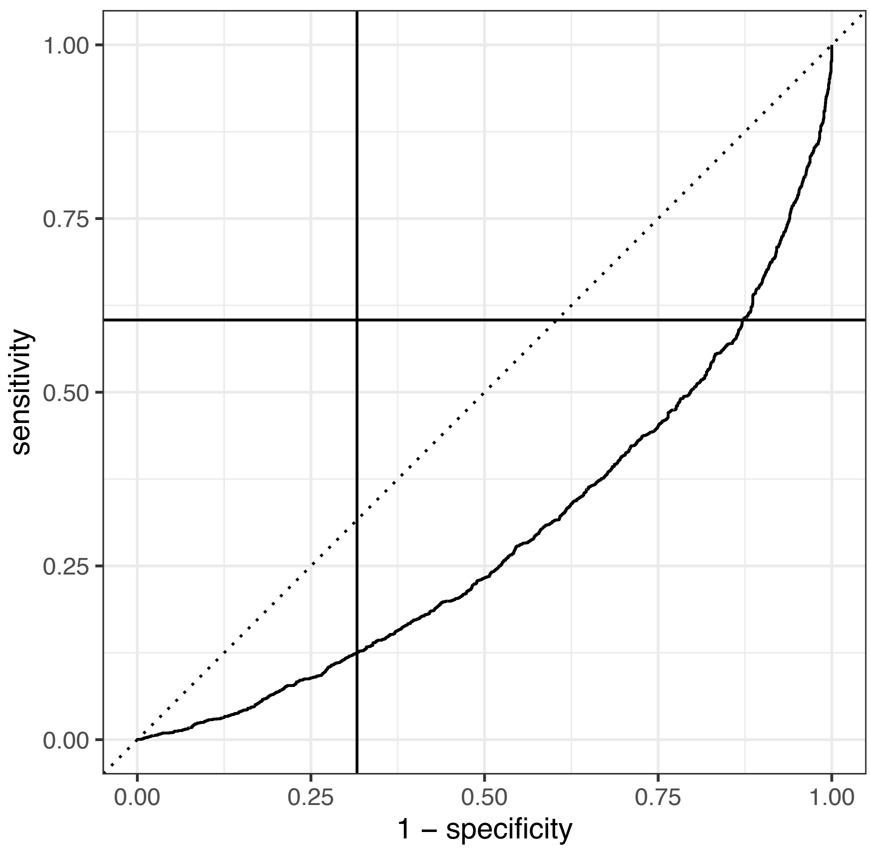
```r
question16_preds %>%
  roc_curve(
    .pred_ta,
    truth = inj,
    event_level = "second"
  ) %>%
  autoplot ()+
  geom_vline(xintercept = 1-specificity)+
  geom_hline(yintercept = sensitivity)
```




```r
question16_preds %>%
  roc_curve(
    .pred_nil,
    truth = inj,
    event_level = "first"
  ) %>%
  autoplot ()+
  geom_vline(xintercept = 1-specificity)+
  geom_hline(yintercept = sensitivity)
```




```
question16_preds %>%
  roc_curve(
    .pred_ta,
    truth = inj,
    event_level = "first"
  ) %>%
  autoplot ()+
  geom_vline(xintercept = 1-specificity)+
  geom_hline(yintercept = sensitivity)
```



17d) The AUC of the correct ROC.

The AUC of 0.706 means that the model has a pretty good level of accuracy, where the closer this number is to 1 means that the score would be more accurate. This results in a pretty good predictive power overall.

```
question16_preds %>%
  roc_auc(
    .pred_ta,
    truth = inj,
    event_level = "second"

  )
```

```
## # A tibble: 1 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>          <dbl>
## 1 roc_auc binary         0.706
```

Q18. Final Predicion

The final model predicts if Lenorado Charming is injured ("ta"), and the liklihood of the injury being 58.6% and not being injured is 41.4%. The probability of 0.50 means that the model is predicting that Charming will definatly be injured at least once during the season. This is due to the set variables of his age, weight, height, and total hours he spent on the field during the season.

```r
question18 <- tibble(
  inj = "ta",
  age = 37,
  weight = 80,
  height = 181,
  hours = (.5)*14,
)
question18_class <- predict(question10_glm, new_data = question18,
                            type = "class")
question18_prob <- predict(question10_glm, new_data = question18,
                           type = "prob")

bind_cols(
  question18_class,
  question18_prob
)
```

```
## # A tibble: 1 x 3
##   .pred_class .pred_nil .pred_ta
##   <fct>           <dbl>    <dbl>
## 1 ta              0.414    0.586
```

The below model made the same calculations, only changing his playing time to see if this may reduce his risk of injury. The time is calculated, whereby Charming is on the field for 30 minutes(0.5 hours) per game, for the 14 games of the season. This resulted in a 45.6% chance of injury, indicating that a reduction in time spent on the field indicate that it reduces the risk of injury, as it has passed the 50% level.

```r
question18 <- tibble(
  age = 37,
  weight = 80,
  height = 181,
  hours = (.25)*14,
)
question18_class <- predict(question10_glm, new_data = question18,
                            type = "class")
question18_prob <- predict(question10_glm, new_data = question18,
                           type = "prob")

bind_cols(
  question18_class,
  question18_prob
)
```

```
## # A tibble: 1 x 3
##   .pred_class .pred_nil .pred_ta
##   <fct>           <dbl>    <dbl>
## 1 nil             0.544    0.456
```

In correlation with he logistic regression model, the prediction of Charming getting injured is 58.6% when he is playing 7 hours per season. When we reduce his time to 3.5 hours per season, it can be clearly seen that the probability of him getting injured reduced to 45.6%. The time is calculated, whereby Charming is on the field for 15 minutes (0.25 hours) per game, for the 14 games of the season.

Therefore my recommendation is to reduce the likelihood that Charming will be injured on the field is to reduce the overall hours he is spent playing. This is proven to have a direct effect of minimizing the major contributors to charming potential injuries.

Assignment 2

Jessica Evans

1/11/2025

Setup

```r
library(tidyverse)
library(tidymodels)
library(moments)
library(caret)
```

Q1. Loading the data

In correlation with the deliverable specifications, it is essential that I choose the correct dataset to my student number modulo 3, resulting in dataset 2, as seen below.

```r
(1876802)%%3
```

```
## [1] 2
```

```r
peileadoir <- read_csv("./a2_data/peileadoir_2.csv")
```

Secondly, the dataset is loaded as a tibble to access the data in the first 10 rows, this is done by using the tidyverse package (this is a form of metapackage), it contains a number of different libraries. The dimensions of the tibble are 2000 rows by 4 columns.

```r
head(peileadoir, 10)
```

```
## # A tibble: 10 x 4
##    `Date of BIRTH` Region  MINS `co$t`
##    <chr>            <dbl> <dbl> <chr>
##  1 17 Iui, 1997       223  839. $25,220.56
##  2 19 Fea, 1998       711  512. $8,258.04
##  3 15 Aib, 1994       467  503. $7,856.46
##  4 05 Sam, 1999       711  389. $4,791.32
##  5 23 Bea, 1995       467  348. $3,835.61
##  6 14 Iui, 1994       676  817. $23,215.23
##  7 24 Mei, 1996       592  737. $19,310.04
##  8 26 Mei, 1995       676  828. $25,283.37
##  9 30 Sam, 1995       223  355. $4,057.64
## 10 17 Bea, 2001       467  532. $8,860.12
```

```r
dim(peileadoir)
```

```
## [1] 2000    4
```

Q2. Adding a column of row numbers

I have added a column on the far right hand side of the tibble called ***GFP_ID***("Gaelic football player identification") that contains the row numbers, this helps to track the necessary changes.

```
question2 <- peileadoir
question2 <- mutate(peileadoir, GFP_ID=c(1:nrow(peileadoir)))
```

This is confirmed when printing out the dimensions, where we can now see that there are 5 columns present.

```
head(question2, 10)
```

```
## # A tibble: 10 x 5
##    `Date of BIRTH` Region  MINS `co$t`     GFP_ID
##    <chr>            <dbl> <dbl> <chr>       <int>
##  1 17 Iui, 1997       223  839. $25,220.56      1
##  2 19 Fea, 1998       711  512. $8,258.04       2
##  3 15 Aib, 1994       467  503. $7,856.46       3
##  4 05 Sam, 1999       711  389. $4,791.32       4
##  5 23 Bea, 1995       467  348. $3,835.61       5
##  6 14 Iui, 1994       676  817. $23,215.23      6
##  7 24 Mei, 1996       592  737. $19,310.04      7
##  8 26 Mei, 1995       676  828. $25,283.37      8
##  9 30 Sam, 1995       223  355. $4,057.64       9
## 10 17 Bea, 2001       467  532. $8,860.12      10
```

```
dim(question2)
```

```
## [1] 2000    5
```

Q3. Randomly Permute all Rows in the Dataset

Random permutation is essential to eliminate poor generalisation and bias.

```
question3 <- question2
set.seed(1876802)
question3 <- sample_n(question2, nrow(question2), replace = FALSE)
question3
```

```
## # A tibble: 2,000 x 5
##    `Date of BIRTH` Region  MINS `co$t`     GFP_ID
##    <chr>            <dbl> <dbl> <chr>       <int>
##  1 01 Fea, 1994       711  556. $9,931.07    1296
##  2 11 DFo, 2002       711  395. $4,911.52     636
##  3 01 Mar, 1998       467  748. $18,763.80    297
##  4 20 Nol, 1994       467  866. $26,960.68   1459
##  5 23 DFo, 1997       223  671. $16,277.37   1820
##  6 27 Ean, 1999       592  332. $3,625.82     236
##  7 26 Sam, 1996       467  683. $16,278.51    264
##  8 05 Iui, 2000       711  326. $3,437.21    1574
##  9 17 Nol, 1994       711  710. $15,932.06   1148
## 10 15 Nol, 1999       592  694. $15,927.68    328
## # i 1,990 more rows
```

Q4. Variable Types Present in the Dataset

It is important to identify the types of variables present for later interpretation.

- **`Date of BIRTH`:** The is depicted as *Categorical Ordinal*, it is presented as a label and has a distinct order. Using the first row as an example, *01,Fea, 1994*,it is prevalent to note that each month is represented in an Irish calendar months, where *Fea* represents Feabhra (February). The row *01 Fea, 1994* is a specific duration in time between 12:00pm, 1 Eanair (January) 1994 - 11:29pm, 31 Nollaig (December) 1998. This is seen whereby comparison each `Date of BIRTH` row is distinctly represented, it is seen as a categorical ordinal variable. This column could later be used to calculate the players age relevant to their experience or to possibility of receiving injury.
- **Region:** This is depicted as *Categorical Nominal* variable, as a label with no specific order. As seen where the names of the regions were removed and replaced with integers that were **randomly assigned**. The reason behind this is to prevent any stereotypes or assumptions to be made about the regions, possibly ones known for being more dangerous or violent, to eliminate bias.
- **MINS:** This is a *Quantitative Discrete* variable, it represents the amount of on-field time, it is noted that this was recorded to exactly 4 decimal places of precision. It would be unreasonable to justify infinitely dividing minutes in this context of quantifying the amount of on-field time in correlation to Coleman's medical costs.
- **`cost`:** This is a *Quantitative Discrete* variable, it is also seen as it contextually wouldn't make sense to be infinitely divided. Cost includes the ambulance and hospital costs for each player. This is seen in the first row as *$9,931.07*, it is represented as AUD and it and been rounded to the nearest cent.
- **GFP_ID:** This is represented as a *Categorical Ordinal*, a label with a specific order, it is just used an identification number for each player.

Q5. Data Taming

It is important to adhere to the data taming conventions to improve the overall productivity and make sure that we have the highest quality data to make informed decisions later on from the randomised dataset. This process first making sure that the naming conventions are correct, all variables identified above are represented correctly. This improves the clarity of the dataset, and ensuring that it is less susceptible to errors.

1. Naming conventions for variable names: We must adhere to the naming conventions for the variable names in the dataset.

```
question5 <- question3
question5 <- rename(question5, dob = `Date of BIRTH`)
question5 <- rename(question5, region = Region)
question5 <- rename(question5, mins = MINS)
question5 <- rename(question5, cost = `co$t`)
question5 <- rename(question5, gfp_id = GFP_ID)
question5
```

```
## # A tibble: 2,000 x 5
##    dob          region  mins cost       gfp_id
##    <chr>         <dbl> <dbl> <chr>       <int>
##  1 01 Fea, 1994    711  556. $9,931.07    1296
##  2 11 DFo, 2002    711  395. $4,911.52     636
##  3 01 Mar, 1998    467  748. $18,763.80    297
##  4 20 Nol, 1994    467  866. $26,960.68   1459
##  5 23 DFo, 1997    223  671. $16,277.37   1820
##  6 27 Ean, 1999    592  332. $3,625.82     236
```

```
## 7 26 Sam, 1996    467  683. $16,278.51    264
## 8 05 Iui, 2000    711  326. $3,437.21    1574
## 9 17 Nol, 1994    711  710. $15,932.06   1148
## 10 15 Nol, 1999    592  694. $15,927.68    328
## # i 1,990 more rows
```

Using the rename function we made sure that all column names are in snake_case to enhance overall readability and continuity.

2. Relocating the dob column:

Next it is important to put the column representing the subjects in the left hand column to enhance readability.

```r
question5 <- relocate (question5, gfp_id, .before=dob)
question5
```

```
## # A tibble: 2,000 x 5
##    gfp_id dob           region  mins cost
##     <int> <chr>          <dbl> <dbl> <chr>
##  1   1296 01 Fea, 1994    711  556. $9,931.07
##  2    636 11 DFo, 2002    711  395. $4,911.52
##  3    297 01 Mar, 1998    467  748. $18,763.80
##  4   1459 20 Nol, 1994    467  866. $26,960.68
##  5   1820 23 DFo, 1997    223  671. $16,277.37
##  6    236 27 Ean, 1999    592  332. $3,625.82
##  7    264 26 Sam, 1996    467  683. $16,278.51
##  8   1574 05 Iui, 2000    711  326. $3,437.21
##  9   1148 17 Nol, 1994    711  710. $15,932.06
## 10    328 15 Nol, 1999    592  694. $15,927.68
## # i 1,990 more rows
```

3. Dates:

Next, it is important to represent the `Date of BIRTH:` column correctly, this involves renaming the Irish calendar months to English. First I Identified all the different months in the column by extracting the alphabetic characters from the strings.

```r
check_dob <- unique(str_match(question5$dob, "([:alpha:]+)")[,1])
check_dob
```

```
##  [1] "Fea" "DFo" "Mar" "Nol" "Ean" "Sam" "Iui" "Lun" "Mei" "Aib" "MFo" "Bea"
```

This was done to make sure there were no typos, and cross reference the Irish to English names (BBC Bitesize, 2025).

```r
question5$dob <- str_replace(question5$dob, "Ean", "Jan")
question5$dob <- str_replace(question5$dob, "Fea", "Feb")
question5$dob <- str_replace(question5$dob, "Aib", "Apr")
question5$dob <- str_replace(question5$dob, "Bea", "May")
question5$dob <- str_replace(question5$dob,  "Mei", "Jun")
question5$dob <- str_replace(question5$dob,  "Iui", "Jul")
question5$dob <- str_replace(question5$dob,  "Lun", "Aug")
question5$dob <- str_replace(question5$dob,  "MFo", "Sep")
question5$dob <- str_replace(question5$dob, "DFo", "Oct")
question5$dob <- str_replace(question5$dob, "Sam", "Nov")
question5$dob <- str_replace(question5$dob, "Nol", "Dec")
```

4. Ordered Factors, Factors, Characters, and Logicals:

Before changing the variable names, it is important to remove all non-integer values in the cost column to reduce errors.

```
question5$mins <- str_replace(question5$mins, ",","")
question5$cost <- str_replace(question5$cost, "\\$", "")%>%
  str_replace_all(",","")
```

Now I can change all the types to correctly represent the variables.

```
question5 <- mutate(question5,
                    gfp_id = as.ordered(gfp_id),
                    dob = dmy(dob),
                    region = as.factor(region),
                    mins = as.numeric(mins),
                    cost = as.numeric(cost))
```

All unnecessary punctuation has been deleted, and standardised abbreviations have been included to shorten variable names, as seen in the output below.

```
head(question5, 10)
```

```
## # A tibble: 10 x 5
##    gfp_id dob        region  mins   cost
##    <ord>  <date>     <fct>  <dbl>  <dbl>
##  1 1296   1994-02-01 711     556.  9931.
##  2 636    2002-10-11 711     395.  4912.
##  3 297    1998-03-01 467     748. 18764.
##  4 1459   1994-12-20 467     866. 26961.
##  5 1820   1997-10-23 223     671. 16277.
##  6 236    1999-01-27 592     332.  3626.
##  7 264    1996-11-26 467     683. 16279.
##  8 1574   2000-07-05 711     326.  3437.
##  9 1148   1994-12-17 711     710. 15932.
## 10 328    1999-12-15 592     694. 15928.
```

```
dim(question5)
```

```
## [1] 2000    5
```

Now that everything has been tamed, it is important to further analyse the data.

Q6. Further Analysis and Updating the Types of Variables

Q6(a) Replacing the Birthdates

Next I replaced the *dob* column with a new variable called *age_gf* that contains each players' age at the start of the year the data was collected.

I subtracted the dob column from the start of the year, and removed the dob column.

```
question5 <- mutate(question5,age_gf = as.numeric(dmy("01-01-2024")- dob)/365)

question5 <- mutate(question5, dob = NULL)
```

Q6(b) Convert the Column containing Time to Hours

To enhance readability, I converted the hours column to minutes to better answer the given question, I also changed the column name hours to minutes.

```r
question5 <- mutate(question5, mins = (mins/60))
question5 <- rename(question5, hours = mins)
```

This improves clarity where the aim is to determine Colmans' medical costs based on the given minutes of on-field time.

Q6(c) Convert Medical Costs to Thousands of AUD

It is also important to convert medical costs to thousands of dollars, in correlation with the need to see if medical costs exceed the cost of 20,000AUD.

```r
question5 <- mutate(question5, cost = ((cost/1000)))
```

Q6(d) Output the First 10 Rows and Dimensions of the Dataset.

All changes are displayed below.

```r
head(question5, 10)
```

```
## # A tibble: 10 x 5
##    gfp_id region hours  cost age_gf
##    <ord>  <fct>  <dbl> <dbl>  <dbl>
##  1 1296   711     9.26  9.93   29.9
##  2 636    711     6.59  4.91   21.2
##  3 297    467    12.5  18.8    25.9
##  4 1459   467    14.4  27.0    29.1
##  5 1820   223    11.2  16.3    26.2
##  6 236    592     5.53  3.63   24.9
##  7 264    467    11.4  16.3    27.1
##  8 1574   711     5.43  3.44   23.5
##  9 1148   711    11.8  15.9    29.1
## 10 328    592    11.6  15.9    24.1
```

```r
dim(question5)
```

```
## [1] 2000    5
```

Q6(e) Updated Variables

While making the above changes in 6 (a), (b) and (c), the variable types also changed, as seen below:

- **age_gf:** In 6(a), this a **Quantitative Continuous** variable, because it is recorded with decimals, as we calculated it by subtracting the "01-01-2024" from the dob of the players. We don't know the number of precision that will be estimated, hence quantitative continuous is the best variable type.
- **hours:** This is converted from minutes in question 6(b), it is represented as a **Quantitative Continuous** variable as mins as represented to exactly 4 decimal places, hence when we divide mins by 60.
- **cost:** This is also **Quantitative Continuous**, it is calculated by dividing the cost by 1000. To allow for more numbers of precision, quantitative continuous is the better option because it can be infinitely divided.

Q7. Subset of the Data

I am taking a subset of the data to fit our model of 1,400 rows. This allows us to focus on a targeted sample allowing performance to be determined.

```r
question7 <- question5
set.seed(1876802)
question7 <- sample_n(question7, 1400, replace = FALSE)
head(question7, 10)
```

```
## # A tibble: 10 x 5
##    gfp_id region hours  cost age_gf
##    <ord>  <fct>  <dbl> <dbl>  <dbl>
##  1 1011   592     9.85 11.7    29.4
##  2 487    711    12.5  18.9    32.3
##  3 1836   223    13.8  23.8    27.9
##  4 1707   467     9.25  9.45   28.2
##  5 1821   223    13.2  20.4    26.8
##  6 594    676    14.8  29.2    26.1
##  7 354    676    11.1  14.7    29.4
##  8 521    223    11.7  16.6    32.6
##  9 344    711    12.7  20.8    27.4
## 10 1532   223     5.26  3.24   27.7
```

```r
dim(question7)
```

```
## [1] 1400    5
```

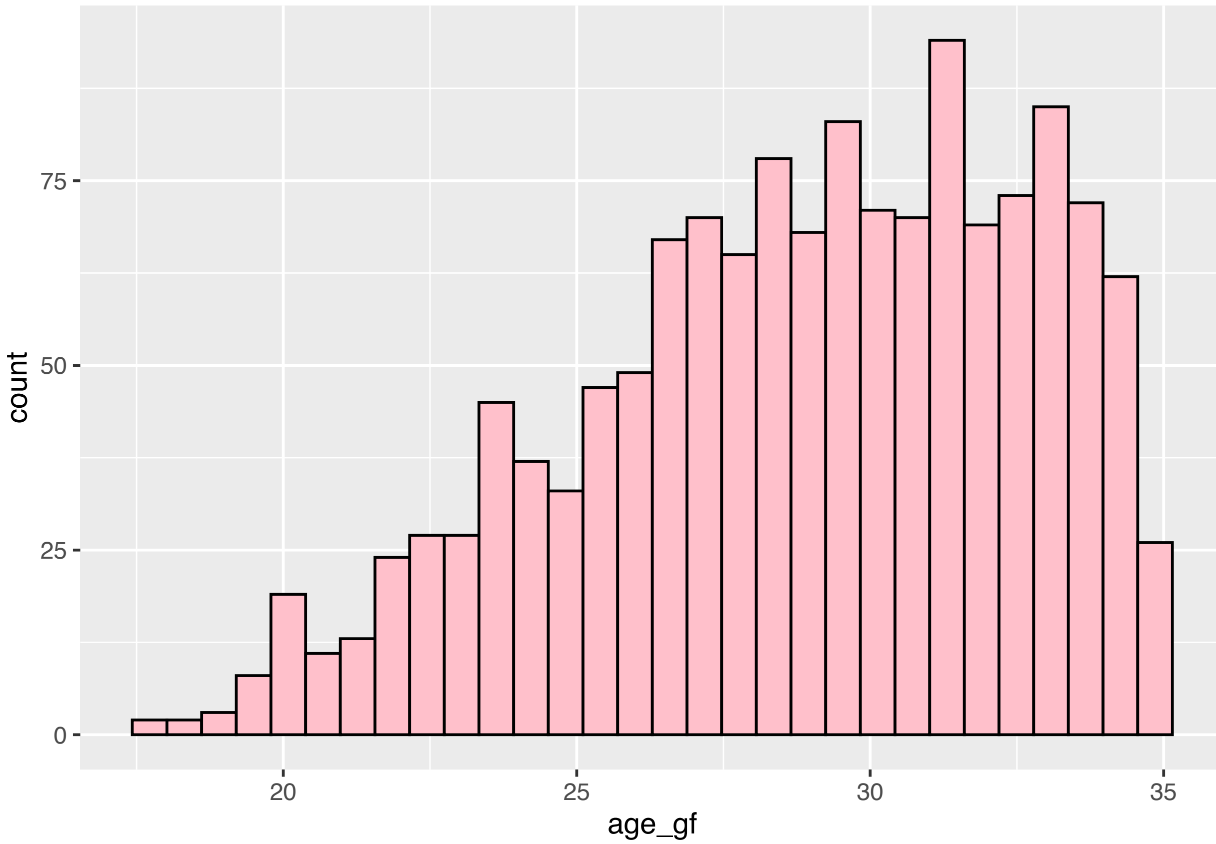
This helps us to better read and interprate the data enabling us to directly focus and explore patterns corresponding to the players age and costs.

Q8. Drawing Conclusions from the Age Data

Q8(a) Histogram of Players Age

The histogram can help us visualise this data, and determine the skewness to conclude if the leans towards younger or older players.

```r
question8 <- question7
ggplot(question8, aes(age_gf))+
  geom_histogram(fill = "pink", color = "black")
```




```r
moments::skewness(question8$age_gf)
```

```
## [1] -0.4879909
```

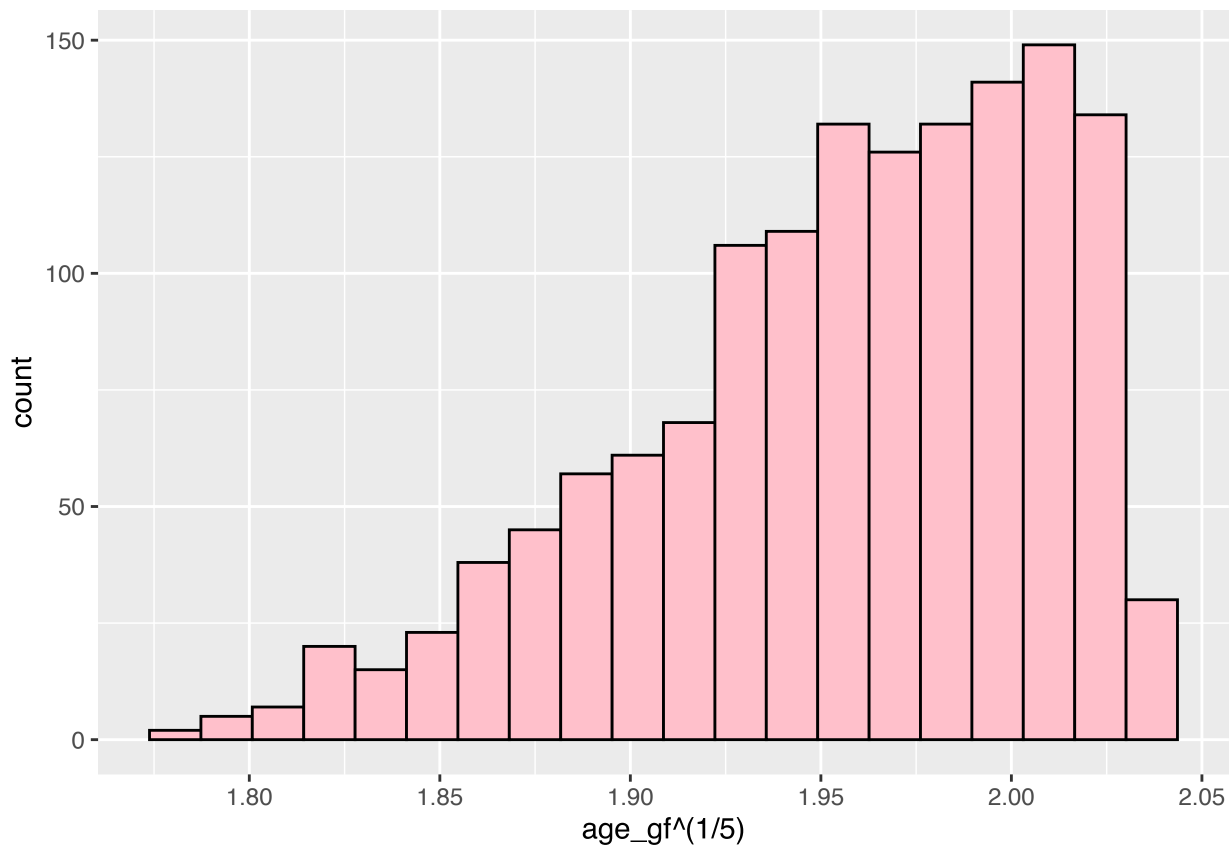
For the data to be symmetrical around mean, the skewness must be close to 0 between (-0.1 and 0.1). It can be concluded that skewness of *-0.4862186* which is approximately *-0.486*. Since the skewness value is negative, this indicates that there are more older players than younger players. This mild skewness indicates that the dataset is not closest enough to 0 to be symmetrical, it is not normally distributed.

In the following question, a sequence of power transformation is applied, to make the data closer to 0 (the normal distribution).

Q8(b) Applying a sequence of Power Transformations

p = 1/5

```r
question8 %>%
  ggplot(aes(x=age_gf^(1/5)))+
  geom_histogram(bins = 20, fill = "pink", color = "black")
```

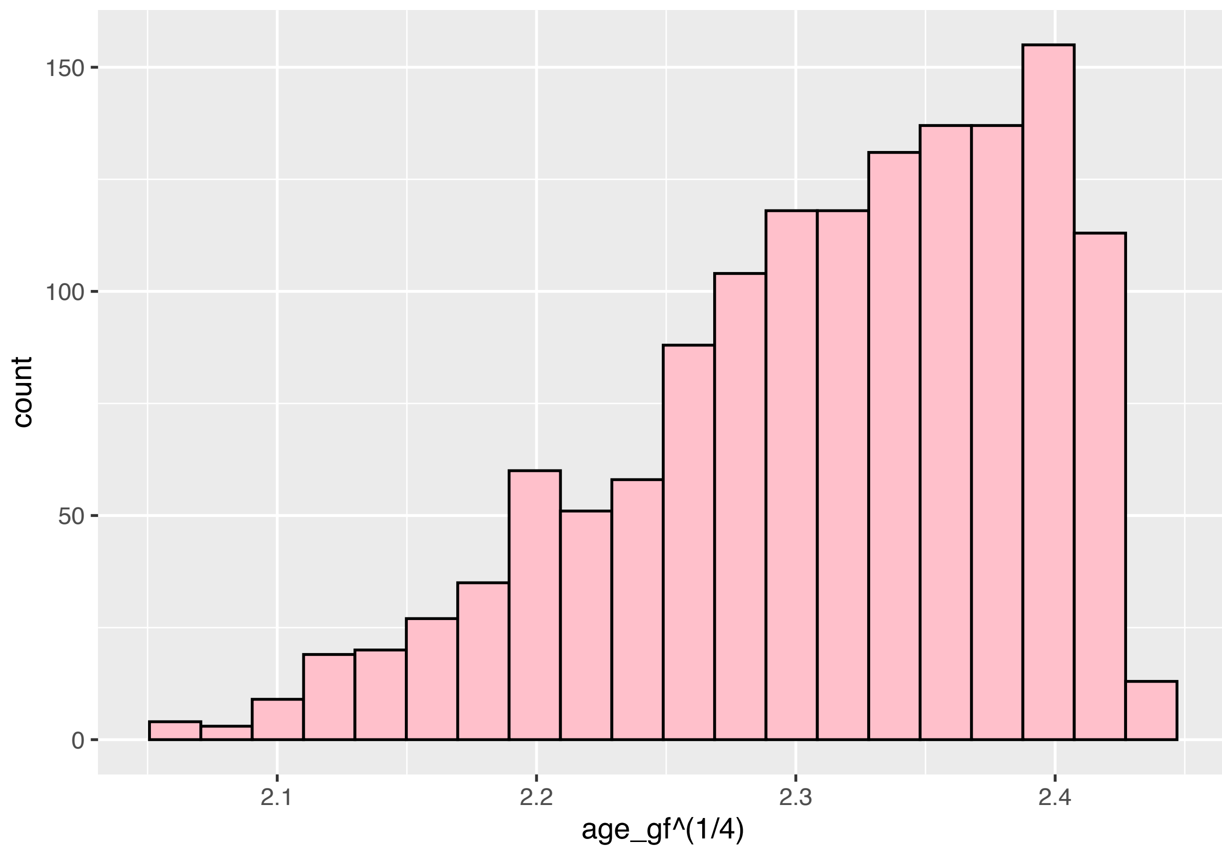



```r
moments::skewness((question8$age_gf)^(1/5))
```

```
## [1] -0.7087055
```

$\mathbf{p = 1/4}$

```r
question8 %>%
  ggplot(aes(x=age_gf^(1/4)))+
  geom_histogram(bins = 20, fill = "pink", color = "black")
```

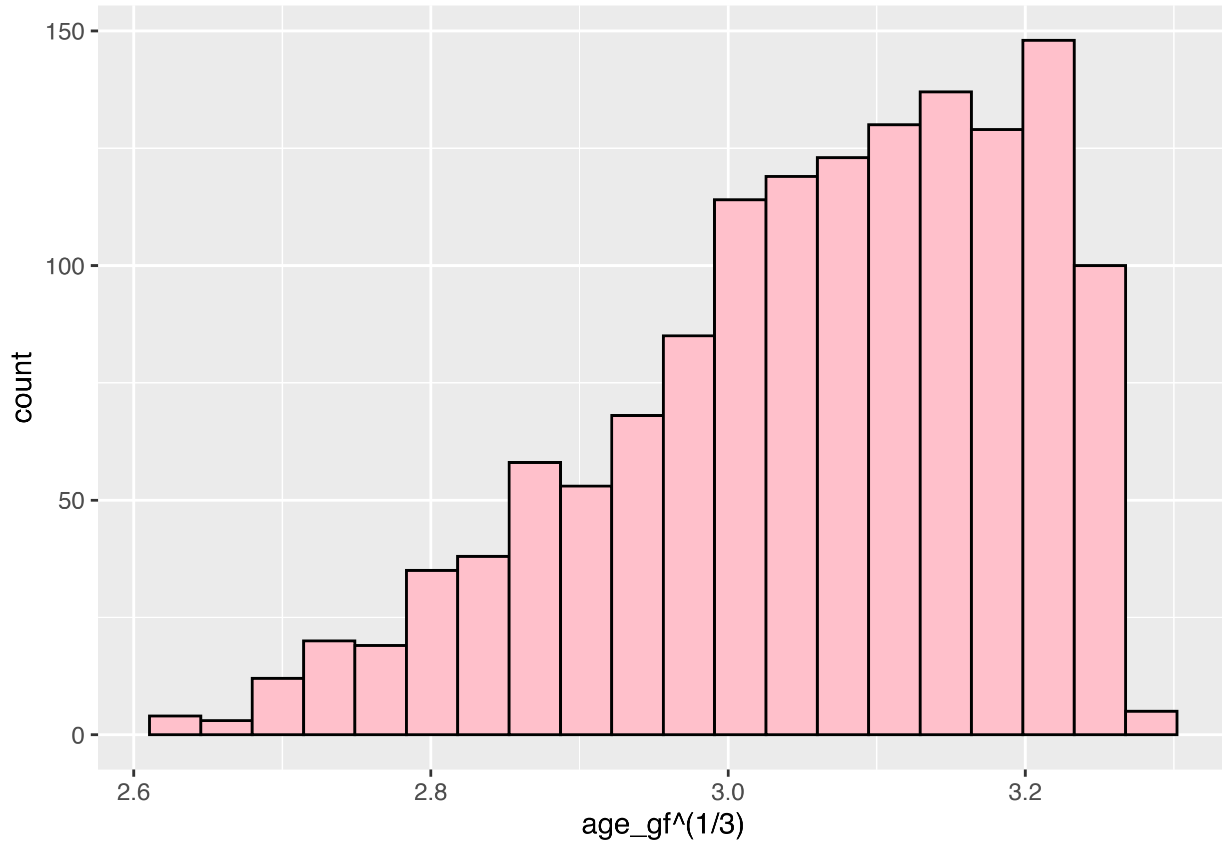



```
moments::skewness((question8$age_gf)^(1/4))
```

```
## [1] -0.6942245
```

p=1/3

```
question8 %>%
  ggplot(aes(x=age_gf^(1/3)))+
  geom_histogram(bins = 20, fill = "pink", color = "black")
```

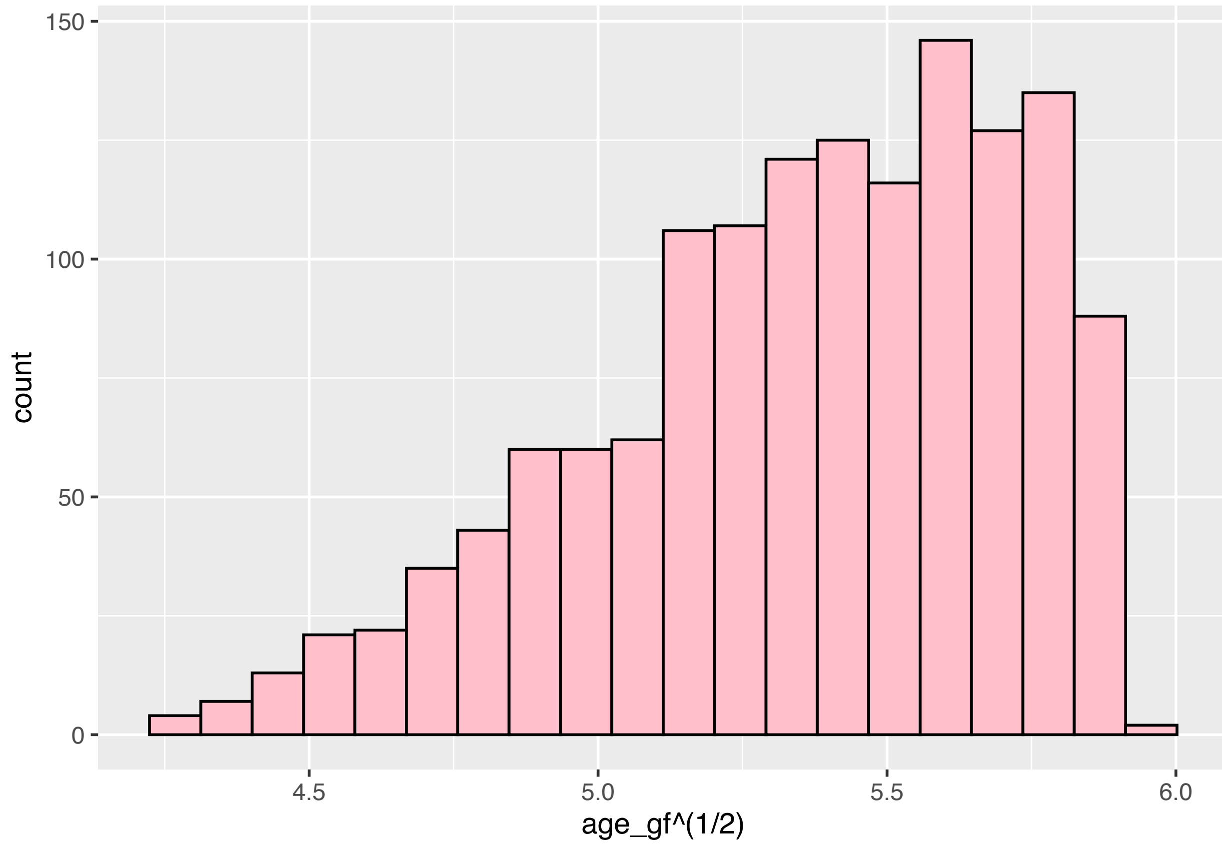



```
moments::skewness((question8$age_gf)^(1/3))
```

```
## [1] -0.6703007
```

p=1/2

```
question8 %>%
  ggplot(aes(x=age_gf^(1/2)))+
  geom_histogram(bins = 20, fill = "pink", color = "black")
```

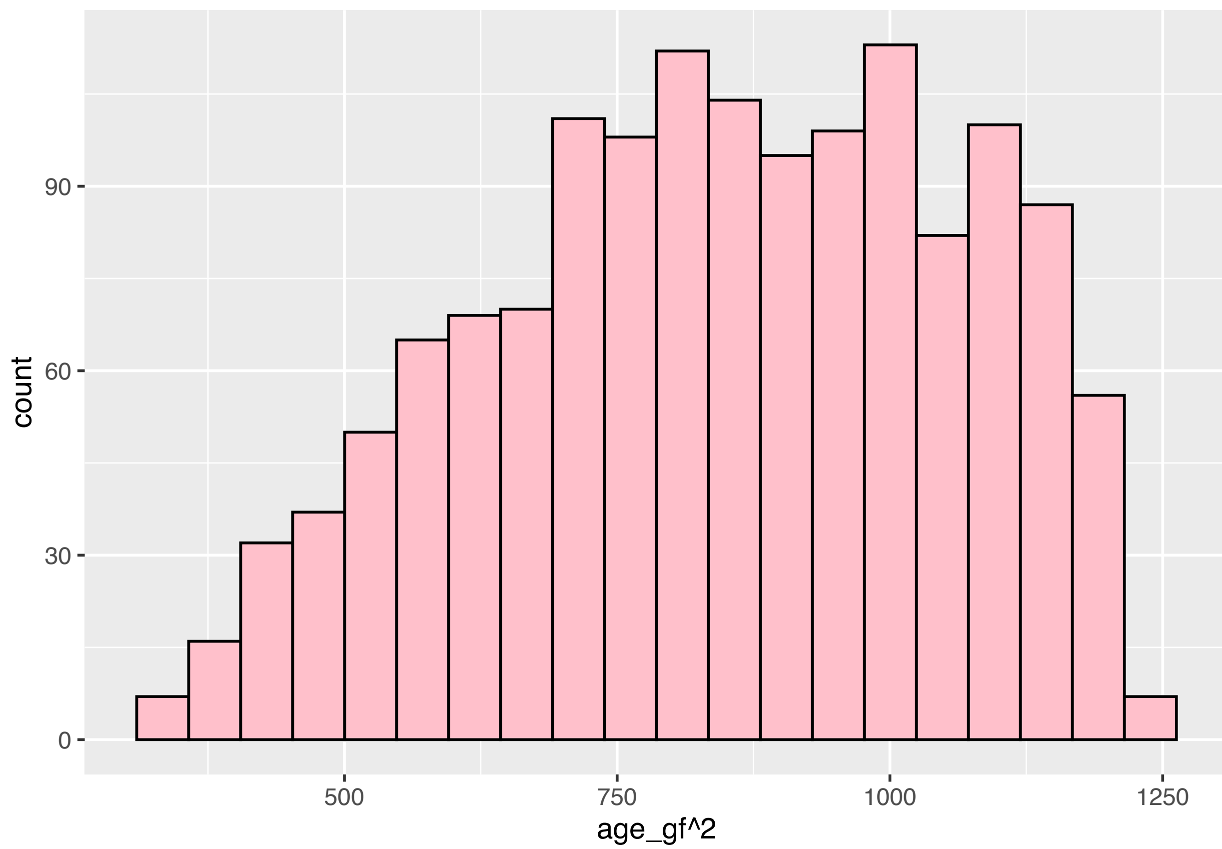



```
moments::skewness((question8$age_gf)^(1/2))
```

```
## [1] -0.6232317
```

p=2

```
question8 %>%
  ggplot(aes(x=age_gf^2))+
  geom_histogram(bins = 20, fill = "pink", color = "black")
```

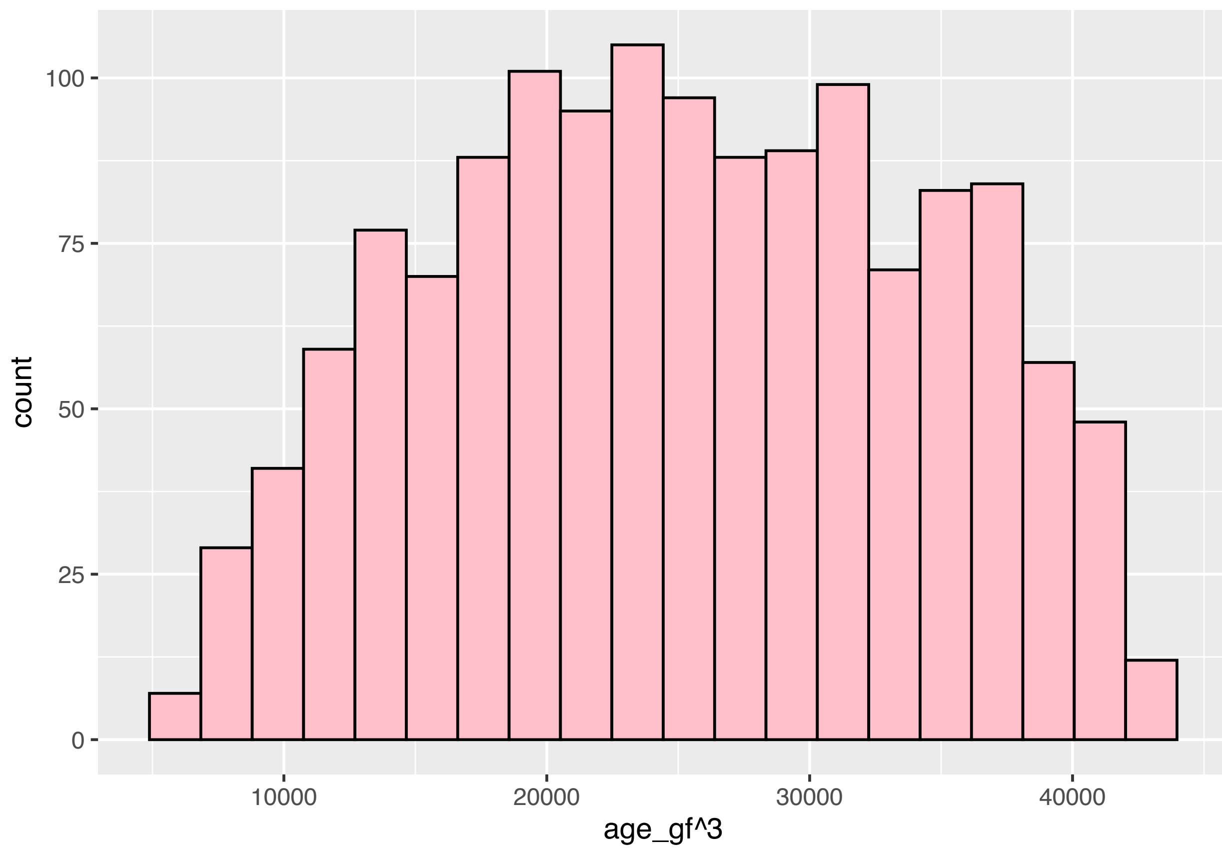



```
moments::skewness((question8$age_gf)^2)
```

```
## [1] -0.2418505
```

p=3

```
question8 %>%
  ggplot(aes(x=age_gf^3))+
  geom_histogram(bins = 20, fill = "pink", color = "black")
```

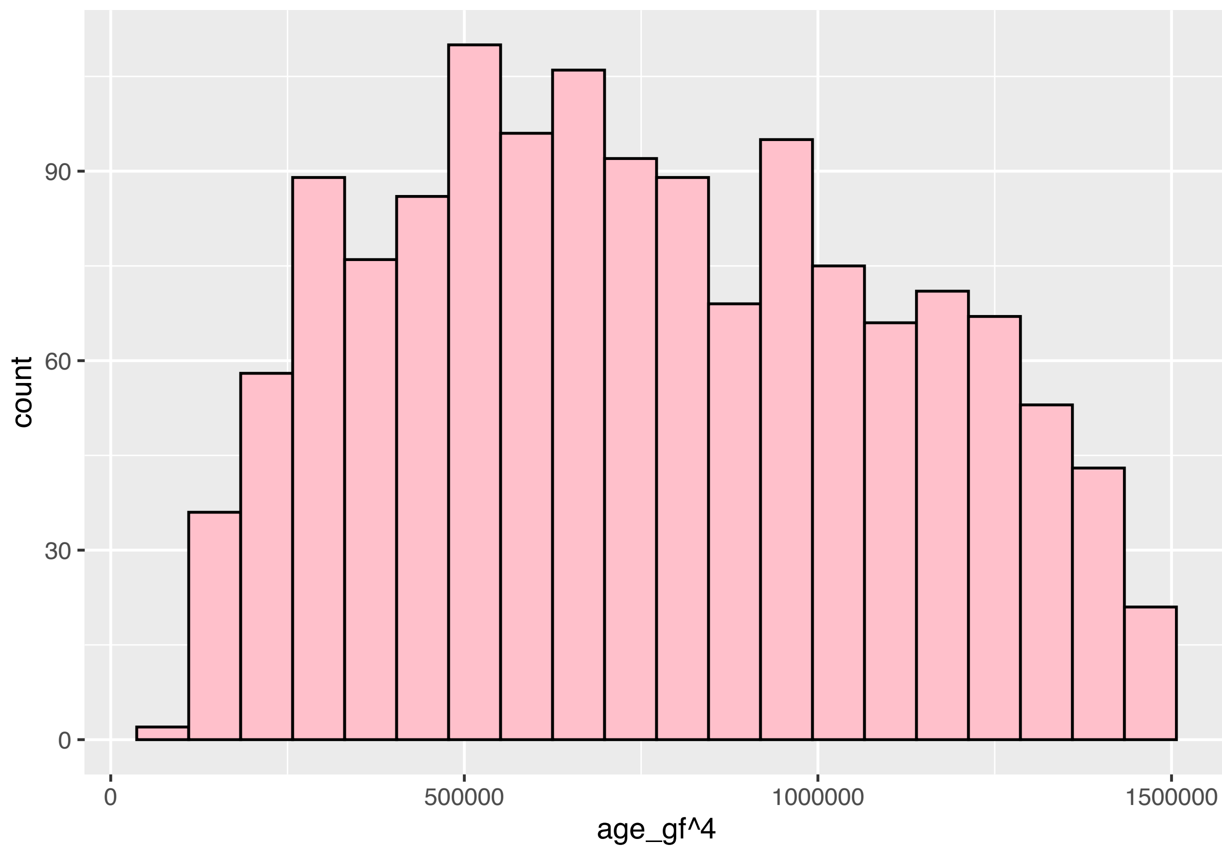



```
moments::skewness((question8$age_gf)^3)
```

```
## [1] -0.02354479
```

p=4

```
question8 %>%
  ggplot(aes(x=age_gf^4))+
  geom_histogram(bins = 20, fill = "pink", color = "black")
```

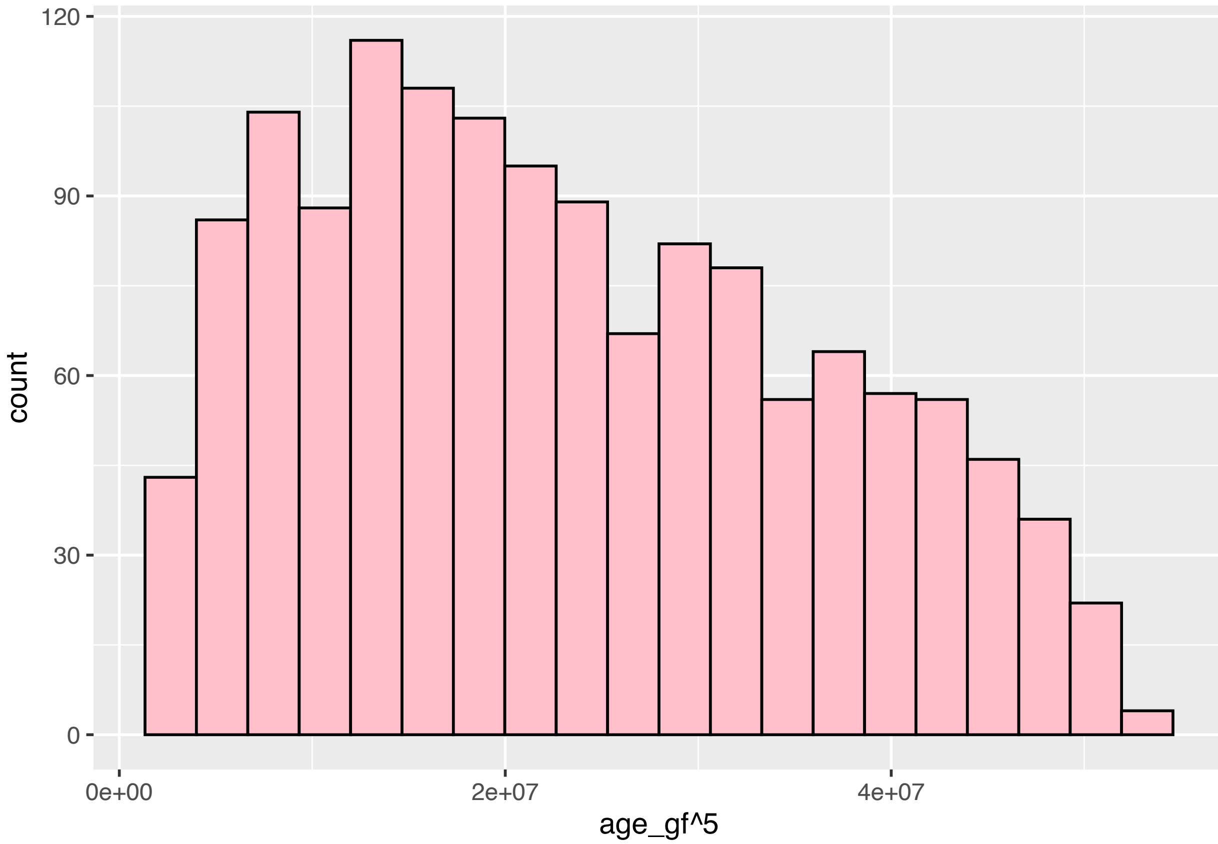



```
moments::skewness((question8$age_gf)^4)
```

```
## [1] 0.1717972
```

p=5

```
question8 %>%
  ggplot(aes(x=age_gf^5))+
  geom_histogram(bins = 20, fill = "pink", color = "black")
```




```r
moments::skewness((question8$age_gf)^5)
```

```
## [1] 0.3482802
```

The above sequence of power transformations has been applied to the age data to produce new histograms and skewness values for each transformation. This is important because it helps us understand how the change in difference in skewness can change using different exponents. The aim of this is to reduce the skewness and bring the it closer to a bell curve.

The power transformations applied include *p = 1/5, 1/4, 1/3, 1/2, 2, 3, 4, 5.* The skewness value should be closet to 0, below we will find out which transformation can help us achieve this.

Q8(c) The Best *p* to Unskew the Data

It can be determined that the *p=3* is the closest to 0 equating to *-0.02354479*, which is approximately *-0.024*. This makes sense, because whole numbers stretch the larger numbers in the dataset to make the histogram more symmetrical.

Q8(d) Box-Cox Value for the Age Data

It is important to find the Box-Cox transformation to the age data to reduce the overall skewness and make the data more normally distributed. The function tested the range of values between -2, and 5 in steps of 0.005.

```r
df_bc <- BoxCoxTrans(y= question8$age_gf, lambda = seq(-2,5,0.005))
df_bc$lambda
```

```
## [1] 2.575
```

The approximate Box Cox value is 2.575. This means that raising the age to this power *^2.575* will provide us with the most symmetric distribution.

Q8(e) Applying the Box-Cox Transform to the Age Data

We will now apply the Box-Cox transformation to the age data, and create a new variable called *tf_age_gf*.

```
question8 <- mutate(question8, tf_age_gf = ((age_gf^df_bc$lambda)-1)/df_bc$
                      lambda)
question8
```

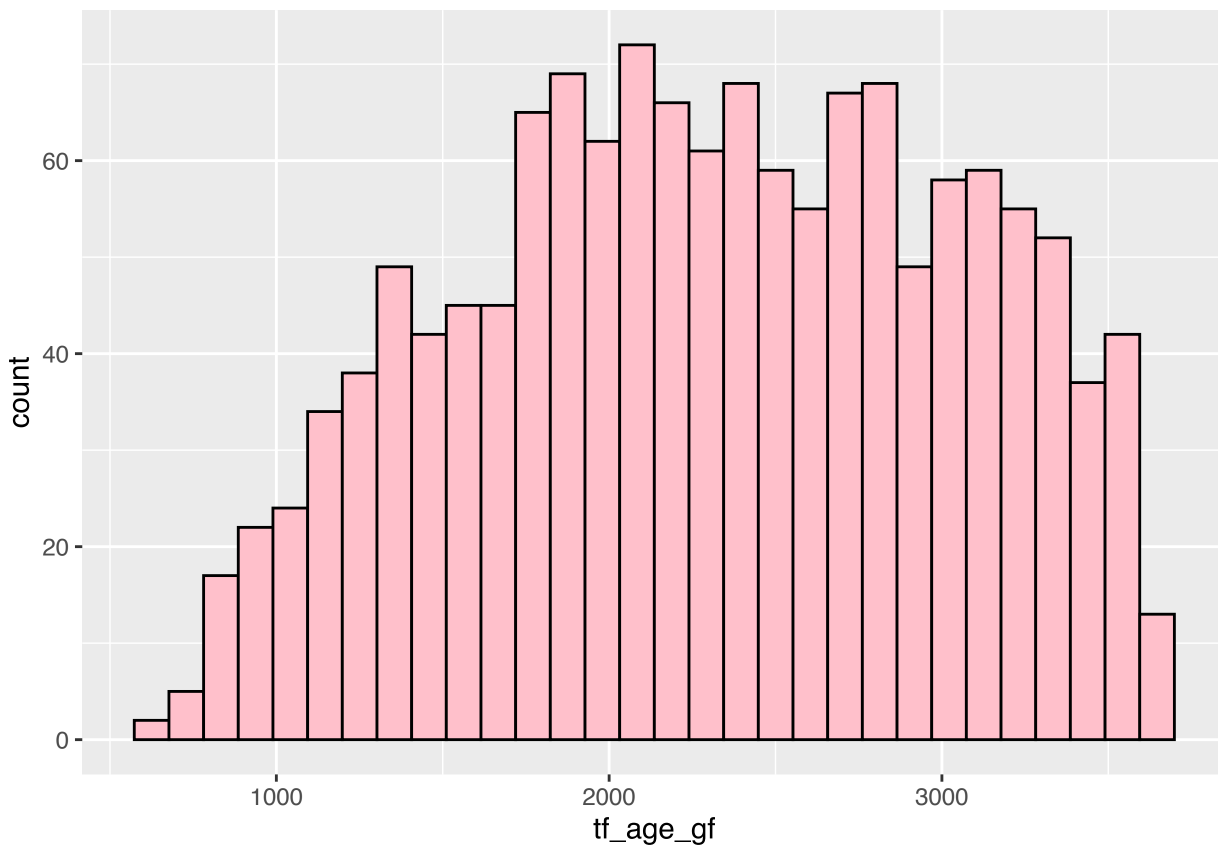
```
## # A tibble: 1,400 x 6
##    gfp_id region hours  cost age_gf tf_age_gf
##    <ord>  <fct>  <dbl> <dbl>  <dbl>     <dbl>
##  1 1011   592     9.85 11.7    29.4     2343.
##  2 487    711    12.5  18.9    32.3     2984.
##  3 1836   223    13.8  23.8    27.9     2058.
##  4 1707   467     9.25  9.45   28.2     2099.
##  5 1821   223    13.2  20.4    26.8     1848.
##  6 594    676    14.8  29.2    26.1     1733.
##  7 354    676    11.1  14.7    29.4     2341.
##  8 521    223    11.7  16.6    32.6     3060.
##  9 344    711    12.7  20.8    27.4     1956.
## 10 1532   223     5.26  3.24   27.7     2008.
## # i 1,390 more rows
```

This adjusted the scale of the age values and altered the skewness to ultimately make the distribution more symmetrical. The new column can be seen above, and the next step is to visualise this transformation to compare the histogram to question *8(a)*.

8(f) Producing a Histogram of the Transformed Age Data

The transformed age data is much more closer to normal distribution as seen in the histogram below.

```
ggplot(question8, aes(tf_age_gf))+
  geom_histogram(fill = "pink", color = "black")
```




```
moments::skewness(question8$tf_age_gf)
```

```
## [1] -0.1132541
```

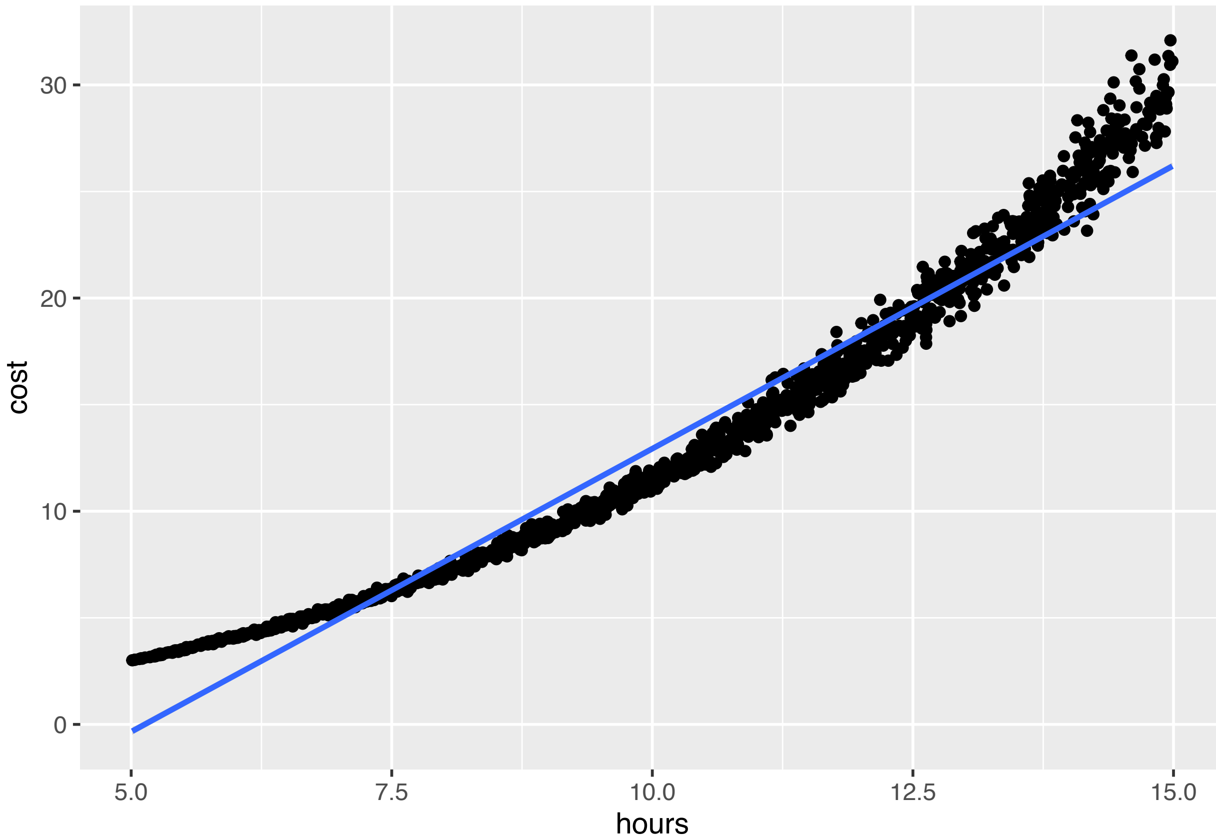
From the untransformed data in question *8(a)* the skewness was -0.488, we havesuccessfully changed the skewness and transformed the age data to have a skewness of *-0.1132541* which is approximately *-0.113*. Applying the sequence of power transformations to the data enabled me to manually see that the ***^3*** reduced the skewness dramatically. This then corresponded to applying the Box-Cox transformation where the more precise value was *2.575*, this was close to *3* therefore was the more suitable value for this transformation.

Q9. Analysis of the Bivariate data of Medical Costs compared to Time on the

9(a) Plot of Medical Costs against Time

The scatterplot below depicts the relationship between the number of hours on the field as the independent variable on the (x-axis) the medical costs as the dependent variable (y-axis). The straight line of best fit helps to visulise this data.

```
question9 <- question8
ggplot(question9, aes(x= hours, y =cost))+
  geom_point()+
  geom_smooth(method = "lm")
```



Hours were chosen as the x-axis value because it directly influences cost, where the longer a team member plays on the field the more susceptible they are to receiving injuries, therefore it is likely that the cost will increase, as the on field time increases.

The data is mostly corresponding to Normality, where the error terms, are mostly along the line of best fit. There is a strong positive trend between hours and cost. It is evident that the longer a team member is on-field, the more likely they are to have higher medical costs.

9(b) Fitting a Linear Model to the Plot

The linear regression model was fitted with cost as the response variable, and hours as the predictor variable.

```
question9_lm <- linear_reg() %>%
  set_engine("lm")%>%
  fit(cost~hours, data = question8)
summary(question9_lm$fit)
```

```
##
## Call:
## stats::lm(formula = cost ~ hours, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.4805 -1.1473 -0.4334  0.9453  6.2357
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -13.62703    0.15445  -88.23   <2e-16 ***
```

```
## hours         2.65619    0.01482  179.18   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.499 on 1398 degrees of freedom
## Multiple R-squared:  0.9583, Adjusted R-squared:  0.9582
## F-statistic: 3.211e+04 on 1 and 1398 DF,  p-value: < 2.2e-16
```

The summary of this linear regression model indicates that every time an hour of time passes on the field, the medical cost is an estimated *2.65619 = $2656.20* AUD. The closer the Multiple R-squared value is to 1 indicated the goodness of fit, the value from question 9 is *0.9583*. This indicates that it is estimated to be *95.8%* of the variation in the response variable (medical costs) are predicted by the hours played.

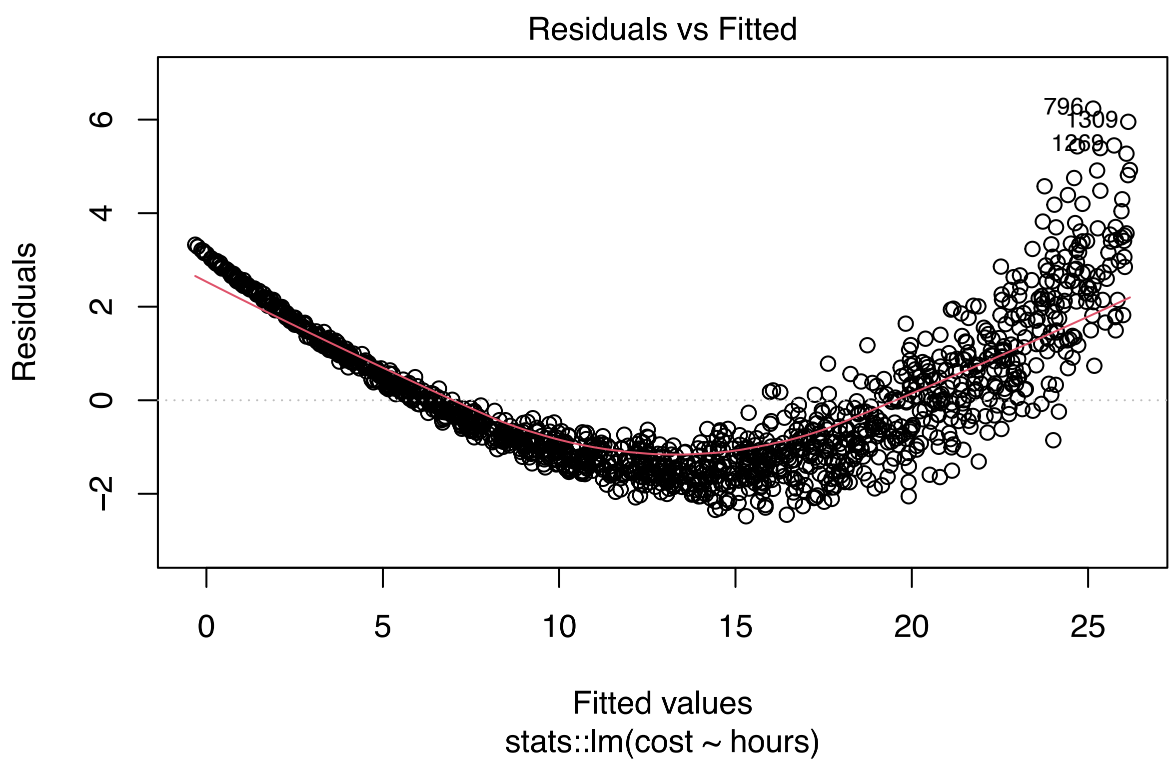
The coefficients depicted are both statistically significant (***), this means that the result is very unlikely due to random chance.

9(c) Checking if the Model Satisfies The Assumptions of Linearity, Homoscedasticity and Normality

Linearity

In correlation with the model below, linearity refers to the residual against fitted values.

```
plot(question9_lm$fit, which = 1)
```



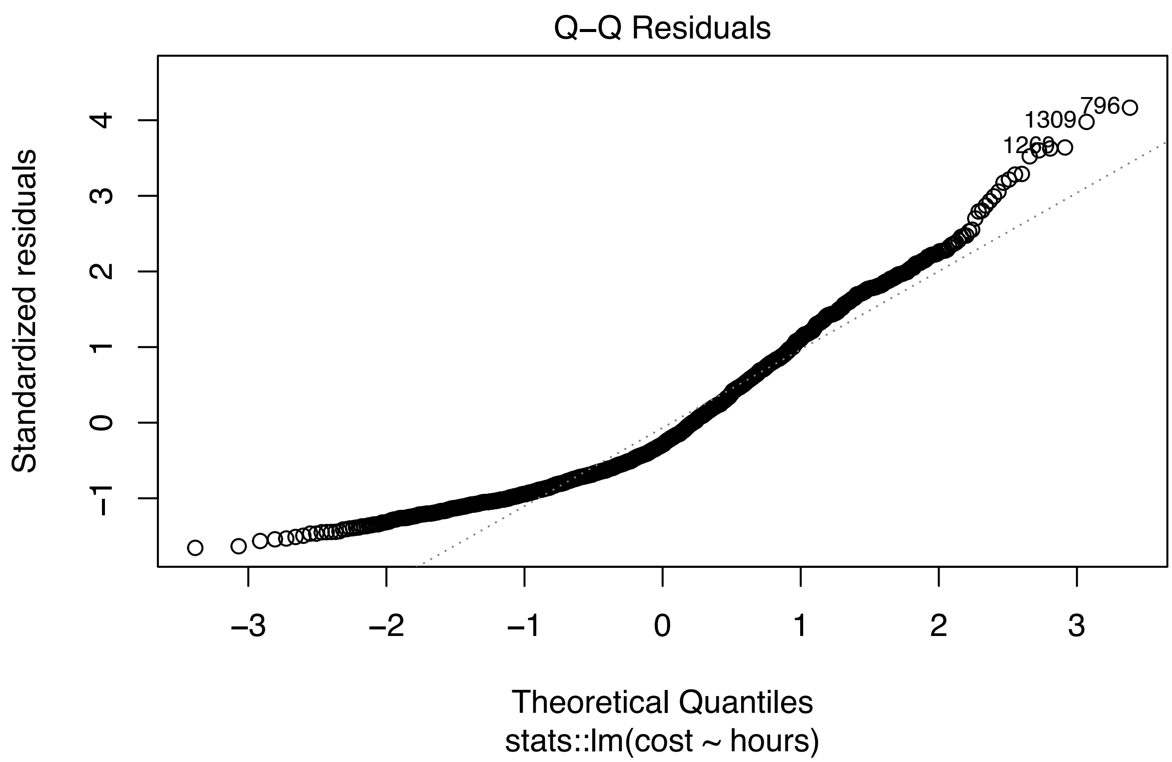
If the red line was straighter, this would indicate that the assumption is satisfied. However this is not the case, there is a curve in the data in the values following a ‘U’ shape, this indicates that linearity would not be a safe assumption.

In this model the horizontal axis (Fitted Values) depicts what the predicted medical costs are based on the hours played. The vertical axis (Residuals) portrays the difference between the actual cost and the predicted cost. A specific point on this graph depicts one players cost and hours. Where if the point in the model is above 0 (the grey line) it indicates that the cost was underestimated. In comparison, if the point is below 0, this indicates that the cost was overestimated. In this case, the assumption is not satisfied, as there is a clear curve seen in this graph.

Normality

In correlation with the model below, Normality is how the residual errors are placed in correlation to the grey line.

```
plot(question9_lm$fit, which = 2)
```



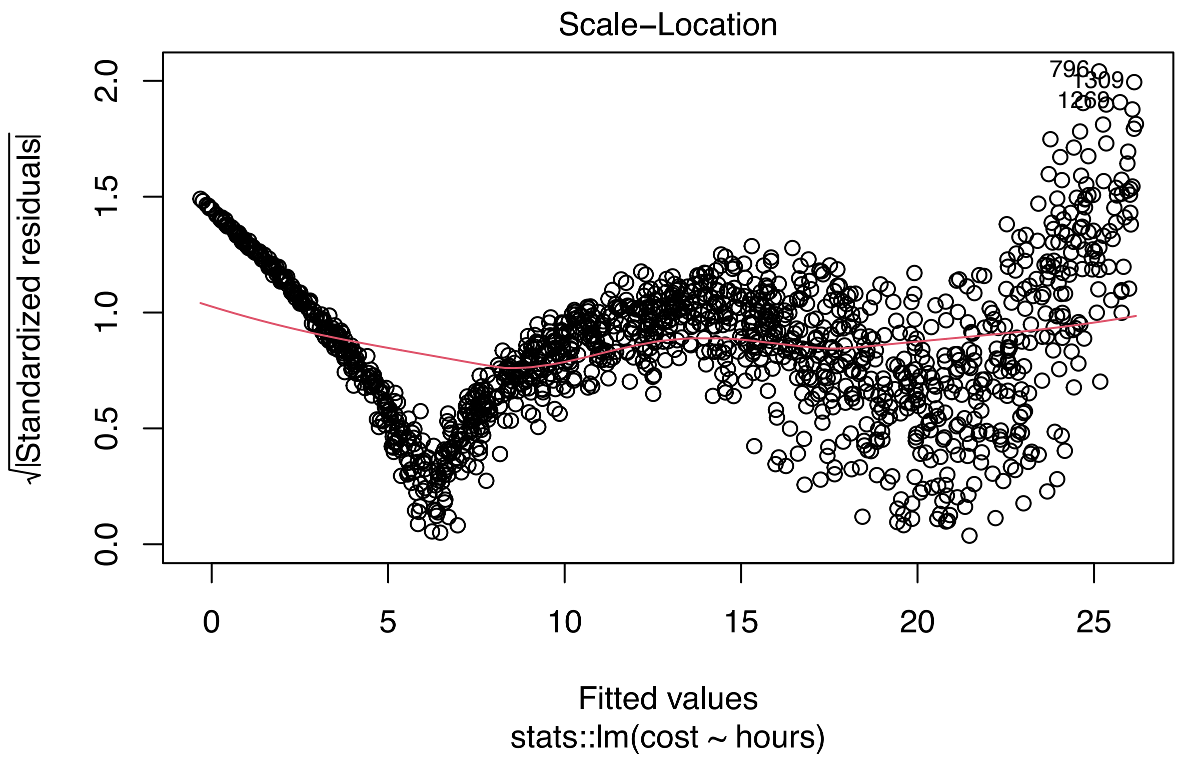
The horizontal axis that is seen as (Theoretical Quantiles) depicts what the residuals should look like if they were normal. The vertical axis (Standardized residuals) represents the specific distribution of the residuals from the model. If the dots were perfectly in line with the line, this would mean that the residuals are normally distributed, and normality is satisfied.

In the model above the dots on this line closely follow the diagonal line except for the points below -1 and above 2 on the horizontal axis. This means that the assumption of normality is mostly satisfied.

Homoscedasticity

In relation to the model below, Homoscedasticity is the even spread of points above and below the red line.

```
plot(question9_lm$fit, which = 3)
```



The horizontal axis (Fitted Values) depict the predicted medical costs in correlation with the on field time on the field. The vertical axis (Standardized Residuals) depict the difference in cost from the predicted cost.

If the data points are evenly spread around the red line, this would indicate that the assumption is satisfied, however this is not the case, indicating that the assumption is not met, as the difference in cost from the predicted cost fluctuates.

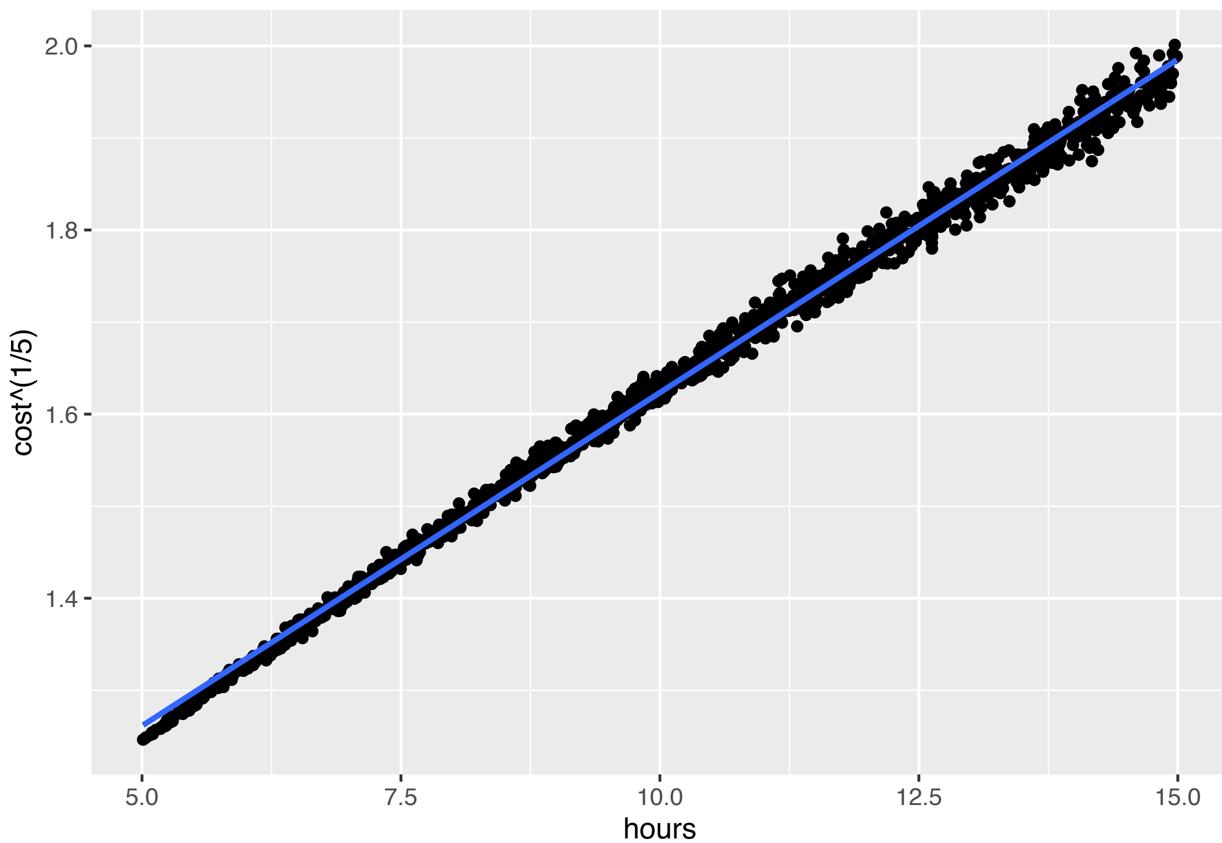
Q10. Improving the linearity of by Transforming our Data

10(a) Applying the Sequence of Transformation to the Medical Cost Data

```
question10 <- question9
```

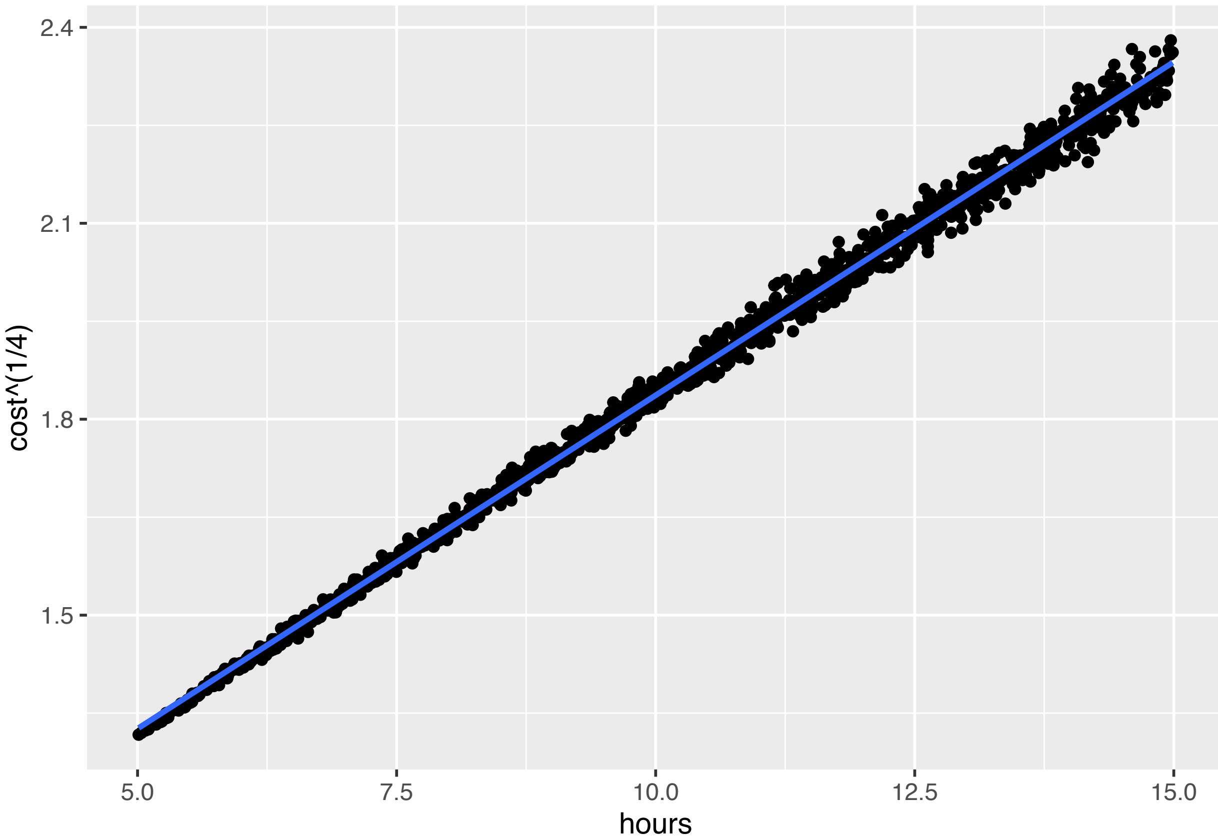
p = 1/5

```
question10 %>%
  ggplot(aes(x=hours, y= cost ^(1/5)))+
  geom_point()+
  geom_smooth(method="lm", se = FALSE)
```



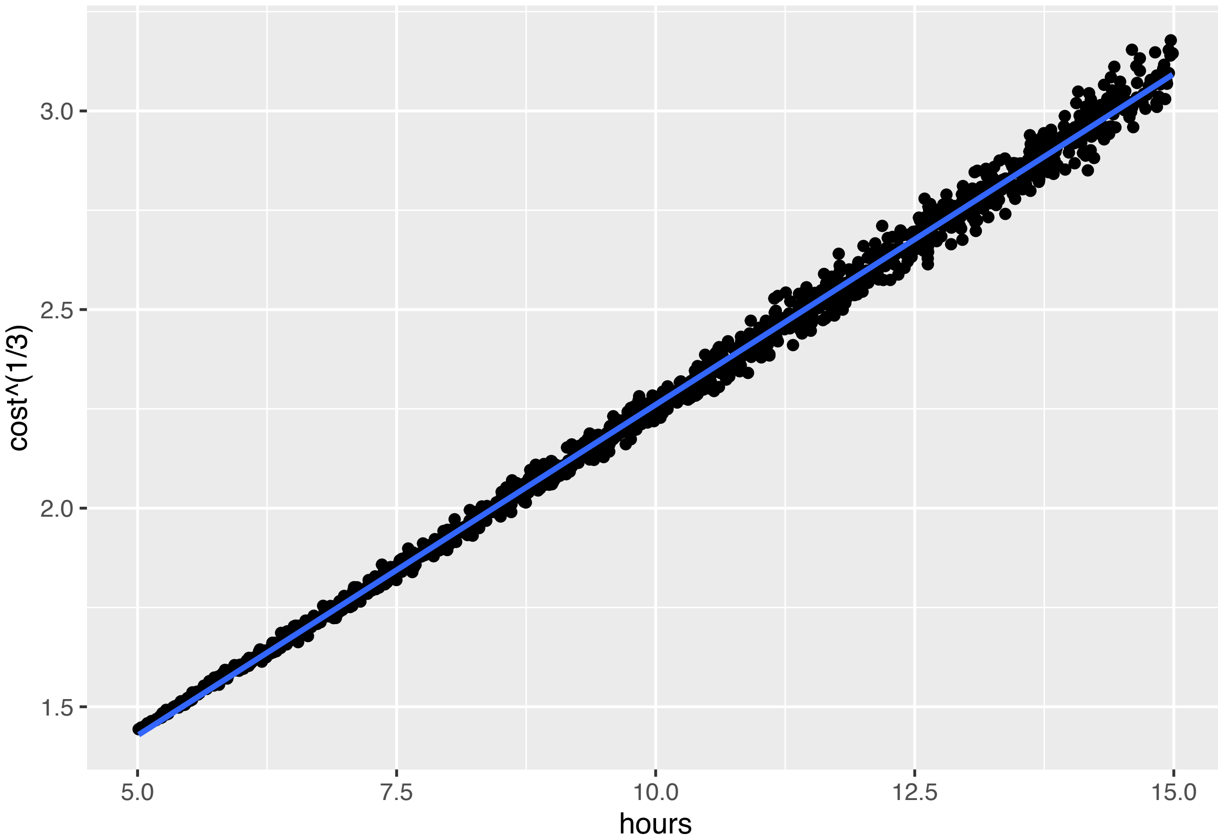
p = 1/4

```
question10 %>%
  ggplot(aes(x=hours, y= cost ^(1/4)))+
  geom_point()+
  geom_smooth(method="lm", se = FALSE)
```



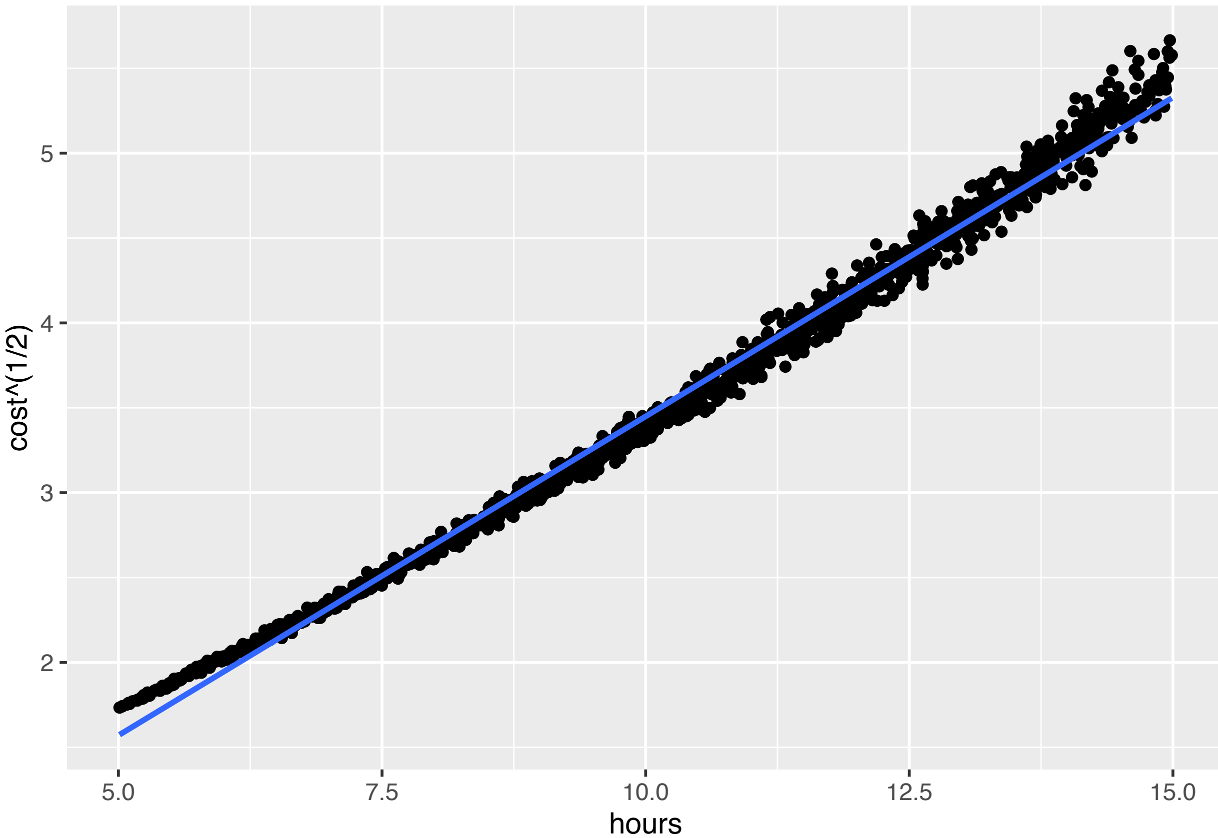
p=1/3

```
question10 %>%
  ggplot(aes(x=hours, y= cost ^(1/3)))+
  geom_point()+
  geom_smooth(method="lm", se = FALSE)
```



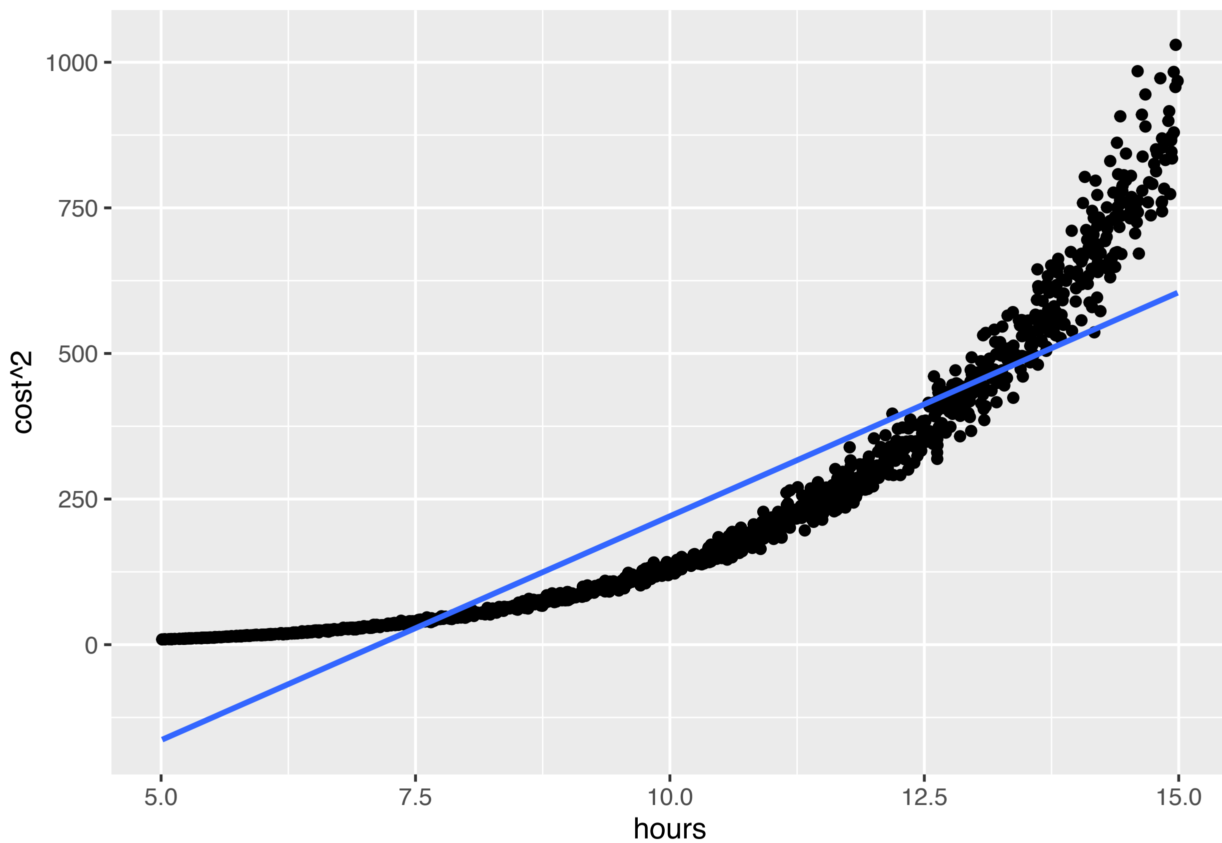
p=1/2

```
question10 %>%
  ggplot(aes(x=hours, y= cost ^(1/2)))+
  geom_point()+
  geom_smooth(method="lm", se = FALSE)
```



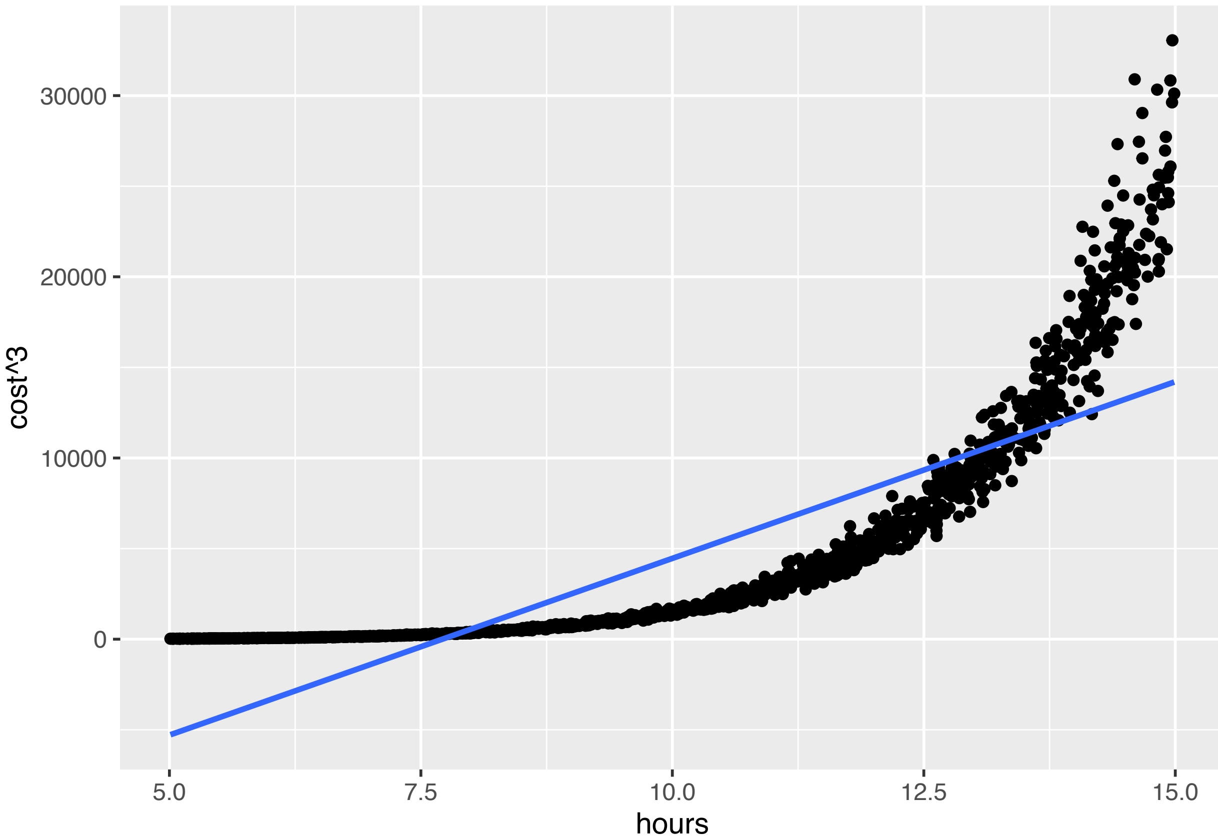
p=2

```
question10 %>%
  ggplot(aes(x=hours, y= cost ^2))+
  geom_point()+
  geom_smooth(method="lm", se = FALSE)
```



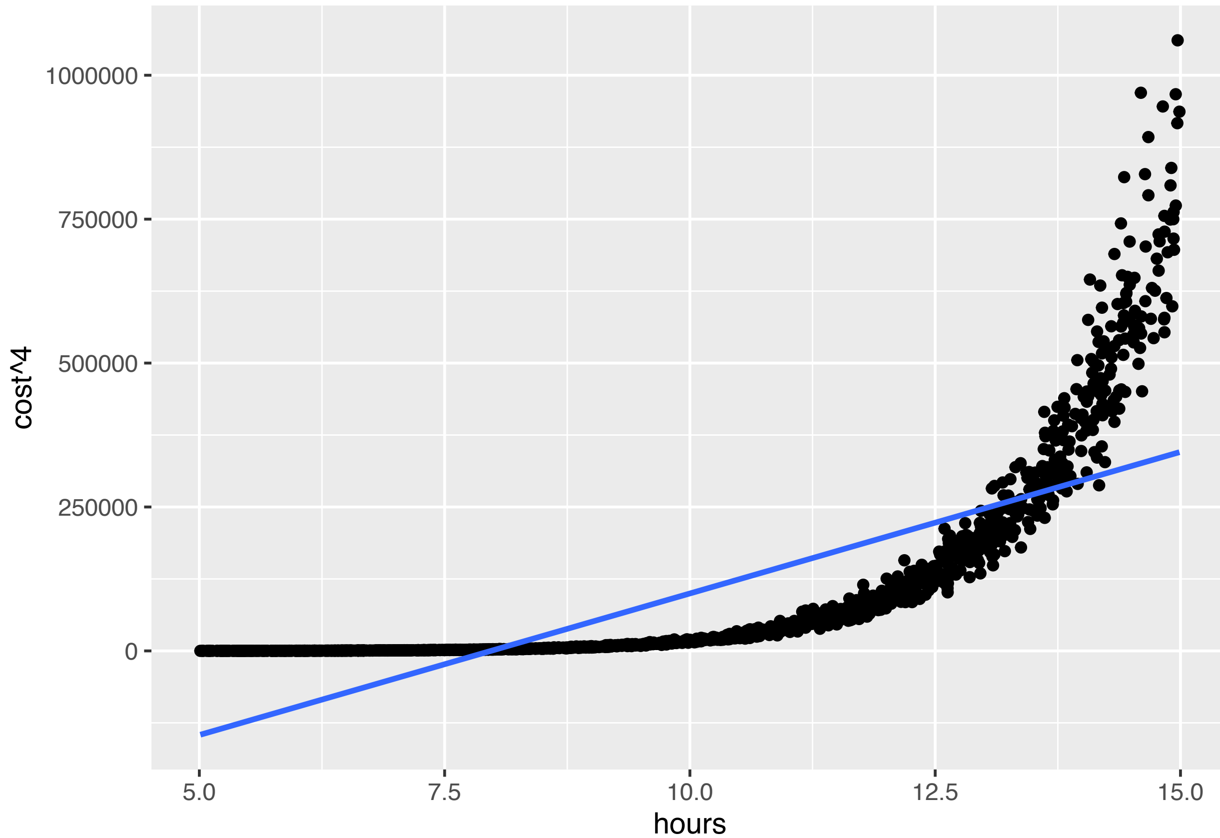
p=3

```
question10 %>%
  ggplot(aes(x=hours, y= cost ^3))+
  geom_point()+
  geom_smooth(method="lm", se = FALSE)
```



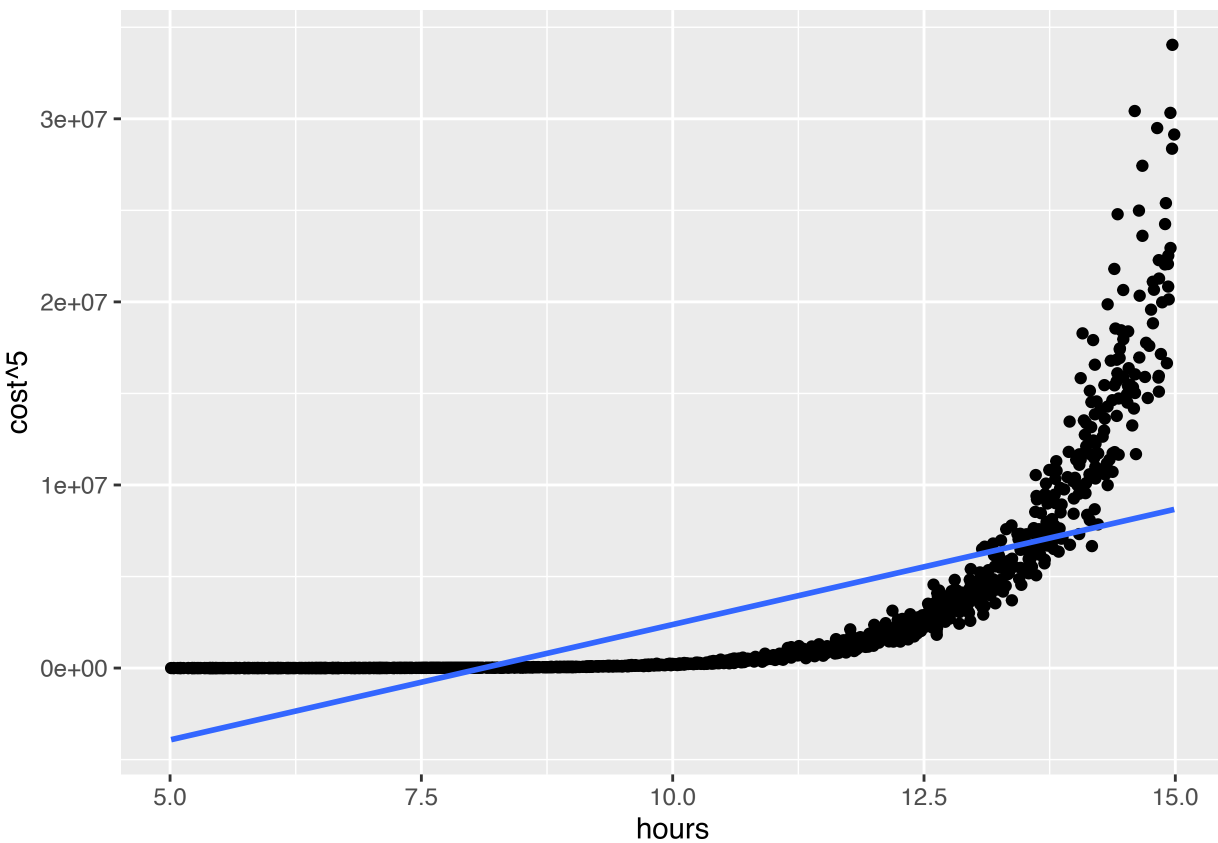
p=4

```
question10 %>%
  ggplot(aes(x=hours, y= cost ^4))+
  geom_point()+
  geom_smooth(method="lm", se = FALSE)
```



p=5

```
question10 %>%
  ggplot(aes(x=hours, y= cost ^5))+
  geom_point()+
  geom_smooth(method="lm", se = FALSE)
```



The aim is to improve the linear model from Question 9, the sequence of power transformations has been applied to the cost variable, and hours remained as the explanatory variable. The powers remain the same as Question 8 p = *1/5, 1/4, 1/3, 1/2, 2, 3, 4, 5*.Scatterplots have been applied to each transformation, the purpose of this is to identify the most linear model reduce curvature.

10(b) The Best p Value

From the above scatterplots, as found earlier the powers less than 1 were the produced the most linear models. The p = *1/4* gave the most linear graph, where the points were closely spread along the line. The larger powers from *2-5* depict curvature in their models, indicating that the linear trend is not as strong, because the cost values are further apart.

The *1/4* was the best p value as it had the most linear relationship between cost and hours.

10(c) Finding the Box-Cox Value

```
df_bc_10 <- BoxCoxTrans(y=question10$cost, x=question10$hours, lambda =
                         seq(-2,5,0.005))
df_bc_10$lambda
```

```
## [1] 0.28
```

The Box Cox transformation was applied to the cost using the values -2, and 5 in steps of 0.005. The optimal lambda value was *0.28*. This means that the cost isstill positively skewed, it will be used in applications to the cost data.

10(d) Applying the Box Cox Transformation to the Cost Data

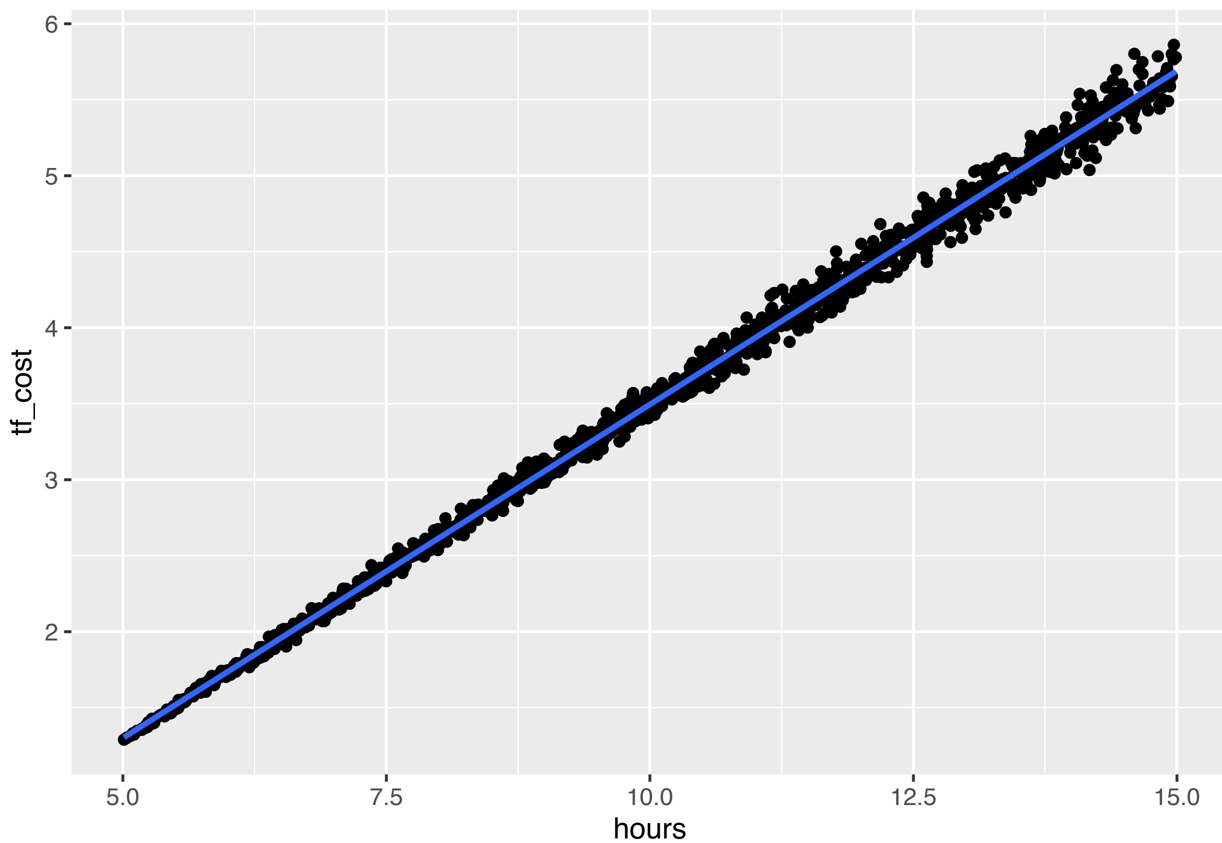
```r
question10 <- mutate(question10, tf_cost = ((cost^df_bc_10$lambda)-1)/
                         df_bc_10$lambda)
relocate(question10, "tf_cost", .after = cost)
```

```
## # A tibble: 1,400 x 7
##    gfp_id region hours  cost tf_cost age_gf tf_age_gf
##    <ord>  <fct>  <dbl> <dbl>   <dbl>  <dbl>     <dbl>
##  1 1011   592     9.85 11.7     3.55   29.4     2343.
##  2 487    711    12.5  18.9     4.56   32.3     2984.
##  3 1836   223    13.8  23.8     5.10   27.9     2058.
##  4 1707   467     9.25  9.45    3.13   28.2     2099.
##  5 1821   223    13.2  20.4     4.74   26.8     1848.
##  6 594    676    14.8  29.2     5.62   26.1     1733.
##  7 354    676    11.1  14.7     4.00   29.4     2341.
##  8 521    223    11.7  16.6     4.27   32.6     3060.
##  9 344    711    12.7  20.8     4.78   27.4     1956.
## 10 1532   223     5.26  3.24    1.39   27.7     2008.
## # i 1,390 more rows
```

The estimate from the above question has been used on the above transformation to the cost data creating a new variable *tf_cost*. This is located to the right of the column containing the untransformed cost data. This transformed data makes the spread more evenly distributed, by reducing the high cost values.

10(e) Scatterplot of the Transformed Bivariate Data

```r
ggplot(question10, aes(x=hours, y = tf_cost))+
  geom_point()+
  geom_smooth(method="lm", se=FALSE)
```



The scatterplot of the transformed bivariate data of *tf_cost* against *hours* depict a stronger linear straight line, this indicates that the transformation was successful.

Now that the data is more linearly related, this makes it more efficient to model linear regression.

10(f) The Effect of the Transformation on the Linearity

The transformation reduced the overall skewness and evenly distributed the spread of the data. The effect on the linearity made the distribution between hours and cost more consistent. This is seen where the hours increase, the costs also increase.

Practically, this makes sense where players are more suceptable to have injuries the longer they are on the field playing soccer. The transformation overall made the model more reliable for prediction.

Q11. Fitting a New Model with our Box-Cox Medical Data

11(a) Fitting the Linear Model

```
question11 <- question10
df_question11 <- linear_reg()%>%
  set_engine("lm")%>%
  fit(tf_cost ~hours, data = question11)
summary(df_question11$fit)
```

```
##
## Call:
## stats::lm(formula = tf_cost ~ hours, data = data)
```

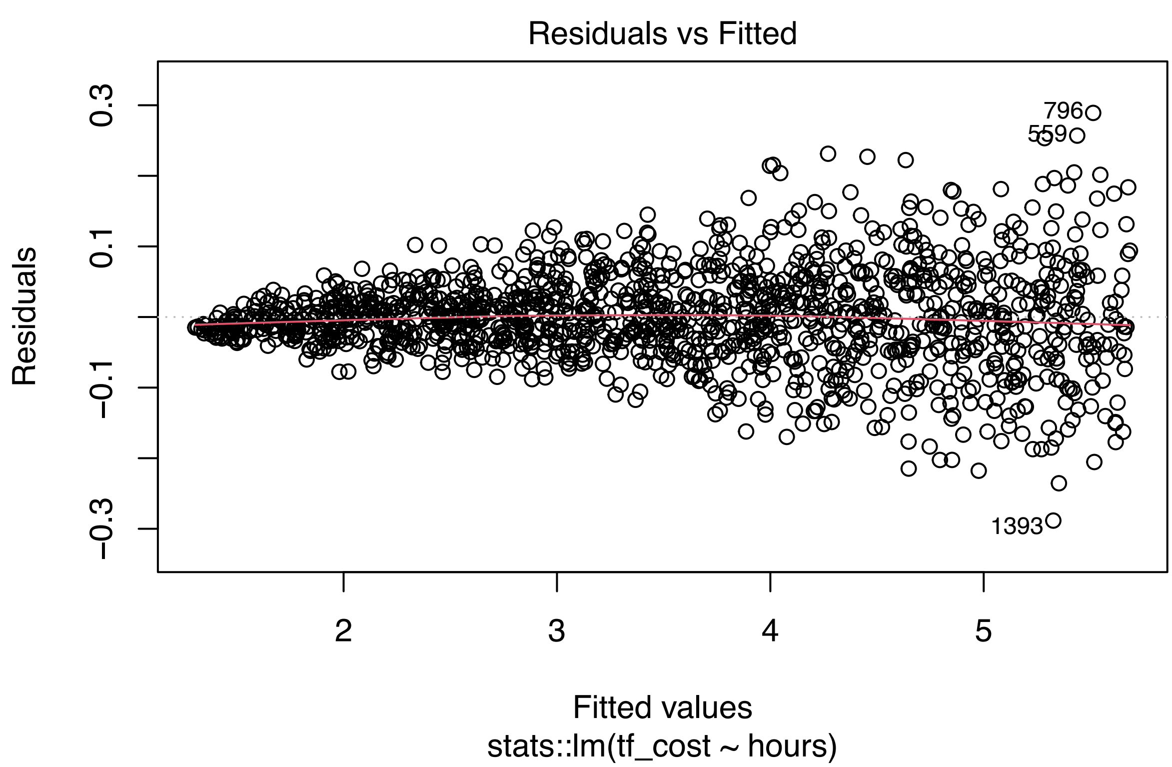
```
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.288373 -0.034041 -0.002934  0.034275  0.289092
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.8949084  0.0068053  -131.5   <2e-16 ***
## hours        0.4390068  0.0006532   672.1   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.06605 on 1398 degrees of freedom
## Multiple R-squared:  0.9969, Adjusted R-squared:  0.9969
## F-statistic: 4.518e+05 on 1 and 1398 DF,  p-value: < 2.2e-16
```

The linear model was fitted to *tf_cost* as the response variable and *hours* as the predictor.

11(b) Checking if the Model Satisfies The Assumptions of Linearity, Homoscedasticity and Normality

Linearity

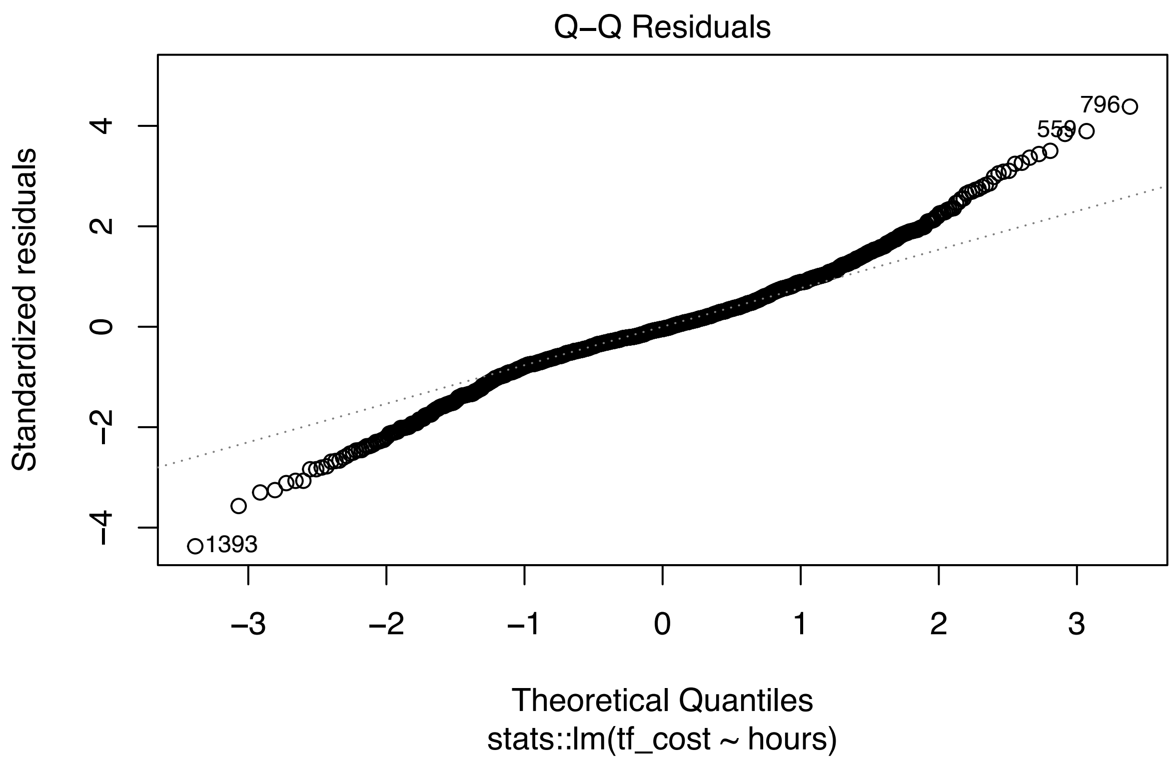
```r
plot(df_question11$fit, which = 1)
```



The plot depicting the linearity with Residual and vs Fitted is satisified because the data are evenly spread above and below the red line. The fitted value is closer to 2 the data is closer to the line in correlation to point 5, as seen in the data 796, and 1393 where the data is further apart, although they are all flat and equal alongside the red line resulting in a satisfied linear assumption.

Normality

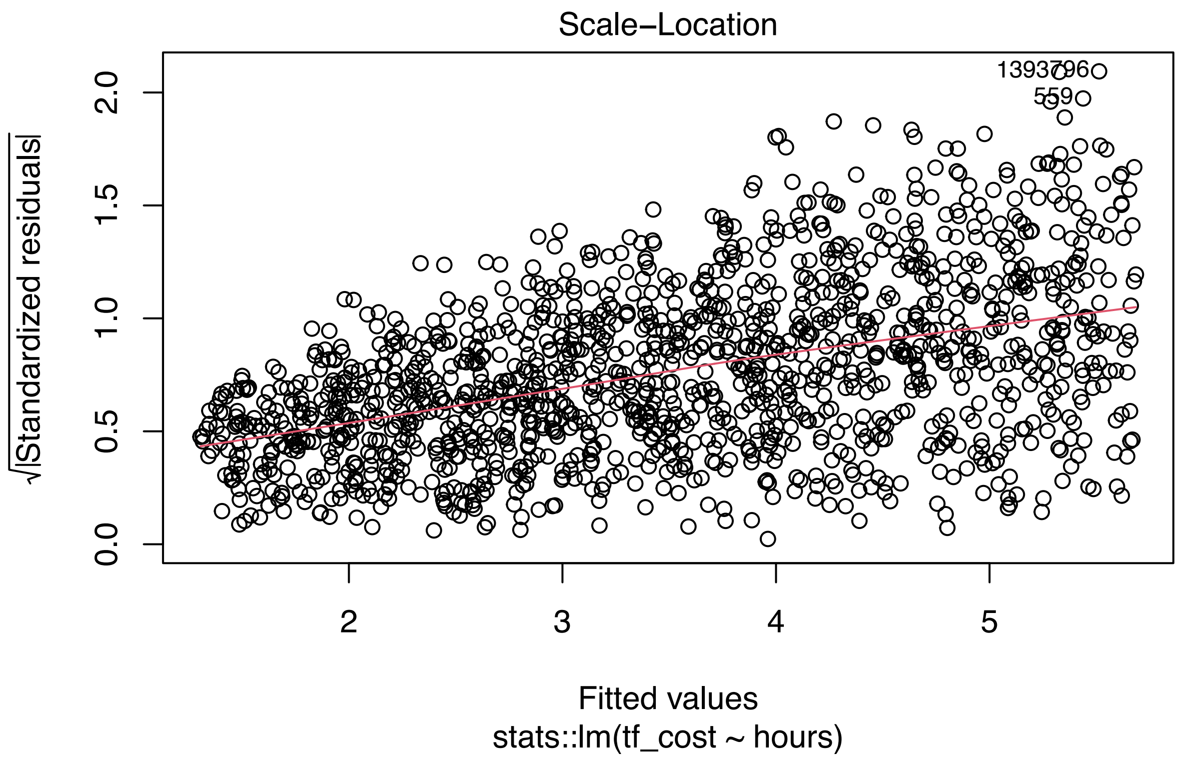
```
plot(df_question11$fit, which = 2)
```



The normality assumption for linearity in the Q-Q Residuals plot, is also satisfied, where the points are close to the grey line, this means that the points are normally distributed. Along the x-axis (Theoretical Quantiles), there are evident outliers of 1393 before -1, and after 2 of 796. Although these outliers shift away from the line, the data is mostly consistent along the line, resulting in a satisfied linear model.

Homoscedasticity

```
plot(df_question11$fit, which = 3)
```



The Scale-Loation graph depicts the (square root Standarized residuals) against the Fitted Values. The residuals are mostly evenly spread alongside the line. It is evident that as the Fitted value increase the residuals shift further away from the line as seen in the outlier 559. Their is a slight upwards slope evident where most of the points are consistent alongside the red line. This indicates that the assumption for the homoscedasity is mostly satisfied.

Independance

Independence suggests that each data point is totally unrelated. The possible problem from the Independence assumption is that their is no evidence that the medical costs are not related.

The cost of one player is independent of the cost of other players in the team this shows that the other variables are independent.

Q12. True Linear Model Equation when Fitted to the Entire Population

12(a) The General Equation for the True Linear Model

The transformed costs is the response variable, and we'll write the general equation for the true linear relationship between hours played and the Box-Cox transformed costs across the entire population. This equation will include the actual parameters and an error term to reflect how things work in the full population.

The Linear Model Equation for the Population is given by:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

where: y_i: Box-Cox transformed medical cost for the ith observation. x_i: the hours played for the ith observation. β_0: the population intercept. β_1: the change in medical costs per additional hour played. ε_i: the random error term for the ith observation, where $\varepsilon_i \sim N(0, \sigma^2)$.

12(b) Writing the Estimated Line of Best Fit

The model summary provides the following estimates for the parameters: Estimated intercept $\hat{\beta}_0 = -0.8949084 \times 10^{-1}$. Estimated slope $\hat{\beta}_1 = 0.4390068 \times 10^{-4}$. Using these estimates, the equation for the line of best fit for the sample population is:

$$\hat{y}_i = -0.8949084 + 0.4390068 x_i$$

where: $\hat{y}_i$: Predicted Box-Cox transformed medical cost for the ith player i. x_i: Hours played for the ith individual i. $\hat{\beta}_0 = -0.2642$: Estimated intercept from the sample. $\hat{\beta}_1 = 0.0008077$: Estimated slope from the sample. The estimated distribution for the errors, based on the residual standard error of *0.06605*.

This means that the every additional hour played, the Box-Cox transform medical cost is predicted to increase *0.4390068* units. There is a strong linear relationship, depicted using the adjusted r squared value of *0.9969*, which equates to 99.7%. This indicates that almost all the difference in medical costs in correlation between how long they are on the field. The other *0.3%* of variation is due to factors including variation in treatment costs, or injury history of the players that may influence the results.

Q13. Predicting the Medical Costs for John Coleman

The prediction of the medical costs for John Coleman was done using the Box-Cox linear regression model, based on the three different scenarios * 150 minutes of on-field time over the whole season * 500 minutes in total. * 700 minutes over the season and finals. These have been converted to hours diving them by 60 as seen below.

The predictions were made using both the 99.9% prediction interval and, 99.9% confidence interval.

```
pred_values<- tibble(
  hours=c(150/60,500/60,750/60)
)
prediction <- predict(df_question11, new_data = pred_values)

predict_int <- predict(df_question11, new_data = pred_values,
                       type = "pred_int", level = 0.999)

confidence_int <- predict(df_question11, new_data = pred_values,
                          type = "conf_int", level = 0.999)

predict_int
```

```
## # A tibble: 3 x 2
##   .pred_lower .pred_upper
##         <dbl>       <dbl>
## 1     -0.0159       0.421
## 2      2.55         2.98
## 3      4.37         4.81
```

```
confidence_int
```

```
## # A tibble: 3 x 2
##   .pred_lower .pred_upper
##         <dbl>       <dbl>
```

```
## 1       0.185       0.220
## 2       2.76        2.77
## 3       4.58        4.60
```

```
predict_cost <- ((prediction$.pred * df_bc_10$lambda) +1)^(1/df_bc_10$lambda)
predict_lower <- ((predict_int$.pred_lower *
                     df_bc_10$lambda +1^(1/df_bc_10$lambda)))
predict_upper <- ((predict_int$.pred_upper *
                     df_bc_10$lambda +1^(1/df_bc_10$lambda)))
confidence_upper <- ((confidence_int$.pred_lower *
                        df_bc_10$lambda +1^(1/df_bc_10$lambda)))
confidence_lower <- ((confidence_int$.pred_upper *
                        df_bc_10$lambda +1^(1/df_bc_10$lambda)))

predict_cost
```

```
## [1]  1.217829  7.743248 19.159349
```

```
predict_lower
```

```
## [1] 0.9955579 1.7127636 2.2249294
```

```
predict_upper
```

```
## [1] 1.117903 1.834786 2.346969
```

```
confidence_upper
```

```
## [1] 1.051887 1.771840 2.283755
```

```
confidence_lower
```

```
## [1] 1.061574 1.775710 2.288144
```

The predicted medical costs for *John Coleman*, was the following: * 150 minutes of on-field time over the whole season there is an estimated *1.22* * 500 minutes in total there is an estimated *7.74*(thousand) * 700 minutes over the season and finals is an estimated *19.16*

The predicted lower and upper values depict how certain the model is about the range of cost. John's cost could potentially fall between ***0.99 - 1.12*** for 150 minutes, *1.71 - 1.83* for 500 minutes, and *2.22 - 2.35* for 700 minutes of on field time.

The predicted upper and lower confidence intervals are the following for John's on field time range between *1.05-1.06* for 150 minutes, *1.77 - 1.78* for 150 minutes, *2.28 - 2.29* for 500 total minutes.

All the estimated values above are in thousands.

The results indicate a strong positive indication between medical costs and on field time. This means that there has been a successful variance in producing accurate outcomes.

Q14. The Expected Medical Costs for John Coleman and what this means for the

Gaelic Football Team.

The expected medical costs for John Coleman show a strong correlation with hours played, with the Adjusted R-squared value of *0.9969* or around *99.7%* the changes in costs between players can be depicted by the number of hours played in the model. This means that only *0.03%* of the variation is due to randomness, indicating the accuracy of our model. This could be to the playerspre-existing injuries, position he is playing on the field and level of intensity, and the age of the player and possible recovery rate.

In correlation with John Coleman's medical costs, the model suggests that their is a strong indication that the longer the hours John plays on the field, the more likely he is to receive an injury and have higher medical costs.

References

BBC Bitesize, 2025, Days and months, viewed 31 October 2025, https://www.bbc.co.uk/bitesize/articles/zr4hhbk